The Untaxables? Effective Capital Gains Taxes with Diversified Portfolios

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Abstract

Capital gains are central to the taxation of the very rich, yet realization taxes allow investors to defer payment. Despite its policy relevance, a measure of long-run effective taxation capturing optimal deferral has been missing because of technical challenges. I close this gap with a model that represents portfolios by distributions of embedded gains and losses. It describes a dynamic trading problem in which today's sales change tomorrow's tax opportunities. I establish when harvesting losses and selling the least appreciated positions is optimal over any horizon. Aggregate downside risk can reverse this rule. I express the consumption benefits of tax-optimal trading as effective tax rates and equivalent returns. The calibration matches historical U.S. market conditions and the OECD average capital gains tax rate. A 25% tax on realized gains and dividends sustains the same consumption perpetuity as an accrual tax of about 14%. The corresponding return premium is 74 basis points: at 25%, an accrual-taxed portfolio needs this additional expected annual return to match consumption under realization taxes. A cohort-model extension predicts that revenue under realization taxes is 49% lower than under accrual taxes. Continued deferral thus limits the revenue and progressivity of realization-based capital gains taxes.

JEL Classification: H21, H24, D14, G11

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I Introduction

Capital gains are central to the income and taxation of the very rich (Piketty et al., 2023, Saez and Zucman, 2021, Scheuer and Slemrod, 2020, 2021). In 2021, net realized capital gains accounted for 58% of income among the richest U.S. taxpayers.[1] Their taxation therefore matters for both public revenue and the distribution of tax burdens. However, the official ("statutory") tax rate on investment profits overstates what these investors effectively pay. Because capital gains are commonly taxed when assets are sold ("realized"), investors can postpone taxes by retaining appreciated asset positions ("deferral") and reduce current taxes by realizing losses ("loss-harvesting"). For wealthy households with broad portfolios, and for a wider population of households using recent fintech innovations[2], these deferral and loss-harvesting incentives are a recurring feature of taxation. Assessing the reach of a capital gains tax therefore requires understanding how investors respond to these incentives.

This paper studies how investors with many assets optimally choose which gains and losses to realize over time. To quantify the resulting advantage of optimal realization timing, I solve a dynamic trading problem in which today's sales change tomorrow's opportunities for avoiding or postponing tax. I find that taxing upon realization rather than accrual has large quantitative effects on investor consumption and government revenue. At the baseline, optimal trading reduces the investor's effective tax rate and the government's revenue flow to little more than half of what the statutory rate suggests. Investors achieve these savings by continually deferring the most appreciated remaining positions.[3]

I find that, at a statutory rate of 25%, optimal trading perpetually sustains the same constant real consumption as an accrual tax of about 14%. I call this consumption-equivalent accrual rate the *effective tax rate*. This effective rate is 44% lower than the statutory rate. The baseline features a 25 percent realization tax, which corresponds to the 2025 OECD average capital gains tax,[4] and market conditions calibrated to historical U.S. asset returns, inflation rates, and corporate dividend payout practices. The gap between statutory and effective rates undermines tax progressivity because capital gains are concentrated among the very rich. To raise the effective rate to the 35.1 percent OECD average tax wedge on labor, one would need a 57 percent realization tax on gains and dividends, more than twice the baseline.[5] Reaching this

[1] Author's calculation from Internal Revenue Service (2021, Table 1.4, tax year 2021). Net realized capital gains comprise taxable net gains and capital-gain distributions, less deductible net capital losses.

[2] Robo-advisors are one example. Reported assets in 2026 include Wealthfront's $93.2 billion in platform assets, Betterment's over $65 billion under management, and Wealthsimple's C$124.8 billion under administration (Nasr, 2026).

[3] This result requires no forgiveness event such as the *basis step-up* for inherited portfolios in the U.S.

[4] Author's calculation for 28 OECD countries with realization taxes, using statutory rates on long-held listed shares for non-substantial shareholdings. Source: Mengden (2025, Table 2). The mean is 24.75 percent.

[5] Single, childless worker earning the national average wage in 2025; income tax and employee and employer social contributions, net of benefits, as a share of total labor costs (OECD, 2026, Table 1.2).

benchmark implies an average pass-through from statutory to effective rates of 0.67. Locally, pass-through is even smaller: a one-percentage-point increase in the statutory rate from the baseline raises the effective rate by about 0.60 percentage points.

The same consumption advantage can also be expressed in expected-return units, connecting the effective tax rate to return-alpha measures in finance. Here, the *return alpha* is the additional annual expected return an accrual-taxed portfolio would need, at the same statutory rate, to sustain the consumption achievable through optimal realization trading. At the baseline, this alpha is 73.8 basis points: 44.4 basis points come from single-asset deferral, 26.3 from tax-lot selection, and 3.1 from their interaction.

Because the effective tax rate and return alpha measure consumption equivalence rather than government collections, I also calculate revenue in a cohort version of my model with accidental bequests. The model aggregates tax payments across portfolios with different tax histories, capturing how continued deferral affects long-run collections. At the baseline, realization taxation collects 49 percent less revenue than taxing gains upon accrual at the same statutory rate.

My quantitative analysis contributes to the literature in two ways. First, it provides unified measures of the realization advantage, connecting effective tax rates in public finance with return alphas in finance. Second, it quantifies this advantage and its sensitivity to statutory tax rates and market conditions, and assesses its implications for government revenue.

Quantifying these effects requires a tractable model of optimal trading. My first theoretical contribution is therefore to identify the relevant state of a large taxable portfolio. An intuitive economic insight applies: within a portfolio, the identity of an individual asset becomes negligible once idiosyncratic risk is diversified. This allows me to represent the portfolio by the distribution of market value across tax lots—the state representation of the Tax Portfolio Model. A tax lot groups assets whose sale realizes the same gain or loss per dollar of value sold; these are precisely the assets that have accrued the same cumulative gross return since purchase. My second theoretical contribution is to use this state to formulate a recursive class of portfolio problems and characterize trading rules that determine which assets should be sold today once future trading opportunities are taken into account.

The two theoretical contributions overcome two obstacles that have long constrained the literature on portfolio choice under realization taxation. The first is computational. Realization taxation requires tracking both market value and tax basis for every asset, while choosing consumption, liquidation, and reinvestment over many periods. The resulting state and choice spaces quickly become too large for conventional numerical methods. The second is theoretical. With finitely many assets, trades motivated by risk and taxes compete: a tax-favorable sale can leave the wrong risk exposure, while a risk-motivated rebalance can trigger taxes. A simple rule for which asset to sell first may therefore not exist. These difficulties help explain why

analytical characterizations often focus on a single asset (Constantinides, 1984), multi-asset studies rely on back-testing and trading heuristics (Fischer and Gallmeyer, 2016, Haugh et al., 2016), and life-cycle models generally retain only one or two risky assets (Dammon et al., 2001a, 2004).

My approach is to let the number of assets grow without bound in a portfolio with many small positions. As asset-level risk diversifies, choosing which securities to sell within a tax lot no longer affects the portfolio's risk or expected return. Security-specific risk diversifies away in this limit, but common aggregate shocks remain. Individual assets can therefore lose value even when the diversified portfolio as a whole earns a nonnegative return. The portfolio can then be represented as a distribution of market value across tax lots. Section II establishes this state-reduction result formally, but the intuition is simple: consider two assets with the same expected return. If the portfolio spreads risk across many small positions, selling either asset has a negligible effect on the risk and expected return of the retained portfolio. Their tax consequences, however, depend on their tax lots. Per dollar sold, the more appreciated asset realizes the larger gain and triggers the larger tax payment. Hence, in the large-portfolio limit, if two liquidation rules each sell the same amount from every tax lot, choosing different securities within a lot does not affect consumption or utility. By contrast, shifting sales across tax lots changes realized gains, current taxes, and which embedded losses remain available to shelter future gains. The distribution of asset value across tax lots is therefore the relevant state of a large taxable portfolio. Intuitively, the investor chooses how much to sell from baskets of securities grouped by their accrued returns.

Representing the portfolio this way makes trading operations intuitive. Let an investor hold a portfolio $w(\cdot)$, a distribution of market value over cumulative gross return bins $r > 0$. Because newly purchased assets have gross return one, investment adds a point mass at tax lot $r = 1$. Selling assets subtracts value from the tax lots sold, and period returns transport the post-trade distribution into the next period. After-tax liquidation value equals the total value sold across tax lots minus taxes, which are calculated lot by lot and summed. This structure accommodates the three tax systems analyzed in this paper, which differ in how they treat realized losses. Full use of losses (FUL) pays an immediate rebate on realized losses. No use of losses (NUL) taxes realized gains and ignores realized losses. Limited use of losses (LUL) offsets realized gains with realized losses and carries any unused loss forward as a tax credit; the carryforward adds a single tax-credit balance to the portfolio state. Section III develops this structure formally.

Using these ingredients, I define two classes of recursive problems, over both finite and infinite horizons. The first class requires finding the largest constant real withdrawal the portfolio can sustain. Its infinite-horizon version—the perpetuity problem—provides the basis for the taxation measures in this paper. By abstracting from life-cycle mechanisms that

make consumption timing specific to each investor, it provides a common benchmark for comparing tax systems. The second class consists of classical time-separable discounted-utility problems, allowing consumption to vary over time. This formulation extends life-cycle models in the spirit of Dammon et al. (2001a) to a diversified continuum of assets. Both classes use the same portfolio state and trading structure. Section IV establishes that these problems are well defined, that backward iteration converges, and that optimal policies are attained for a broad class of tax systems.

To solve either recursive problem, and hence to construct effective taxation and return alpha measures, I establish what optimal trading actually looks like. The analysis generalizes a rule familiar to investors: selling loss positions and reinvesting the proceeds, then meeting cash needs by selling the least appreciated remaining positions. This prescription goes back to Constantinides (1983, 1984), appears in standard investments curricula (Bodie et al., 2013), and is automated by robo-advisors. Formally, the class of loss-harvesting cutoff liquidation (LHCL) strategies harvests a lower interval of losses and funds consumption by selling remaining positions in ascending order of cumulative return. Liquidation stops at a cutoff tax lot once the required cash target is reached. I characterize when an optimal trade can be found within this class in Section IV.

To prove optimality, I use arguments that compare trades that fund the same current consumption and ask which trade leaves the investor with a continuation portfolio better suited to fund future consumption. The relevant comparison under full use of losses taxation is the right-domination order: one portfolio right-dominates another if it offers at least as much after-tax liquidation value and contains at least as much market value in the right tail, i.e., above every cumulative-return cutoff. The first condition ensures that the dominant portfolio can generate weakly more liquidity in the short run; the latter preserves wealth in the more appreciated tax lots. Keeping these positions invested postpones the largest embedded tax bills. Moreover, for a given proportional increase in market value, the tax liability as a share of market value rises less for a more appreciated position.[6] The recursive argument shows that the continuation value function is increasing in this order: a right-dominant continuation portfolio is worth at least as much to the investor as any dominated one. Thus, among trades that fund the same current consumption, the investor prefers a trade whose continuation

[6]Consider a capital gains tax rate $0 < \tau < 1$. Portfolio A holds $1/(1-\tau)$ dollars in tax lot $r = \infty$: the limiting case of an infinite cumulative gross return and zero cost basis. Its entire market value is therefore taxable upon liquidation. Portfolio B holds one dollar of assets at cost, in tax lot $r = 1$. Both portfolios yield one dollar upon immediate liquidation, but A right-dominates B: it contains more market value, all in a more appreciated tax lot. If both portfolios subsequently appreciate by a factor $R > 1$, A remains in tax lot $r = \infty$ and yields $(1-\tau)R/(1-\tau) = R$ dollars upon liquidation. B instead moves to tax lot $r = R$ and yields $1 + (R-1)(1-\tau)$ dollars. A therefore delivers $\tau(R-1) > 0$ more dollars after tax, despite the portfolios' identical initial liquidation values. With full refunds for realized losses, a gross return $0 < R < 1$ reverses this comparison, illustrating why downside returns matter.

portfolio right-dominates that of the alternative.

LHCL strategies produce the required portfolio ordering in two steps. Harvesting losses for reinvestment brings their tax benefits forward and shifts market value from low-return tax lots ($r < 1$) into newly purchased positions ($r = 1$). Raising additional cash from the least appreciated remaining positions preserves market value in the right tail. Aggregate risk, however, introduces a qualification. To guarantee that right dominance survives subsequent returns, the diversified portfolio must earn a nonnegative net period return in every aggregate state. If an aggregate downturn can make the portfolio lose value, two portfolios that were ordered immediately after trading need not remain so after returns are drawn. Under this return-support condition, each feasible trade can be matched by an LHCL trade that funds the same consumption and guarantees a right-dominant continuation portfolio. Thus, trades from the LHCL class are optimal under full use of losses and the support condition. Reversal examples show why the qualifications matter: first, aggregate downside risk can favor selling appreciated positions and retaining losses because doing so reduces exposure to the aggregate shock. For the same consumption, selling more appreciated tax lots removes more market value; their tax liability per dollar of market value also falls less for a given proportional price decline. Second, when realizing losses generates neither refunds nor tax credits (as under NUL), the investor may prefer to keep them invested as a shield against future appreciation. In both cases, LHCL strategies are no longer guaranteed to be optimal. In contrast, reinvestment costs can make harvesting small losses unattractive and shrink the harvest region, but leave LHCL optimality unaffected. Finally, my results extend to limited use of losses with perfect carryforward and frictionless reinvestment, under the same return-support assumptions and a dominance order that also accounts for unused loss credits.

Once the optimal policy is known, effective taxation becomes computable. Section V does this in the setting without aggregate risk, where LHCL strategies are optimal, and studies how the effective rate changes under alternative policy configurations and market conditions. The baseline allows losses to offset gains, with perfect carryforward and frictionless reinvestment.[7] The value of realization discretion then depends on the opportunities to fund consumption while preserving appreciated positions. Greater return dispersion creates more losses and small gains from which the investor can fund consumption, lowering effective taxation (-0.033 pp per 1% increase in dispersion). Dividends force taxable payouts ($+0.12$ pp per 1 pp increase in the payout share), while inflation creates taxable nominal gains without increasing real returns ($+1.31$ pp per 1 pp increase in inflation). All effects are evaluated locally at the baseline. Allowing investors to borrow leaves the effective tax rate unchanged if interest is not deductible,

[7] The numerical benchmark uses a 5-percent expected real return, 30-percent return dispersion, 2-percent inflation, a 30-percent dividend payout rate, and a 25-percent statutory tax rate. The investor can carry unused loss credits forward without expiry and reinvest without cost.

while deductible interest acts like a loss credit and can lower it sharply.

The effective tax rate captures the total value of realization discretion but does not separate its sources. Section VI uses return alpha to distinguish the contribution of single-asset deferral from that of tax-lot selection and their interaction. It compares optimal trading of a single diversified asset, such as a mutual fund mirroring the market, with optimal trading across many assets. A single diversified asset offers deferral and risk diversification, while holding many assets also allows the investor to select which gains and losses to realize. The resulting total alpha is similar in magnitude to empirical estimates in the literature. Chaudhuri et al. (2020), for example, estimate a historical tax-loss-harvesting alpha of 1.08 percent before transaction costs. Across papers, however, estimates differ in the tax provisions and trading frictions included, the return data and tax rates used, and the benchmark against which the tax advantage is measured. These differences limit direct comparisons and motivate a portable theoretical benchmark. The return alphas developed here provide such a benchmark: they express the consumption benefit of optimal realization trading as an equivalent annual expected return under explicit tax and market assumptions.

In sum, this paper isolates the investor-side trading margin and provides a flexible framework for analyzing the benefits investors derive from the discretion built into realization-based capital gains taxation. To focus on this margin, I hold the return process fixed and study how investors choose among tax lots over time.[8] The large-portfolio result assumes that assets share a common expected return and that security-specific risk diversifies across many balanced positions (Appendix Section A extends the model to heterogeneous, nondiversifiable factor exposures when the distribution of factor loadings is independent of current tax lots). The framework therefore applies most directly to investors with broad taxable portfolios and access to tax-aware trading. Belda (2026) studies the complementary asset-pricing margin in an equilibrium with a single risky asset. The framework also provides a foundation for studying life-cycle consumption and portfolio choice with many risky assets. In follow-up work, Bennhoff and Scheuer (2026) use it to analyze a quantitative life-cycle model with many assets and aggregate risk, comparing alternative treatments of capital gains at death.

Related Literature

This paper studies investors whose taxable portfolios are broad enough to make tax-lot selection a recurring economic choice. A recent literature documents that such investors matter. Capital gains are central to the income and taxation of the very rich (Piketty et al., 2023, Saez and Zucman, 2021, Scheuer and Slemrod, 2020, 2021). Bach et al. (2020) and Fagereng et al. (2020)

[8] Embedded gains are primarily investor-specific state variables because they depend on individual purchase histories. They therefore need not aggregate into an economy-wide state. Their first-order effects on equilibrium asset prices are not mechanically determined by the portfolio problem studied here.

document persistent heterogeneity in returns to wealth. Balloch and Richers (2023) study the portfolios of wealthy households, while Gabaix et al. (2023) document their security-level holdings, flows, and rebalancing. Msall and Næss (2026) show that gains held until death have direct fiscal consequences through their treatment at death. How investors should trade such portfolios is the question taken up by the portfolio-choice literature on realization taxation.

The modern portfolio-choice literature on realization taxation begins with Constantinides (1983, 1984). Constantinides (1983) derives the canonical logic of realizing losses and deferring gains in an environment in which investors can offset appreciated positions without realizing them.[9] Constantinides (1984) studies differential taxation of short- and long-term gains and the option to reset an asset's holding period. Dybvig and Koo (1996) formulate a dynamic exact-basis portfolio problem and solve numerical examples with small portfolios. DeMiguel and Uppal (2005) solve terminal-wealth problems with exact basis for short-sale-constrained investors holding a small number of risky assets. Other contributions develop numerical methods for richer portfolios and tax environments. Haugh et al. (2016) analyze dynamic tax-aware policies under limited loss offsets, Fischer and Gallmeyer (2016) study heuristic trading rules with transaction costs, and Moehle et al. (2021) incorporate exact tax lots and wash-sale restrictions into a convex-optimization approach. These papers substantially broaden the class of tax-aware portfolio problems that can be solved numerically.

A larger life-cycle literature jointly studies consumption, investment, and tax-basis dynamics. Dammon et al. (2001a,b) analyze portfolios with one or two risky assets, and Dammon et al. (2004) add tax-deferred accounts and asset-location choices. Fischer et al. (2013) find small welfare gains from tax-optimal trading over simpler policies in a life-cycle model with one risky asset and labor income. My quantitative results, however, show that optimal realization choices can substantially reduce effective taxation and raise sustainable portfolio consumption, highlighting the importance of these incentives for the richest households, whose resources depend heavily on financial wealth. This branch connects to the earlier asset-location analyses of Auerbach and King (1983), Black (1980), Tepper (1981), and to the later work of Garlappi and Huang (2006), Huang (2008). Later work isolates particular features of the tax environment: short-selling costs, wash-sale constraints, limits on loss rebates, loss carryovers, tax-deferred accounts, transaction costs, and year-end taxation.[10] Exact-basis models must track asset- and lot-specific state variables, so the state expands rapidly with the number of assets and purchase

[9] Constantinides' many-asset characterization permits "shorting against the box": short selling an appreciated asset for liquidity while retaining the long position and its deferred gain. I exclude this transaction because it separates liquidity from tax realization. Tax codes may also neutralize it; in the United States, for example, IRC §1259 generally treats it as a constructive sale.

[10] Gallmeyer et al. (2006) study two risky assets and costly short selling; Jensen and Marekwica (2011) analyze wash-sale constraints; Marekwica (2012) limits rebates on realized losses; Ehling et al. (2018) introduce carryover losses; Fischer and Gallmeyer (2017) combine limited loss use with taxable and tax-deferred accounts; and Dai et al. (2026) study transaction costs and year-end taxation.

histories. This helps explain why explicit life-cycle models generally remain limited to one or two risky assets. This paper replaces asset-specific basis histories with a distribution of value across tax lots and uses that distribution to characterize long-horizon trading. Bennhoff and Scheuer (2026) build on the Tax Portfolio Model to study a quantitative life-cycle problem with many assets and aggregate risk and compare alternative treatments of capital gains at death.

The state representation also connects to theories of large portfolios and economies represented by distributions. Chamberlain and Rothschild (1983), Ross (1976) establish approximate factor structures under which security-specific risk can be diversified. Al-Najjar (1998) develops factor pricing in a large asset economy, while Khan and Sun (1997, 2003) extend arbitrage pricing to economies with an asset continuum. In these theories, systematic exposures remain relevant for pricing. This paper applies their large-portfolio logic under common expected returns and diversifiable residual risk.

A separate public-finance literature studies realization responses, effective capital-gains taxation, and government revenue. Poterba (2002) surveys how taxation affects household portfolio behavior, Alan et al. (2010) document portfolio responses, and Agersnap and Zidar (2021) estimate realization elasticities and revenue-maximizing tax rates. Ehling et al. (2026) directly connect optimized portfolio choice to aggregate realizations and revenue by simulating investors under historical returns and capital-gains tax rates. The empirical and reduced-form branches of this literature generally study how total realizations respond without deriving which assets investors optimally sell. This paper supplies that position-selection mechanism. For the revenue comparison, the cohort version of my model draws on Yaari's (1965) treatment of uncertain lifetimes and Blanchard's (1985) cohort structure with a constant mortality rate. I use this demographic structure to aggregate the tax payments generated by the model's optimal perpetuity policies in an economy with accidental bequests.

An effective-tax literature translates realization taxation into an accrual-equivalent rate. Glenday and Davies (1990), Poterba (1989), Protopapadakis (1983) infer such rates from accrued and realized gains or by equating terminal wealth, present values, or internal rates of return; Glenday and Davies (1990) also distinguish investor-side equivalence from revenue equivalence. The effective tax rate and closed-cohort rate extend these two criteria to an optimally managed many-asset perpetuity. A finance literature expresses the benefits of tax-aware trading in annual-return units: Chaudhuri et al. (2020) estimate the historical alpha from tax-loss harvesting, while Mainardi (2026) measures tax alpha from differences in household tax expenses. The return alphas developed here likewise translate realization benefits into expected-return units, but distinguish single-asset deferral, tax-lot diversification, and their combined value. The total alpha is dual to the effective tax rate, so the two measures connect the effective-rate language of public finance to the alpha language of finance within the same optimal-trading problem.

II Tax Motives Dominate in Large Portfolios

This section formalizes and proves the claim that in large portfolios studied in this paper, liquidation decisions are governed at first order by tax considerations rather than by security-specific risk management. I show that, as the number of assets grows, the distribution of portfolio value across *tax lots*—positions grouped by the same gross cumulative return—becomes the only relevant state variable for the limiting portfolio problem.

Consider a two-period investor who chooses which assets to liquidate from a portfolio of n securities. Denote by $\boldsymbol{\alpha}^n = (\alpha_1^n, \ldots, \alpha_n^n) \in [0,1]^n$ the vector of liquidation shares, where α_i^n is the fraction of asset i's current market value liquidated at date 1. The investor has expected utility

$$U^n(\boldsymbol{\alpha}^n) := \mathbb{E}\Big[u\big(c_1^n(\boldsymbol{\alpha}^n), c_2^n(\boldsymbol{\alpha}^n; Z, \boldsymbol{R}^n)\big)\Big], \tag{1}$$

Utility is defined over date-1 and date-2 consumption, c_1^n and c_2^n. Both depend on the date-1 liquidation choice $\boldsymbol{\alpha}^n$, while c_2^n also depends on the aggregate state Z and the vector of asset-level returns $\boldsymbol{R}^n$. The function u is continuous, increasing, concave, and continuously differentiable. At date 1, the investor is endowed with n securities with current market values $\omega_i^n \geq 0$, collected in $\boldsymbol{\omega}^n$, which amount to total market value $\sum_{i=1}^n \omega_i^n = M$. Each security i also has an exogenously given cumulative gross return $r_i > 0$ since purchase, so its cost basis is ω_i^n / r_i. Assets with the same accrued return carry the same embedded gain per unit of market value and are therefore in the same tax lot.

The liquidated market value, realized gain, and date-1 consumption are

$$\mathrm{M}_1^n(\boldsymbol{\alpha}^n) := \sum_{i=1}^n \alpha_i^n \omega_i^n, \qquad \mathrm{CG}_1^n(\boldsymbol{\alpha}^n) := \sum_{i=1}^n \alpha_i^n \omega_i^n \Big(1 - \frac{1}{r_i}\Big), \qquad c_1^n = \mathrm{M}_1^n - \mathrm{T}_1(\mathrm{CG}_1^n),$$

whereby date-1 capital gains taxes are determined by the continuous map T_1. This formulation takes realized gains as the primitive tax base. The map T_1 summarizes date-1 taxation, while the date-2 map T_2 will allow taxes to depend both on current realized gains and on gains realized at date 1.

The unsold shares are liquidated at date 2. Between dates 1 and 2, each asset earns a positive gross return R_i^n. Asset returns have a common finite mean $\mathbb{E}[R_i^n] = 1 + \mu > 1$ for every $i \leq n$, but their marginal distributions and variances may otherwise differ. Asset-specific variances and cross-asset dependence are summarized by the positive semidefinite covariance matrix Σ^n. In addition, an independent aggregate factor $Z \sim f_Z$ with $\mathbb{E}[Z] = 1$ and $\mathbb{E}[Z^2] < \infty$ is common to all assets and cannot be diversified away. Date-2 liquidated market value and realized gain are therefore

$$\mathrm{M}_2^n(\boldsymbol{\alpha}^n; Z, \boldsymbol{R}^n) := \sum_{i=1}^n (1 - \alpha_i^n)\omega_i^n Z R_i^n, \qquad \mathrm{CG}_2^n(\boldsymbol{\alpha}^n; Z, \boldsymbol{R}^n) := \mathrm{M}_2^n - \sum_{i=1}^n \frac{(1-\alpha_i^n)\omega_i^n}{r_i}.$$

Given continuous date-2 taxes $\mathrm{T}_2(\mathrm{CG}_2, \mathrm{CG}_1)$, future consumption is

$$c_2^n = \mathrm{M}_2^n - \mathrm{T}_2(\mathrm{CG}_2^n, \mathrm{CG}_1^n).$$

Random variates and taxes are assumed well-behaved, such that date-2 tax payments are bounded *a.s.* and that the tax levied on a realization never exceeds the asset's market value.

Now consider a sequence of such utility-maximization problems indexed by n, in which the investor gains access to an ever larger asset universe while the total market value of her endowment remains fixed at M. For simplicity, assume that the set of embedded cumulative returns that appears along the sequence, $\mathcal{R} := \{r_i : i \in \mathbb{N}\}$, is finite (Section III removes this requirement). Also assume that, for every $r \in \mathcal{R}$, the total market value invested in lots indexed by r converges to some number $w(r) \geq 0$.[11] For each r, define the function $w : \mathcal{R} \to \mathbb{R}_+$ via

$$w(r) := \lim_{n\to\infty} \sum_{i \in I^n(r)} \omega_i^n, \qquad I^n(r) := \{i \leq n : r_i = r\}.$$

Thus, w is the limiting distribution of value across tax lots, and $w(r)$ is the value of all lots whose embedded cumulative return equals r. By construction, the total market value is $M = \sum_{r\in\mathcal{R}} \sum_{i\in I^n(r)} \omega_i^n$, and therefore, in the limit, $M = \sum_{r\in\mathcal{R}} w(r)$.

Two regularity conditions ensure that the expanding asset universe indeed diversifies idiosyncratic risk. First, *balancedness* requires that no single position dominates: call the triangular array $\{\boldsymbol{\omega}^n\}_{n\in\mathbb{N}}$ balanced if $\max_{i\leq n} \omega_i^n / M \to 0$. As more securities become available, the portfolio broadens with them rather than remaining concentrated in a few positions. Second, *diversifiability* requires that the new assets represent genuinely new securities: call the covariance sequence $\{\Sigma^n\}_{n\in\mathbb{N}}$ diversifiable if the corresponding sequence of largest eigenvalues is uniformly bounded. This rules out adding assets whose variance explodes with n, or assets that become nearly perfectly correlated with the existing ones, as would happen if asset n were a mere linear combination of existing ones.[12]

The next theorem formalizes this reduction. Two investors whose portfolios share the same asymptotic tax-lot distribution $w(r)$ achieve asymptotically equivalent consumption and utility if their liquidations generate the same aggregate liquidated market value $\mathrm{M}_1 := \lim \mathrm{M}_1^n$ and realized gain $\mathrm{CG}_1 := \lim \mathrm{CG}_1^n$. Which combinations of M_1 and CG_1 can be achieved depends on how much value sits in the lots at each r, making the function w the relevant portfolio state. However, neither the security-level composition of the portfolio nor the identity of the liquidated assets nor the covariance structure among returns matters in the limit. Aggregate risk remains present, since "perfect diversification" here refers only to asset-level residual risk.

[11] This is a well-posedness condition for the limiting portfolio problem: although each individual position becomes negligible, new positions are added as n grows, so one requires that the lot-level sums settle down.

[12] One could weaken the technical condition by allowing the largest eigenvalue to grow slowly with n; the boundedness assumption keeps the statement transparent. Balancedness and diversifiability are analogous to the conditions ensuring well-diversified portfolios in Ross (1976) and the approximate factor structure of Chamberlain and Rothschild (1983).

Theorem 1 (Asymptotic Sufficiency of the Tax-Lot Distribution)**.** *Consider two balanced and diversifiable sequential portfolio problems with endowments and covariance matrices* $\{(\boldsymbol{\omega}^n, \Sigma^n)\}$ *and* $\{(\bar{\boldsymbol{\omega}}^n, \bar{\Sigma}^n)\}$*. Let their return vectors* $\{\boldsymbol{R}^n\}$ *and* $\{\bar{\boldsymbol{R}}^n\}$ *satisfy* $\mathbb{E}[\boldsymbol{R}^n] = \mathbb{E}[\bar{\boldsymbol{R}}^n] = (1+\mu)\mathbf{1}_n$ *for every* n *and some common* $\mu > 0$*; component distributions and covariance structures may otherwise differ across assets and between the two portfolio problems. Suppose* $\{\boldsymbol{\omega}^n\}$ *and* $\{\bar{\boldsymbol{\omega}}^n\}$ *have the same limiting tax-lot distribution* w *and let* $\{\boldsymbol{\alpha}^n\}$ *and* $\{\bar{\boldsymbol{\alpha}}^n\}$ *be any two sequences of liquidation shares such that, for every* $r \in \mathcal{R}$*, the lot-level liquidation limits exist:*

$$w_\alpha(r) := \lim_{n\to\infty} \sum_{i \in I^n(r)} \alpha_i^n \omega_i^n, \qquad \bar{w}_{\bar{\alpha}}(r) := \lim_{n\to\infty} \sum_{i \in I^n(r)} \bar{\alpha}_i^n \bar{\omega}_i^n.$$

Let (c_1^n, c_2^n, U^n) *and* $(\bar{c}_1^n, \bar{c}_2^n, \bar{U}^n)$ *denote consumption and utility under* $(\{\boldsymbol{\alpha}^n\}, \boldsymbol{\omega}^n, \Sigma^n)$ *and* $(\{\bar{\boldsymbol{\alpha}}^n\}, \bar{\boldsymbol{\omega}}^n, \bar{\Sigma}^n)$*, respectively. Then, i) implies ii):*

i) The limiting liquidated market value and realized gain coincide:

$$\sum_{r\in\mathcal{R}} w_\alpha(r) = \mathrm{M}_1 = \sum_{r\in\mathcal{R}} \bar{w}_{\bar{\alpha}}(r), \qquad \sum_{r\in\mathcal{R}} w_\alpha(r)\big(1 - \tfrac{1}{r}\big) = \mathrm{CG}_1 = \sum_{r\in\mathcal{R}} \bar{w}_{\bar{\alpha}}(r)\big(1 - \tfrac{1}{r}\big).$$

ii) The two sequences are asymptotically equivalent in consumption and utility:

$$\lim_{n\to\infty} |c_1^n - \bar{c}_1^n| = 0, \quad \operatorname*{plim}_{n\to\infty} |c_2^n - \bar{c}_2^n| \to 0, \quad \lim_{n\to\infty} \left|U^n - \bar{U}^n\right| = 0.$$

Proof. See Appendix A. ∎

A transparent way to implement the theorem's matching conditions on $\boldsymbol{\alpha}^n$ and $\bar{\boldsymbol{\alpha}}^n$ is to choose the same limiting liquidation $\tilde{w}(r) \in [0, w(r)]$ from each tax lot. Because both portfolios converge to the same distribution w, any such lot-level liquidation feasible for one investor is feasible for the other. It leaves both with the same post-liquidation distribution $w - \tilde{w}$, and therefore with the same current proceeds, realized gains, and retained tax basis. Conditional on the aggregate state, security-specific return differences diversify away, so the portfolios have the same limiting date-2 market value; if all remaining assets are then liquidated, their taxable gains and consumption coincide as well. This construction also provides the formal foundation for the *Tax Portfolio Model* developed in the next section. Under a common return density, identical post-liquidation distributions evolve into identical next-period distributions. Matching subsequent liquidations lot by lot can therefore match realized gains and portfolio states at every later date. The distribution $w(\cdot)$ and the liquidation function $\tilde{w}(\cdot) \le w(\cdot)$ are thus the natural state and control objects of the Tax Portfolio Model.

Theorem 1 is an asymptotic equivalence and is silent on how fast security-level selection becomes irrelevant. The next corollary supplies the rate: reshuffling *which* securities are sold within tax lots moves consumption and expected utility at rate $n^{-1/2}$, while moving liquidation *across* tax lots changes tax payments by $O(1)$. This is the quantitative content of the claim that tax motives dominate in large portfolios.

Corollary 1 ($\sqrt{n}$-decay of security-selection effects). *In the setting of Theorem 1, fix one endowment sequence* $\{(\omega^n, \Sigma^n)\}$, *strengthen balancedness to* $\max_{i\le n} \omega_i^n \le CM/n$ *for some* $C \ge 1$, *and bound the largest covariance eigenvalues by* $\bar{\lambda} := \sup_n \lambda_{\max}(\Sigma^n) < \infty$. *Let taxes be linear,* $T_t(CG) = \tau CG$, *and let* $\{\alpha^n\}$ *and* $\{\bar{\alpha}^n\}$ *be two liquidation sequences with identical lot-level liquidations at every* n:

$$\sum_{i\in I^n(r)} \alpha_i^n \omega_i^n = \sum_{i\in I^n(r)} \bar{\alpha}_i^n \omega_i^n \qquad \text{for all } r \in \mathcal{R}.$$

Define the relative date-2 consumption gap by $g_2^n := (\bar{c}_2^n - c_2^n)/M$. *Then* $c_1^n = \bar{c}_1^n$, $\mathbb{E}[g_2^n \mid Z] = 0$, *and*

$$\mathrm{sd}(g_2^n) \;\le\; (1-\tau)\sqrt{\bar{\lambda}\, C\, \mathbb{E}[Z^2]}\;\frac{1}{\sqrt{n}}.$$

If, in addition, marginal utility in the second argument is bounded by L, *then* $|U^n - \bar{U}^n| \le LM\mathbb{E}|g_2^n| \le LM\,\mathrm{sd}(g_2^n) = O(n^{-1/2})$.

The bound implies small residual security-selection risk even in moderately sized portfolios. Consider an equally weighted portfolio of 100 assets with independent Gaussian residual returns of 30 percent volatility, no aggregate shock, and a 25 percent tax rate. For two trading rules implementing the same tax-lot liquidations, the standard deviation of their consumption difference is at most 2.25 percent of initial portfolio wealth. To express the remaining security-selection risk in consumption units, take proportional liquidation within each tax lot as the benchmark: sell the same fraction of every security within a given lot. Under these assumptions, this minimizes consumption variance among trading rules implementing the same tax-lot liquidations. With CARA utility over date-2 consumption and absolute risk aversion $a = 5/M$, the certainty-equivalent consumption loss from using another rule is at most 0.13 percent of initial portfolio wealth for 100 assets and 0.03 percent for 500 assets.[13]

In sum, large portfolios permit us to ignore the identities of the assets, as long as we keep track of their distribution across tax lots. The preceding bound suggests that this approximation can also be useful for moderately sized portfolios. Under the common-expected-return assumption, historical embedded returns contain no information about expected future performance, while differences in marginal return distributions and variances diversify away (provided the covariance sequence remains diversifiable). Asset identity can matter again when expected returns or nondiversifiable factor exposures are heterogeneous. Supplementary Online Appendix Section A considers an extension with heterogeneous factor exposures.

[13] Let $\bar{c}_2^n$ denote consumption under proportional liquidation. Liquidation differences sum to zero within each tax lot, so independent residual returns with identical variances imply $\mathrm{Cov}(g_2^n, \bar{c}_2^n) = 0$. Joint Gaussianity then makes the consumption gap independent of benchmark consumption. Under $u_2(c) = -e^{-ac}$, the certainty-equivalent loss, expressed as a share of initial wealth, is therefore $\frac{aM}{2}\,\mathrm{var}(g_2^n) \le 2.5\,\frac{0.225^2}{n}$. This gives 0.13 percent at $n = 100$ and 0.03 percent at $n = 500$. The choice $aM = 5$ corresponds to relative risk aversion of 5 at consumption M.

Section III now uses the insights of Theorem 1 to build a model of tradable, taxable, and fully diversified portfolios.

III The Tax Portfolio Model

In this section, I introduce the core objects of the *Tax Portfolio Model* (TPM): portfolios, liquidation choices, tax operators, and the portfolio law of motion. I then show how aggregate risk fits into the framework. For the remainder of the paper, unless stated otherwise, idiosyncratic period returns have a common and continuous marginal density f on support $D \subseteq \mathbb{R}_{>0}$.[14]

III.A Portfolios, Trading, and Taxation

Assets, Returns and Portfolios. Let ω_i denote the current market value of asset $i \in [0,1]$ and $\breve{\omega}_i$ its market value at purchase. The cumulative gross return of asset i is $r(i) := \omega_i/\breve{\omega}_i \in (0,\infty)$. The common marginal return $R \sim f$ satisfies $\mathbb{E}[R] - 1 = \mu \geq 0$, while dependence across assets is mostly unrestricted; see Section II. Theorem 1 allows me to suppress asset identities and describe a portfolio by its distribution of market value across tax lots. I represent this distribution by a nonnegative, Lebesgue-integrable function $w : (0,\infty) \to \mathbb{R}_{\geq 0}$, and denote the set of portfolios by $\mathcal{W}$. Whenever I take a portfolio as the endowment for an optimization problem, I assume that its market value, liquidation value, and tax liability, defined below, are finite.

In lot-level examples, $\boldsymbol{\delta}_r$ denotes a Dirac portfolio that places one unit of market value in the tax lot defined by accrued return r. Note that the tax lot represented by $\boldsymbol{\delta}_r$ still aggregates infinitely many underlying assets with the same accrued return.

The quantity $w(r)\ \mathrm{d}r$ represents the market value held in tax lots with cumulative gross returns between r and $r + \mathrm{d}r$. Integrating over these lots, $\int_0^{\bar{r}} w(s)\ \mathrm{d}s$, gives the value of all assets with cumulative gross returns at most $\bar{r}$. Setting $\bar{r} = \infty$ yields the market value of the entire portfolio. Dividing $w(r)$ by the cumulative gross return r converts market value into its corresponding cost basis. Integrating $w(r)/r$ over tax lots gives the total cost basis, and subtracting this from market value gives the embedded net capital gain. These three portfolio statistics are

$$\mathrm{M}(w) := \int_0^\infty w(r)\ \mathrm{d}r, \qquad \mathrm{B}(w) := \int_0^\infty \frac{w(r)}{r}\ \mathrm{d}r, \qquad \mathrm{CG}(w) := \mathrm{M}(w) - \mathrm{B}(w).$$

For example, the portfolio $w = 2\boldsymbol{\delta}_{0.8} + \boldsymbol{\delta}_1 + 3\boldsymbol{\delta}_{1.5}$ has market value 6, basis $2/0.8 + 1 + 3/1.5 = 5.5$, and an embedded net gain of 0.5. Thus, the coefficients on the Dirac lots are dollar market values, while their subscripts carry the tax-basis information.

[14] In two-period problems, it is enough that investors lack asset-level information that would let them select securities with systematically different marginal return distributions.

Asset Sales and Purchases. Investors generate liquidity by choosing which parts of their portfolio to sell. A feasible liquidation portfolio $\widetilde{w}$ cannot contain negative sales or exceed the holdings in any tax lot. Thus,

$$\widetilde{w} \in [0, w], \qquad [0, w] := \{v \in \mathcal{W} : 0 \leq v(r) \leq w(r) \text{ for every } r > 0\}.$$

This set of feasible liquidations is the order interval between the zero portfolio and w. The unsold portion $w - \widetilde{w} \in \mathcal{W}$ is the post-liquidation portfolio.

Returning to this example, selling $\widetilde{w} = \boldsymbol{\delta}_{0.8} + \boldsymbol{\delta}_{1.5}$ from w raises $\mathrm{M}(\widetilde{w}) = 2$ dollars before tax and realizes the net gain $\mathrm{CG}(\widetilde{w}) = 2 - (1/0.8 + 1/1.5) = 1/12$. Panel (a) of Figure 1 shows this liquidation: orange denotes the portions sold, while dark gray denotes the post-liquidation portfolio $w - \widetilde{w} = \boldsymbol{\delta}_{0.8} + \boldsymbol{\delta}_1 + 2\boldsymbol{\delta}_{1.5}$.

To compute the liquidity generated by a sale, one must know the taxes it triggers. Start with the simple FUL-case of a constant statutory rate $\tau \in (0, 1)$. Selling market value $\widetilde{w}(r)$ from the lots at r triggers tax

$$\tau\big(\widetilde{w}(r) - \widetilde{w}(r)/r\big)\,\mathrm{d}r = t_{\mathrm{FUL}}(r)\widetilde{w}(r)\,\mathrm{d}r,$$

where $t_{\mathrm{FUL}}(r) = \tau\frac{r-1}{r}$ is the tax share per unit sold from a lot at r. Integrating over lots yields total tax liability, $\mathrm{T}_{\mathrm{FUL}}(\widetilde{w}) = \int t_{\mathrm{FUL}}(r)\widetilde{w}(r)\,\mathrm{d}r = \tau\,\mathrm{CG}(\widetilde{w})$. More generally, one can take either a tax share function $t : r \mapsto t(r) < 1$ satisfying the no-tax-at-zero-gain condition $t(1) = 0$, or the tax operator T itself, as primitive. Definition 1 introduces this object together with liquidation value.

Definition 1 (Tax Liability and Liquidation Value)**.** *A tax operator* T *is a linear functional which maps a portfolio w to its 'tax liability'* $\mathrm{T}(w)$*. The functional* $\mathrm{K} := \mathrm{M} - \mathrm{T}$ *yields the 'liquidation value'* $\mathrm{K}(w)$*. Additionally, if* T *is defined through a tax share function, $t(r)$ with $t(1) = 0$, then $k(r) = 1 - t(r)$ is the 'payoff function' and*

$$\mathrm{T}(w) := \int_0^\infty w(r)t(r)\,\mathrm{d}r, \qquad \mathrm{K}(w) = \int_0^\infty w(r)k(r)\,\mathrm{d}r. \tag{2}$$

The integral representation in (2) covers the canonical tax systems that operate through lot-specific tax shares such as FUL and NUL (no use of losses). Later, when I turn to LUL, the relevant tax object can no longer be written in this simple way, and T will be taken as primitive, with a second argument for potential carry-forwards.

Generally, the investor's problem is to choose a liquidation portfolio $\widetilde{w}$ that delivers the required current liquidity $\mathrm{K}(\widetilde{w})$ while leaving an optimal post-liquidation portfolio $w - \widetilde{w}$. The current sale determines taxes and cash today through $\mathrm{K}(\widetilde{w})$, while the residual market value $\mathrm{M}(w - \widetilde{w})$ determines how much principal is carried into the next period. Only in some cases does solving this problem amount to maximizing $\mathrm{M}(w - \widetilde{w})$ subject to a liquidity target.

In addition to selling assets, an investor can purchase them. Investment adds value to the lot at $r = 1$, where acquisition price and tax basis coincide. Formally, investing $I > 0$ adds a

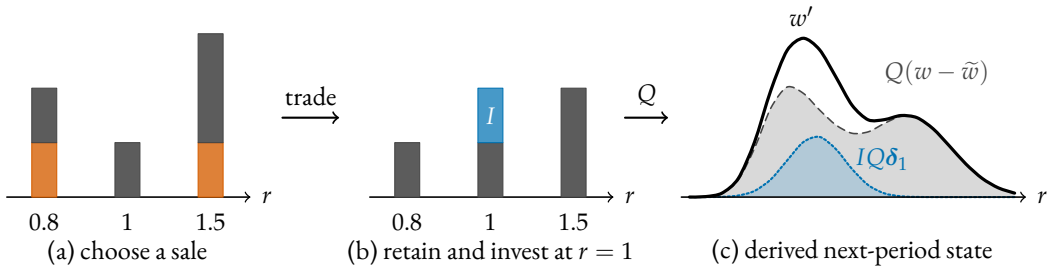


Fig. 1. TPM mechanics for $w = 2\boldsymbol{\delta}_{0.8} + \boldsymbol{\delta}_1 + 3\boldsymbol{\delta}_{1.5}$. Panel (a) selects the liquidation portfolio $\widetilde{w} = \boldsymbol{\delta}_{0.8} + \boldsymbol{\delta}_{1.5}$, panel (b) adds investment $I = 1$ at $r = 1$, and panel (c) applies the portfolio law of motion. Dark gray denotes retained value, orange sold value, and blue new investment.

Dirac mass $(\delta_I I)\boldsymbol{\delta}_1$ to w, where $\delta_I \in [0,1]$ is an investment friction. The law of motion then diffuses the point mass into the next-period portfolio. Liquidation plus (re-)investment is the only way to shift value across lots.[15] In the running example, frictionless investment of $I = 1$ adds $\boldsymbol{\delta}_1$ to the post-liquidation portfolio. Panel (b) of Figure 1 shows the resulting post-trade portfolio, with the new investment in blue at $r = 1$.

Given a portfolio w and exogenous liquid resources $y \geq 0$, a feasible control consists of a liquidation portfolio $\widetilde{w} \in [0, w]$ and a nonnegative investment I that does not exceed available liquidity. The feasible-control correspondence is thus

$$\Gamma(w) := \{(\widetilde{w}, I) : \widetilde{w} \in [0, w], \quad 0 \leq I \leq y + \mathrm{K}(\widetilde{w})\}.$$

A trading-strategy class specifies which feasible controls are allowed at each portfolio: it is a correspondence Γ^S satisfying $\Gamma^S(w) \subseteq \Gamma(w)$ for every $w \in \mathcal{W}$.

A natural generalization of the canonical trading strategy in Constantinides (1983) is to harvest losses up to a threshold and, if additional liquidity is required, sell the least appreciated remaining tax lots. I call these loss-harvesting cutoff-liquidation (LHCL) strategies. I additionally define strict LHCL (sLHCL) strategies as those with no "round trips": there are no simultaneous investments and realizations of positive gains.

Definition 2 (LHCL and sLHCL Strategies)**.** *Fix a portfolio w and a harvest interval $(0, h]$, $h \geq 0$. For a cutoff $a \in [h, \infty]$, current consumption under liquidation $w\mathbf{1}_{(0,a]}$ is $y + \mathrm{K}(w\mathbf{1}_{(0,a]}) - I$. The LHCL correspondence indexed by h is*

$$\Gamma_h^L(w) := \{(w\mathbf{1}_{(0,a]}, I) \in \Gamma(w) : a \in [h, \infty]\}.$$

The sLHCL strategy class imposes the no-roundtrip restriction

$$\Gamma_h^{sL}(w) := \{(w\mathbf{1}_{(0,a]}, I) \in \Gamma_h^L(w) : a > h \Rightarrow I = 0\}.$$

[15] Note that, in a finite-asset description, repeated purchases of the same security would also raise lot-identification rules such as FIFO or LIFO. These constraints do not bind in the TPM because latent asset richness lets sold and retained positions be treated as marginal slices of a large asset pool.

The Portfolio Law of Motion. A law of motion maps today's liquidation and investment choices into tomorrow's portfolio, w'. Fix a post-liquidation portfolio $w^p \in \mathcal{W}$. Value currently in tax lot s is scaled by the realized period return R and therefore appears tomorrow in tax lot sR. Averaging over the common marginal return distribution yields the next-period tax-lot distribution. Before adding new investment or aggregate risk, the value of all assets with total return at most r in the next period is

$$\int_0^\infty \int_0^{r/R} w^p(s) R \, \mathrm{d}s \, f(R) \, \mathrm{d}R = \int_0^r (Q w^p)(t) \, \mathrm{d}t,$$

where Q is the transition operator. Differentiating gives the pointwise law of motion

$$Q w^p(r) = \int_0^\infty w^p\left(\frac{r}{R}\right) f(R) \, \mathrm{d}R.$$

Applied to new investment, Q gives $I\delta_I \chi(r)$, where $\chi(r) := Q\boldsymbol{\delta}_1(r) = rf(r)$. Definition 3 collects Q, introduces the additional aggregate-risk operator A_z, and combines liquidation, investment, and aggregate risk in the transition maps G_z. Aggregate risk causes all assets to appreciate by the same factor and thus shifts mass uniformly between tax lots.

Definition 3 (Portfolio Law of Motion)**.** *For any portfolio $w \in \mathcal{W}$, define*

$$Q w(r) := \int_0^\infty w(r/R) f(R) \, \mathrm{d}R.$$

For aggregate risk Z with finite support $\mathcal{Z} \subset (0, \infty)$, write $p_z := \mathbb{P}(Z = z)$ and define $A_z : \mathcal{W} \to \mathcal{W}$ by $A_z w(r) := w(r/z)$. Without aggregate risk, take $\mathcal{Z} = \{1\}$ and drop z-subscripts. Given $\widetilde{w} \in [0, w]$ and $I \geq 0$, the next-period portfolio conditional on $z \in \mathcal{Z}$ is

$$G_z(w, \widetilde{w}, I) := A_z\big(Q(w - \widetilde{w}) + I\delta_I \chi\big); \qquad \chi(r) := rf(r) = Q\boldsymbol{\delta}_1(r). \tag{3}$$

Panel (c) of Figure 1 completes the example. Without aggregate risk and with frictionless investment, the next-period state is $w' = Q(w - \widetilde{w}) + IQ\boldsymbol{\delta}_1$. The dashed gray curve is the transitioned post-liquidation portfolio $Q(w - \widetilde{w})$, the dotted blue curve is the transitioned investment $IQ\boldsymbol{\delta}_1$, and the solid black curve is their sum w'.

Thus, the TPM reduces security-level dynamics to linear operators on tax-lot distributions: diffuse the post-liquidation portfolio across lots according to f, add the contribution of new purchases, and scale lot returns by the aggregate realization. This yields closed-form laws of motion for market value and liquidation value under FUL taxation, shown in Lemma 1.

Lemma 1. *Consider $\mathrm{T}_{\mathrm{FUL}}$ with statutory rate $\tau \in [0, 1]$. Given any portfolio w, liquidation portfolio $\widetilde{w} \in [0, w]$ and investment $I \geq 0$, set $w^p = w - \widetilde{w}$. Conditional on aggregate realization $z \in \mathcal{Z}$, write $w'_z := G_z(w, \widetilde{w}, I)$. The laws of motion for market value, basis, capital gains, and liquidation value are*

$$\begin{aligned} \mathrm{M}(w'_z) &= z\mathbb{E}[R](\mathrm{M}(w^p) + \delta_I I), \qquad \mathrm{B}(w'_z) = \mathrm{B}(w^p) + \delta_I I, \\ \mathrm{CG}(w'_z) &= \mathrm{M}(w'_z) - \mathrm{B}(w'_z), \quad \textit{and} \end{aligned} \tag{4}$$

$$\mathrm{K}_{\mathrm{FUL}}(w'_z) = \mathrm{M}(w'_z) - \tau\,\mathrm{CG}(w'_z). \tag{5}$$

Intuitively, post-liquidation market value grows at rate $\mathbb{E}[R]$ and is then scaled by the realized aggregate shock z, while the cost basis changes only through sales and new purchases. Capital gains are the difference between the two. The identities in (4) do not depend on the tax system; (5) additionally uses the linearity of $\mathrm{T}_{\mathrm{FUL}}$.

Three Tax Operators. I now define three tax systems within this framework: i) full-use-of-losses (FUL) taxation $\mathrm{T}_{\mathrm{FUL}}$, ii) taxation without use of losses (NUL) $\mathrm{T}_{\mathrm{NUL}}$, and iii) limited-use-of-losses (LUL) taxation $\mathrm{T}_{\mathrm{LUL}}$. i) is the simplest of all realization-based tax systems, where every capital gain $w(r) - w(r)/r$ is taxed at a constant rate, τ. This case gives rise to a tax operator $\mathrm{T}_{\mathrm{FUL}}$ with $t_{\mathrm{FUL}}(r) = \tau\frac{r-1}{r}$ (recall Definition 1 and the preceding example). The associated payoff share is $k_{\mathrm{FUL}}(r) := 1 - \tau + \tau/r$, so $\mathrm{K}_{\mathrm{FUL}}(w) = \int_0^\infty w(r)k_{\mathrm{FUL}}(r)\,\mathrm{d}r$. Importantly, a negative capital gain leads to a negative tax. This fact is sometimes recast as a feature, with authors claiming that the negative capital gains tax reflects a tax offset 'elsewhere'. However, the TPM is meant to model all assets subject to capital gains taxation jointly. Hence, unless capital losses can be offset against other income classes at the same rate, $\mathrm{T}_{\mathrm{FUL}}$ misstates refundability. Two points mitigate this concern. First, when portfolios are fully appreciated, which may hold for long-term investors, loss refunds are irrelevant. Second, one can bound the other extreme with operator ii) $\mathrm{T}_{\mathrm{NUL}}$: *no use of losses*. The tax function for this operator is then $t_{\mathrm{nc}}(r) = \tau\frac{r-1}{r}\mathbf{1}_{\{r\geq 1\}}$. Optimal liquidation rules and investor behavior can differ markedly across $\mathrm{T}_{\mathrm{FUL}}$ and $\mathrm{T}_{\mathrm{NUL}}$.

Case iii) is a closer representation of the U.S. and most OECD systems which feature limited use of losses: realized losses can offset realized gains, but do not generate cash refunds. Additionally, if *carry-forward* is allowed, unused losses from past periods can be used to offset gains in the current period. This is modeled by introducing a state variable, $F \geq 0$. If initially at zero, liquidating $\widetilde{w}$ carrying a net loss increases the carry-forward to $F = -\mathrm{CG}(\widetilde{w})$. Next period, liquidating some $\hat{w}$ with $\mathrm{CG}(\hat{w}) \geq F > 0$, uses up all carry-forward losses and triggers a tax payment of $\tau(\mathrm{CG}(\hat{w}) - \delta_F F)$. Here, I introduce $\delta_F \in [0,1]$ as a reduced-form carry-forward parameter. The benchmark $\delta_F = 1$ corresponds to perfect carry-forward. Values $\delta_F < 1$ capture any force that lowers the future value of current unused losses, including explicit time limits, offset restrictions, or even a small probability of regulatory change. In this sense, $\delta_F = 1$ is a knife-edge case, while $\delta_F = 0$ represents regimes in which unused losses expire completely.[16]

[16] Perfect carry-forward systems—where capital losses can generally be carried forward indefinitely and offset against future capital gains without major structural restrictions—include Australia, Canada, Ireland, Israel, Norway, and the United Kingdom. Imperfect carry-forward systems include Germany, where loss utilization is effectively constrained by the "minimum taxation" rule limiting the share of income that can be offset in

Accordingly, I define the *limited-use-of-losses tax operator* featuring a constant statutory tax rate τ and carry-forward tax credits as $\mathrm{T}_{\mathrm{LUL}}(w,F) := \left[\tau(\mathrm{CG}(w) - \delta_F F)\right]_+$ (where $[\,\cdot\,]_+$ is the maximum of its argument and zero). Its liquidation functional is $\mathrm{K}_{\mathrm{LUL}}(w,F) := \mathrm{M}(w) - \mathrm{T}_{\mathrm{LUL}}(w,F)$. The aggregate transition extends G_z by adding the future carry-forward state, $F_{\mathrm{LUL}}(\widetilde{w},F) := \left[F - \mathrm{CG}(\widetilde{w})\right]_+$. Given a state (w,F) and control $(\widetilde{w},I)$, define the law of motion and feasible correspondence by:

$$G_{\mathrm{LUL},z}(w,F,\widetilde{w},I) := \big(G_z(w,\widetilde{w},I), F_{\mathrm{LUL}}(\widetilde{w},F)\big),$$
$$\Gamma(w,F) := \{(\widetilde{w},I) : \widetilde{w} \in [0,w],\ 0 \le I \le y + \mathrm{K}_{\mathrm{LUL}}(\widetilde{w},F)\}.$$

Note that, if the tax operator is clear from context, I drop the subscripts on the laws of motion and correspondences to avoid clutter. With the main definitions now in place, I illustrate their economic content with the examples in the following section.

III.B Examples and Failure of Canonical Trading Rules

The familiar loss-harvesting prescription is to sell depreciated tax lots, reinvest the proceeds, and raise any additional cash from the least appreciated remaining positions. Definition 2 makes this prescription precise through the LHCL strategy classes Γ_h^L and their no-roundtrip sLHCL subclasses Γ_h^{sL}. Under symmetric loss offsets, frictionless investment, and sufficiently favorable aggregate returns, the recursive results in Section IV show that the Bellman problems studied there can be restricted to these strategy sets without loss. The following numerical examples show the economics behind each qualification. Throughout the examples, set the asset-level return to $R = 1$ and let all return variation enter through the aggregate gross return Z, which is deterministic unless stated otherwise.

1. No current loss offset. Let the statutory rate be $\tau = 1/2$, let $Z = 2$, and start with $w = \boldsymbol{\delta}_{1/2} + \boldsymbol{\delta}_{3/4}$. Under $\mathrm{T}_{\mathrm{NUL}}$, either tax lot raises one dollar today because a realized loss receives no refund. A lower-cutoff rule sells $\boldsymbol{\delta}_{1/2}$. The retained $\boldsymbol{\delta}_{3/4}$ becomes a two-dollar position at $r = 3/2$ and has continuation liquidation value

$$2k_{\mathrm{NUL}}(3/2) = 2\left(\frac{1}{2} + \frac{1}{3}\right) = \frac{5}{3}.$$

Selling $\boldsymbol{\delta}_{3/4}$ instead preserves $\boldsymbol{\delta}_{1/2}$, which becomes a two-dollar position exactly at basis, $r = 1$. It therefore liquidates for $2 > 5/3$ tomorrow. The investor optimally sells the *less* depreciated position and preserves the deeper loss as a future tax shield. The reversal comes entirely from

very high-income years, and the United States, where capital losses are separated into short- and long-term buckets before cross-netting and where excess losses are subject to specific offset ordering rules. Time-limited carry-forward regimes, where the usefulness of losses depends on the investor's realization horizon, include Japan (3 years), Spain (4 years), Finland, Greece, Poland, Portugal, and Czechia (typically 5 years), and France and Mexico (10 years).

the no-offset kink of k_{NUL} at $r = 1$.

2. Reinvestment friction. Under symmetric loss offsets, reinvestment friction shrinks the harvest region. Set $\tau = 1/2$, let the deterministic aggregate gross return be $Z = z > 1$, and consider one dollar in tax lot r, represented by $\boldsymbol{\delta}_r$. Let $L(r)$ and $H(r)$ denote next-period liquidation value when the position is left in place or harvested and reinvested. Then

$$\begin{aligned} L(r) &= z k_{\mathrm{FUL}}(rz) = \frac{zr+1}{2r}, \\ H(r) &= \delta_I k_{\mathrm{FUL}}(r)\left[z k_{\mathrm{FUL}}(z)\right] = \frac{\delta_I (r+1)(z+1)}{4r}, \\ H(r) - L(r) &= \frac{\delta_I (r+1)(z+1) - 2(zr+1)}{4r}. \end{aligned}$$

If $\delta_I > 2/(z+1)$, harvesting is weakly optimal exactly when

$$r \leq r_H(\delta_I, z) := \frac{\delta_I (z+1) - 2}{2z - \delta_I (z+1)}.$$

Lowering δ_I shrinks the cutoff because only sufficiently deep losses generate a tax subsidy large enough to cover the value destroyed in reinvestment. If $\delta_I \leq 2/(z+1)$, the harvest region is empty. Frictionless investment, $\delta_I = 1$, gives $r_H = 1$ for all z. Otherwise, whether it is ex post optimal to harvest a given tax lot at r depends on the aggregate realization Z. Thus, only lots $r \leq \min_{z \in \mathrm{supp}(Z)} r_H(\delta_I, z)$ are guaranteed to be optimal to harvest. Section IV.C formalizes this logic and computes the corresponding harvest region under linear taxation for general aggregate-return support and reinvestment friction.

3. Leaky carryforward. Carryforward leakage can reverse the lower-cutoff prescription even when reinvestment is frictionless. As in the NUL case in Example 1, realizing a loss can weaken its value as a future tax shield. Here, the loss becomes a carryforward credit that may lose value before it can offset taxable gains. The investor may therefore prefer to realize the smaller loss and preserve the deeper loss.

4. Aggregate downside and risk aversion. Consider $\mathrm{T}_{\mathrm{FUL}}$ with $\tau = 1/2$ and $w = 3\boldsymbol{\delta}_1 + 4\boldsymbol{\delta}_2$. Selling either $3\boldsymbol{\delta}_1$ or $4\boldsymbol{\delta}_2$ raises exactly three dollars today because $\mathrm{K}_{\mathrm{FUL}}(4\boldsymbol{\delta}_2) = 3$. If the aggregate realization is $Z = 1/2$, selling $3\boldsymbol{\delta}_1$ and retaining $4\boldsymbol{\delta}_2$ leaves continuation liquidation value $4\,\mathrm{K}_{\mathrm{FUL}}(A_{1/2}\boldsymbol{\delta}_2) = 2$. Selling $4\boldsymbol{\delta}_2$ instead and retaining $3\boldsymbol{\delta}_1$ leaves $3\,\mathrm{K}_{\mathrm{FUL}}(A_{1/2}\boldsymbol{\delta}_1) = 9/4$. Hence the better trade sells the right tail. Intuitively, selling $4\boldsymbol{\delta}_2$ uses four dollars of market value to raise three, leaving only three rather than four dollars exposed to the aggregate shock.

The same force matters under risk aversion. Let Z be positive and nondegenerate with $\mathbb{E}[Z] = 1$, and let date-2 utility be strictly increasing and concave. Selling $3\boldsymbol{\delta}_1$ and retaining $4\boldsymbol{\delta}_2$ gives $c_2^{\mathrm{sell}\,\delta_1} = 2Z + 1$, whereas selling $4\boldsymbol{\delta}_2$ and retaining $3\boldsymbol{\delta}_1$ gives $c_2^{\mathrm{sell}\,\delta_2} = 3Z/2 + 3/2$.

Both expressions have expected value three, but the former is a mean-preserving spread of the latter. A risk-averse investor therefore sells $4\boldsymbol{\delta}_2$. This example can be easily generalized to arbitrary portfolios $w \in \mathcal{W}$.

5. One-step versus recursive trading. Finally, under $\mathrm{T}_{\mathrm{NUL}}$, a liquidation ranking that maximizes next-period value need not maximize the value left several dates later. The no-offset kink at $r = 1$ means that a ranking can change as retained tax lots compound relative to cost basis. Set $\tau = 1/2$, keep $R = 1$, and let the deterministic aggregate gross return be $Z = 3$ in each period. Start from $w = \boldsymbol{\delta}_{1/2} + (7/5)\boldsymbol{\delta}_{7/3}$ and require one unit of consumption at each of the first two dates. The position $(7/5)\boldsymbol{\delta}_{7/3}$ has current liquidation value one because $(7/5)k_{\mathrm{NUL}}(7/3) = 1$. If retained for one period, it would be worth $12/5$, below the loss tax lot's $5/2$. A rule that sells to maximize date-2 liquidation value therefore sells the appreciated position first. At date two it sells $6/5$ units from the compounded loss tax lot to finance the second unit. Liquidating what remains at date three gives

$$\left(3 - \frac{6}{5}\right) 3k_{\mathrm{NUL}}(9/2) = \frac{33}{10}.$$

The alternative sells $\boldsymbol{\delta}_{1/2}$ first, keeps the larger appreciated position, and at date two sells $7/4$ units from it. Its date-three value is

$$\left(\frac{21}{5} - \frac{7}{4}\right) 3k_{\mathrm{NUL}}(21) = \frac{77}{20}.$$

Thus the strategy that maximizes date-2 liquidation value leaves less market value at date two and only 3.30 at date three, whereas reversing its first trade leaves 3.85. This failure is driven by $\mathrm{T}_{\mathrm{NUL}}$: the no-offset kink prevents the one-period liquidation ranking from being preserved recursively. Section IV instead asks when a liquidation ranking survives recursive continuation. Under $\mathrm{T}_{\mathrm{FUL}}$—and under perfect-carryforward $\mathrm{T}_{\mathrm{LUL}}$ in the corresponding frictionless benchmark—the recursive results restore LHCL optimality when the aggregate-return support condition holds.

Together, these examples delimit the canonical trading rule. Absence of current-loss offsets can make a deeper loss worth preserving; reinvestment costs shrink the harvest region; carryforward leakage changes which loss should be realized; aggregate downside and risk aversion can favor selling appreciated tax lots; and, without restrictions on T, a longer horizon can reverse a one-step ranking. Section IV identifies conditions under which the lower-cutoff ranking survives these recursive comparisons.

IV The Recursive Framework

This section formulates two classes of recursive portfolio problems. The perpetuity problem determines the largest constant real consumption level that a portfolio can sustain indefinitely under a given tax system and market conditions. These consumption levels are used in the quantitative applications. The discounted-utility problem allows consumption to vary over time and accommodates life-cycle preferences.

Readers mainly interested in the quantitative results may proceed directly to Sections V and VI. For those applications, this section establishes that backward iteration converges to the maximal perpetuity and that optimal policies exist. In the setting without aggregate risk or reinvestment costs, and with perfect loss carryforward under LUL, optimal policies can be found among strict loss-harvesting cutoff liquidation (sLHCL) strategies. These results justify the trading restriction used to calculate effective tax rates and return alphas.

For a first reading of the theoretical analysis, the regularity conditions in Section IV.A and the convergence and attainment arguments in the latter part of Section IV.B can be skimmed. Read the two recursive formulations and the perpetuity interpretation at the beginning of Section IV.B, through Corollary 2 (the sup-min limit equals the maximal perpetuity when aggregate risk is absent), then proceed to the optimal-trading analysis in Section IV.C. Its central steps are the right-dominance comparison, the condition under which this comparison survives returns, and Theorem 3 (optimal Bellman choices can be found within the LHCL class). Corollaries 3–4 carry the restriction through finite-horizon iteration and stationary limits and sharpen it when reinvestment is frictionless. The detailed construction of harvest regions can be left for a closer reading. Section IV.D explains how unused loss credits modify the analysis under limited use of losses. Figure 2 summarizes the logical dependencies among the results that establish the recursive restriction.

The following technical arguments use the recursive methods of Stokey and Lucas (1989) and Bertsekas and Shreve (1978). Weighted sup-norm contractions follow Bertsekas (2012); the duality, weak-$*$ compactness, and maximum-theorem arguments use the standard results collected by Aliprantis and Border (2006).

IV.A Regularity Conditions

All Tax Portfolio Model primitives are inherited from Section III: portfolios w, order intervals $[0, w]$, operators M, B, CG, K, Q, A_z, G_z, parameters f, χ, δ_I, and feasible set Γ. The recursive analysis only restricts these objects to a weighted state space and records the regularity conditions needed for the dynamic programming arguments. To define the state space, fix exponents $\alpha < 0 < 1 < \gamma$ and let $\phi(r) := r^{\alpha}\mathbf{1}_{(0,1]}(r) + r^{\gamma}\mathbf{1}_{(1,\infty)}(r)$ for $r > 0$.

The portfolio space on which we interpret the TPM is the positive cone of the weighted function space $\mathcal{L}^\infty_\phi$, equipped with the *portfolio norm* $\|\cdot\|_{\infty,\phi}$:

$$\mathcal{W} := \mathcal{L}^\infty_{\phi,+}, \qquad \|w\|_{\infty,\phi} := \operatorname*{ess\,sup}_{r>0} |w(r)|\phi(r),$$
$$\mathcal{L}^\infty_\phi := \Big\{ w : (0,\infty) \to \mathbb{R} \text{ measurable} : \|w\|_{\infty,\phi} < \infty \Big\}.$$

The portfolio law of motion maps $\mathcal{W}$ into itself if return distributions do not send too many assets close to $r = 0$ or $r = \infty$, which would imply unbounded liquidation values even under linear taxes. Assumption 1 precludes this and is easy to satisfy—for example, with log-normal returns. To stay in $\mathcal{W}$ at all times, the Dirac-measure investments are as before passed through Q and create the investment profile χ.

Assumption 1 (Law of motion, aggregate risk, and investment profile)**.** *The return density and investment profile satisfy* $m_Q := \mathbb{E}[\phi(R)] < \infty$ *and* $\|\chi\|_{\infty,\phi} < \infty$. *For the finite-support aggregate-risk distribution from Definition 3 and* $p \in [0,1]$, *set* $m_{A,p} := \sum_z p_z \phi(z)^p$ *and* $m_{A,0} := 1$.

For tax operators T in the tax-share family, liquid resources are generated by a lot-specific payoff share $k = 1 - t$, so $\mathrm{K}(x) = \int_0^\infty x(r)k(r)\,dr$. The fixed-point and attainment results below use only the regularity conditions collected in Assumption 2. Under these conditions, market value and liquidation value are positive and bounded on $\mathcal{L}^\infty_\phi$, with $\mathrm{M}(w) \le c_\mathrm{M}\|w\|_{\infty,\phi}$ and $\mathrm{K}(w) \le c_\mathrm{K}\|w\|_{\infty,\phi}$; see Proposition 6 in the Supplementary Online Appendix.

Assumption 2 (Tax-share liquidation regularity)**.** *The tax operator T belongs to the tax-share family, and its tax-share payoff function $k = 1 - t$ is nonnegative and belongs to $\mathcal{L}^1_{1/\phi}$. Equivalently, the induced liquidation functional $\mathrm{K} : \mathcal{L}^\infty_{\phi,+} \to [0,\infty)$ is positive, linear and weak-$*$ continuous. (Hence satisfies $0 \le \mathrm{K}(x) \le c_\mathrm{K}\|x\|_{\infty,\phi}$ for some $c_\mathrm{K} < \infty$, with $c_\mathrm{K} = \int_0^\infty k(r)\phi(r)^{-1}\,dr$.)*

Although we are working with value functions which evaluate the sup of a functional maximization problem, many are fixed points of an ordinary contracting Bellman operator in the $\mathcal{B}_W$-family of weighted function spaces. For any state space $\mathcal{S}$ and weight function $W : \mathcal{S} \to [1,\infty)$, define the weighted *value function norm* $\|\cdot\|_W$ and the corresponding function space by

$$\mathcal{B}_W(\mathcal{S}) := \big\{ V : \mathcal{S} \to \mathbb{R} : \|V\|_W := \sup_{s\in\mathcal{S}} |V(s)|/W(s) < \infty \big\}.$$

Contraction properties crucially depend not only on the chosen norm, but also on the components of the associated Bellman equation. Whenever applicable, we thus make Assumption 3, which determines admissible utility functions.

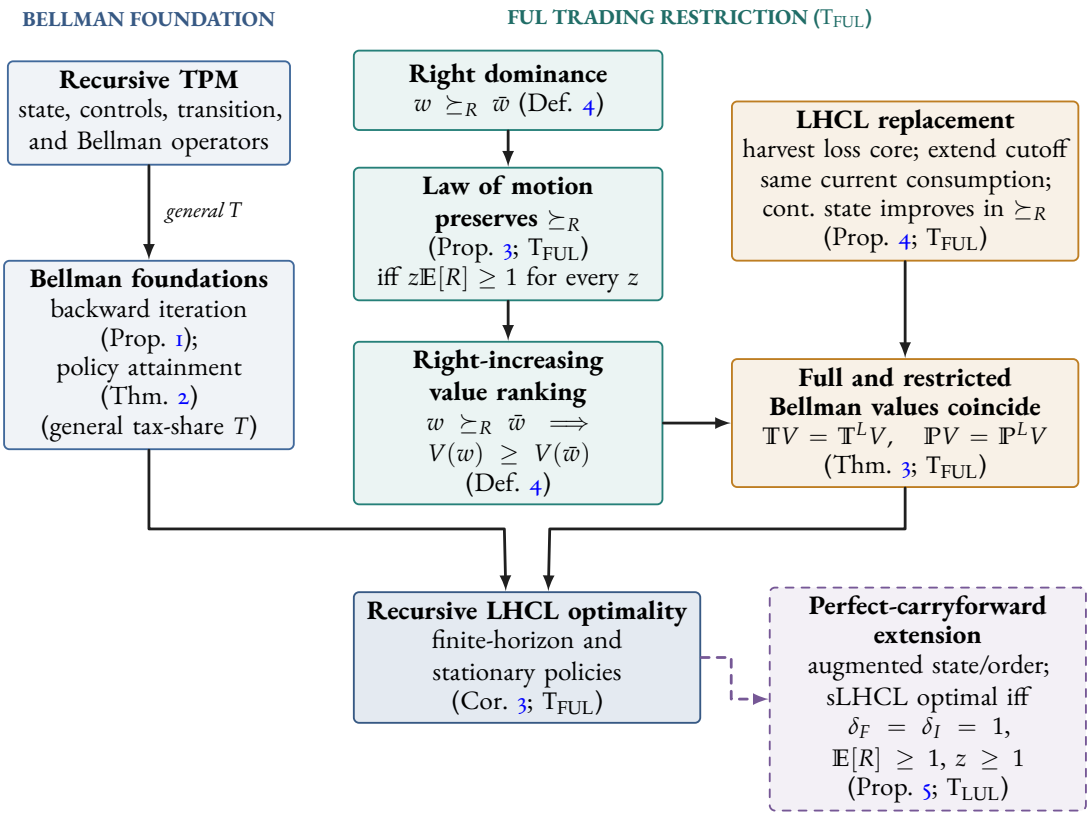


Fig. 2. Proof architecture of the recursive analysis. Bellman foundations apply to the general tax-share family. The solid trading-restriction chain is for full-use-of-losses taxation T_{FUL}. The green boxes trace right dominance, its stability under the support condition, and the ranking of right-increasing continuation values. LHCL replacement produces an ordered continuation state. Together, these steps imply equality of the full and restricted Bellman operators. Combining that restriction with the Bellman foundations yields finite-horizon and stationary LHCL optimality. The dashed branch is the perfect-carryforward extension under T_{LUL} and the conditions shown in the box. The NUL system T_{NUL} is not covered because its kink prevents right dominance from being preserved under the law of motion.

Assumption 3 (Period utility). *The period utility $u : [0,\infty) \to \mathbb{R}$ is continuous, increasing, and strictly concave. If $u(0) = -\infty$, we use the affine extension below a small threshold $\underline{c} > 0$. On the unaffected interval, it is one of the following:*

i) bounded: there is $\bar{u} < \infty$ such that $|u(c)| \leq \bar{u}$ for all $c \geq 0$;

ii) logarithmic: $u(c) = \log c$ on the economically relevant consumption region;

iii) sublinear: there are constants $a_u, b_u \geq 0$ and $p \in (0,1]$ such that $|u(c)| \leq a_u + b_u c^p$ for $c \geq 0$.

Let $\lambda_{I,p}$ be the contraction coefficient for discounted dynamic programming problems, and define the weight function of the value function space to be $W_p(w) := \xi + \|w\|^p_{\infty,\phi}$ with $\xi \geq 0$ for $p > 0$ and $W_p(w) := 1$, otherwise. All parameters α, γ, ξ, p are ingredients of $\lambda_{I,p}$, of which the first three can be chosen freely subject to their respective one-sided inequality. The possible choices of p are dictated by Assumption 3. Thus, $\lambda_{I,p}$ can be made close to its

infimum, $\inf_{\alpha,\gamma,\xi,p} \lambda_{I,p} = \underline{\lambda}_I$, which determines whether a given Bellman equation can be guaranteed to be a contraction mapping on $\mathcal{B}_W(\mathcal{L}^{\infty}_{\phi,+})$.[17]

IV.B Two Bellman Operators

I consider two types of Bellman operators under tax-share taxation systems, i.e., under Assumption 2. The first is the TPM analogue of the canonical discounted-utility Bellman equation. Given exogenous income $y \geq 0$, a feasible control $(\widetilde{w}, I) \in \Gamma(w)$ yields current consumption $c(\widetilde{w}, I) := y + \mathrm{K}(\widetilde{w}) - I$. Continuation states are the maps $G_z(w, \widetilde{w}, I)$. The discounted-utility ($\beta < 1$) Bellman operator is

$$(\mathbb{T}V)(w) := \sup_{(\widetilde{w},I)\in\Gamma(w)} \Big\{ u(c(\widetilde{w}, I)) + \beta \sum_z p_z V\big(G_z(w, \widetilde{w}, I)\big) \Big\}. \tag{6}$$

Second, I consider the following sup-min operator:

$$(\mathbb{P}V)(w) := \sup_{(\widetilde{w},I)\in\Gamma(w)} \min\Big\{ c(\widetilde{w}, I),\ \sum_z p_z V\big(G_z(w, \widetilde{w}, I)\big) \Big\}. \tag{7}$$

The sup-min operator in equation (7) abstracts from utility curvature and discounting, while retaining the same income term inside $c(\widetilde{w}, I)$. It maximizes the minimum of current consumption and expected continuation value. Define this operator's canonical finite-horizon sequence by

$$V^{\mathbb{P}}_0(w) := \sup_{(\widetilde{w},I)\in\Gamma(w)} c(\widetilde{w}, I) = y + \mathrm{K}(w), \qquad V^{\mathbb{P}}_{n+1} := \mathbb{P}V^{\mathbb{P}}_n.$$

Proposition 1 shows that this sequence decreases pointwise to a fixed point $V^{\mathbb{P}}$. When aggregate risk is absent, $V^{\mathbb{P}}_n(w)$ is the maximum constant consumption that can be sustained in the current period and for n future periods. Corollary 2 then identifies $V^{\mathbb{P}}(w)$ with the supremum of the set of perpetuities feasible from w:

$$\mathcal{C}(w) := \left\{ c \geq 0 : \begin{array}{l} \exists (w_t, \widetilde{w}_t, I_t)_{t\geq 0} \text{ with } w_0 = w,\ (\widetilde{w}_t, I_t) \in \Gamma(w_t), \\ w_{t+1} = G(w_t, \widetilde{w}_t, I_t),\ c \leq c(\widetilde{w}_t, I_t) \text{ for all } t \geq 0 \end{array} \right\}. \tag{8}$$

Focusing on perpetuities isolates the gains from tax-optimal trading and abstracts from life-cycle preferences. In practice, one can think of a manager administering a third-party trust fund with a mandate to sustain a constant real payout indefinitely. While the sup-min objective provides a preference-free measure of sustainable consumption, the discounted Bellman problem remains available for welfare questions and life-cycle applications.

[17] For example, the tax-share transition has portfolio-growth bound $m_Q + \delta_I c_{\mathrm{K}} \|\chi\|_{\infty,\phi}$ and income bound $\delta_I y \|\chi\|_{\infty,\phi}$ (Lemma 3). The first coefficient is the part of the law of motion that scales the current portfolio; the second is an additive term generated by exogenous income invested in the new tax lot. The contraction coefficient associated with eq. 6 is $\lambda_{I,p} := \max\{ m_{A,p}(m_Q + \delta_I c_{\mathrm{K}} \|\chi\|_{\infty,\phi})^p,\ 1 + m_{A,p}(\delta_I y \|\chi\|_{\infty,\phi})^p / \xi \}$ and $1 + m_{A,p}(\delta_I y \|\chi\|_{\infty,\phi})^p$ is absorbed for large ξ. Additionally, if p can be chosen arbitrarily close to 0 (log-case), then all remaining terms collapse to 1 and $\underline{\lambda}_I = 1$.

Corollary 2 (Perpetuity interpretation without aggregate risk)**.** *Suppose the hypotheses of Theorem 2 hold and there is no aggregate risk. For the* $\mathbb{P}$*-iteration in Proposition 1 (below),* $V_n^{\mathbb{P}}(w)$ *is the largest constant payoff sustainable in the current period and for n future periods from w, and* $V^{\mathbb{P}}(w) = \sup \mathcal{C}(w)$.

Backward Iteration and Fixed Points

The first standard proposition for the recursive TPM is Proposition 1 and guarantees that backward iteration approaches a stationary value function fixed point for both operators. While $\mathbb{T}$ yields a unique canonical value function $V^{\mathbb{T}}$, the case for $V^{\mathbb{P}}$ is more nuanced. The limiting value function $V^{\mathbb{P}}$ has a perpetuity interpretation only if iteration starts from the canonical $V_0^{\mathbb{P}}$. Indeed, $\mathbb{P}$ is never a contraction mapping since other fixed points exist (for example, $0 = \mathbb{P}0$).

Proposition 1 (Backward iteration)**.** *Suppose Assumptions 1 and 2 hold.*

i) Under Assumption 3(i), $\mathbb{T}$ *is a contraction on* $\mathcal{B}_{W_0}(\mathcal{W})$ *with modulus* β *(set* $p = 0$*). Under Assumption 3(ii),* $\mathbb{T}$ *is a contraction in* $\mathcal{B}_{W_p}(\mathcal{W})$ *with modulus approaching* β*, if one sets* $p > 0$ *small enough. Under Assumption 3(iii),* $\mathbb{T}$ *is a contraction if* $\beta \underline{\lambda}_I < 1$*. In each case,* $\mathbb{T}$ *has a unique fixed point* $V^{\mathbb{T}}$ *and the* $\mathbb{T}$*-iterates* $V_{n+1} := \mathbb{T}V_n$ *converge to* $V^{\mathbb{T}}$ *in the corresponding norm.*

ii) For $\mathbb{P}$*-iteration, let* $V_0^{\mathbb{P}}(w) := \sup_{(\widetilde{w},I)\in\Gamma(w)} c(\widetilde{w}, I) = y + \mathrm{K}(w)$*, and set* $V_{n+1}^{\mathbb{P}} := \mathbb{P}V_n^{\mathbb{P}}$ *for* $n \geq 0$*. Then* $V_{n+1}^{\mathbb{P}} \leq V_n^{\mathbb{P}}$ *for every* $n \geq 0$*, so* $V^{\mathbb{P}}(w) := \lim_{n\to\infty} V_n^{\mathbb{P}}(w) = \inf_{n\geq 0} V_n^{\mathbb{P}}(w)$ *exists pointwise. Moreover,* $V^{\mathbb{P}} = \mathbb{P}V^{\mathbb{P}}$.

The hypotheses of Proposition 1(i) for a contraction are satisfied under some of the most commonly used utility functions: CARA is bounded, CRRA is bounded for a coefficient of relative risk aversion $\sigma > 1$ (a common choice for life-cycle models is $\sigma = 2$) and sublinear for any $p > 0$ in the logarithmic case ($\sigma = 1$). For a formal statement, see Corollaries 5 and 6 in the Supplementary Online Appendix.

In addition to their fixed point properties, both operators preserve monotonicity and concavity, as in Proposition 2. Because $\mathbb{T}$ is a contraction mapping under the hypotheses above, these properties are immediately preserved for its fixed point.

Proposition 2 (Monotonicity and concavity)**.** *Suppose Assumptions 1, 2, and 3 hold.*

i) If V is increasing, then $\mathbb{T}V$ *and* $\mathbb{P}V$ *are increasing.*

ii) If V is concave, then $\mathbb{T}V$ *and* $\mathbb{P}V$ *are concave.*

Consequently, whenever Proposition 1 yields the discounted-utility fixed point $V^{\mathbb{T}}$*, it is increasing and concave.*

Despite their conceptual similarity to finite-dimensional dynamic programs, the considered recursive problems require different compactness and continuity arguments. Theorem 2 applies weak-$*$ compactness and Berge's maximum theorem to both Bellman problems. For the discounted problem, convergence in the weighted value function norm guarantees uniform convergence on every set bounded in the portfolio norm $\|\cdot\|_{\infty,\phi}$, preserving weak-$*$ continuity there. For the sup-min problem, decreasing pointwise convergence instead yields weak-$*$ upper semicontinuity. Together with compactness of feasible controls, these properties ensure statewise policy attainment.

Theorem 2 (Policy attainment)**.** *Suppose Assumptions 1, 2 and (if applicable) 3 hold. Let* $\mathbb{H} \in \{\mathbb{T}, \mathbb{P}\}$. *If* $\mathbb{H} = \mathbb{T}$, *initialize iteration with any weak-$*$ continuous* $V_0^{\mathbb{T}}$ *in the relevant space* $\mathcal{B}_{W_p}$ *and set* $V_{n+1}^{\mathbb{T}} := \mathbb{T}V_n^{\mathbb{T}}$. *If* $\mathbb{H} = \mathbb{P}$, *initialize from* $V_0^{\mathbb{P}}(w) := \sup_{(\widetilde{w},I)\in\Gamma(w)} c(\widetilde{w}, I) = y + \mathrm{K}(w)$ *and set* $V_{n+1}^{\mathbb{P}} := \mathbb{P}V_n^{\mathbb{P}}$. *Write the corresponding sequence as* $V_n^{\mathbb{H}}$. *Then, for every* $n \in \mathbb{N}$*:*

i) the restriction of $V_n^{\mathbb{H}}$ *to every norm-bounded state set is weak-$*$ continuous;*

ii) the supremum defining $V_0^{\mathbb{P}}$ *in the* $\mathbb{P}$ *case and the suprema defining* $V_{n+1}^{\mathbb{H}} = \mathbb{H}V_n^{\mathbb{H}}$ *are maxima;*

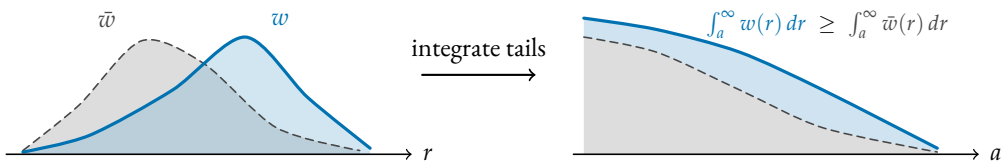
iii) the corresponding argmax correspondences are nonempty-valued, weak-$*$ compact-valued, and upper hemicontinuous on norm-bounded state sets.

The fixed point $V^{\mathbb{T}}$ *is weak-$*$ continuous on every norm-bounded state set and the stationary* $\mathbb{T}$*-argmax correspondence is nonempty-valued and weak-$*$ compact-valued. For* $\mathbb{P}$*-iteration initialized at* $V_0^{\mathbb{P}}$, *the fixed point* $V^{\mathbb{P}}$ *is weak-$*$ upper semicontinuous on every norm-bounded state set, and the stationary* $\mathbb{P}$*-argmax correspondence is nonempty-valued and weak-$*$ compact-valued.*

In sum, the results so far concern the unrestricted recursive model. No trading strategy has yet been imposed, and the tax-share operator T is restricted only by Assumption 2. Thus backward iteration, shape preservation, and policy attainment hold before any LHCL restriction or linear-tax specialization is introduced. Although limited use of losses taxation $\mathrm{T}_{\mathrm{LUL}}$ is not itself a tax-share system, the same Bellman foundations extend after unused losses are added to the state; Section IV.D sketches this extension. The next subsection specializes to $\mathrm{T}_{\mathrm{FUL}}$ to obtain an economically interpretable restriction on optimal trading.

IV.C Recursive Optimal Trading under Linear Taxation

Each trading-strategy class specifies a subset $\Gamma^S(w) \subseteq \Gamma(w)$ of feasible controls at each portfolio w. Under the conditions below, the Bellman problems under $\mathrm{T}_{\mathrm{FUL}}$ attain the same values when controls are restricted to the LHCL class. To establish this result, I use a partial order



(a) tax-lot distributions (b) tail test for every a

Fig. 3. Right dominance. Solid blue denotes w and dashed dark gray $\bar{w}$. Integrating each distribution above a threshold a gives panel (b); the blue curve lies above the gray curve at every threshold, so w passes the tail test. It right-dominates only if it also satisfies $\mathrm{K}(w) \geq \mathrm{K}(\bar{w})$. Curves and shading are schematic.

that I call *right dominance*. Under it, a portfolio is better if it preserves more market value in the right tail and has at least as much current liquidation value. Given linear taxes and the support condition $z\mathbb{E}[R] \geq 1$ for every aggregate realization, z, this order is stable under the law of motion. Consequently, value functions that are increasing in right-dominance admit LHCL optimal controls, because LHCL controls induce right-dominant continuation states. If both investment frictions and aggregate risk are present, the size of the harvest interval of these LHCL controls depends on preferences and cannot be determined explicitly. However, even in those cases there exists a computable *core loss region* which is always optimal to harvest.

Dominance on portfolios. Throughout the subsection, $\mathrm{K} = \mathrm{K}_{\mathrm{FUL}}$ with $k_{\mathrm{FUL}}(r) = 1 - \tau + \tau/r$ and $\tau \in [0, 1)$. This linear specialization satisfies Assumption 2. Supplementary Online Appendix Section B.D.1 gives sufficient conditions for right dominance to be preserved under returns for more general tax-share kernels. The right-dominance order is the accounting order used below.

Definition 4 (Right dominance). *For $w, \bar{w} \in \mathcal{W}$, write $w \succeq_R \bar{w}$ if*

$$\int_a^\infty w(r)\,dr \geq \int_a^\infty \bar{w}(r)\,dr \qquad \text{for every } a > 0, \tag{9}$$

$$\mathrm{K}(w) \geq \mathrm{K}(\bar{w}). \tag{10}$$

A value function V is right-increasing if $w \succeq_R \bar{w}$ implies $V(w) \geq V(\bar{w})$.

Figure 3 illustrates both components of the order. The tail condition (9) says that the dominant portfolio must have at least as much market value in the right tail as the comparison portfolio; eq. (10) requires that it offers at least as much after-tax value. The intuition is clearest at infinity: one unit of after-tax wealth can be held as $\boldsymbol{\delta}_\infty/(1-\tau)$. Since assets at $r = \infty$ no longer move between tax lots, future growth of $\mathrm{M}(w)$ and $\mathrm{K}_{\mathrm{FUL}}(w)$ coincide there, whereas $\mathrm{K}(\boldsymbol{\delta}_r)$ grows by less than $\mathrm{M}(\boldsymbol{\delta}_r)$ at finite r, making it less preferable.

The right-dominance order is only then recursively useful if it is stable under the law of motion in the sense of Proposition 3. The upper-tail component is stable because idiosyncratic and aggregate returns move tax lots by positive multiplicative factors. The liquidation-value

component is more delicate. A unit in tax lot r satisfies $\mathrm{K}_{\mathrm{FUL}}(A_z Q\boldsymbol{\delta}_r) = k_{\mathrm{FUL}}(r) + (1-\tau)(z\mathbb{E}[R]-1)$. Hence the future liquidation-value comparison equals the current $\mathrm{K}_{\mathrm{FUL}}$ comparison plus a market-value comparison multiplied by $(1-\tau)(z\mathbb{E}[R]-1)$. Since the upper-tail component implies $\mathrm{M}(w) \geq \mathrm{M}(\bar{w})$, the whole right-dominance order is stable under the law of motion whenever $z\mathbb{E}[R] \geq 1$ in every aggregate state. The next proposition formalizes this condition.

Proposition 3 (Stability of right dominance under the law of motion)**.** *Fix an aggregate realization z with $z\mathbb{E}[R] \geq 1$. If $w \succeq_R \bar{w}$, then $A_z Q w \succeq_R A_z Q\bar{w}$. Consequently, for every $I \geq 0$, $A_z(Qw + I\chi) \succeq_R A_z(Q\bar{w} + I\chi)$. If $z\mathbb{E}[R] \geq 1$ for every $z \in \mathcal{Z}$, this holds state by state on the whole aggregate support.*

The support condition is sharp. In the aggregate-downside example of Section III.B, $4\boldsymbol{\delta}_2$ right-dominates $3\boldsymbol{\delta}_1$ before returns are realized, but the liquidation-value comparison reverses when $z\mathbb{E}[R] < 1$. Thus, even under $\mathrm{T}_{\mathrm{FUL}}$, right dominance need not be stable over time when the aggregate-return support includes sufficiently adverse realizations.

A distinct failure occurs under $\mathrm{T}_{\mathrm{NUL}}$. The no-offset kink can reverse the liquidation-value comparison even while the upper-tail ranking survives. For example, let $\tau = 1/2$ and idiosyncratic returns be deterministic at $R = 2$, and write $K_{\mathrm{NUL}}(w) := \int k_{\mathrm{NUL}}(r)w(r)\,\mathrm{d}r$. The portfolio $\bar{w} = (4/3)\boldsymbol{\delta}_2$ right-dominates $w = \boldsymbol{\delta}_{1/2}$: it places more mass in every upper tail, while $K_{\mathrm{NUL}}(\bar{w}) = 1 = K_{\mathrm{NUL}}(w)$. After the return, however, $Q\bar{w} = (8/3)\boldsymbol{\delta}_4$ and $Qw = 2\boldsymbol{\delta}_1$, but $K_{\mathrm{NUL}}(Q\bar{w}) = 5/3 < 2 = K_{\mathrm{NUL}}(Qw)$. Thus right dominance is not stable under the NUL law of motion, even with positive deterministic net returns. The tax operator and return support must jointly ensure that the liquidation-value comparison is preserved over time. Under $\mathrm{T}_{\mathrm{FUL}}$ and $z\mathbb{E}[R] \geq 1$, Proposition 3 establishes this preservation.

Dominance-inducing trades. Right dominance is deliberately weaker than a pointwise comparison of market values because it is tailored to trades that induce dominating continuation states. Harvesting low-r tax lots and reinvesting the proceeds shifts mass right. Any additional liquidation is also best taken from the left: among all ways to raise a given amount of after-tax cash, selling an interval $(0, a]$ preserves the most right-tail market value in the residual portfolio.

Right dominance can be applied state by state to identify a harvest region that does not depend on preferences. Because aggregate returns affect this comparison, the region may depend on the realization z. Harvesting a unit from tax lot $r \leq 1$, $\boldsymbol{\delta}_r$, and reinvesting the after-tax proceeds contributes $A_z(\delta_I k_{\mathrm{FUL}}(r)\chi)$ to the continuation state, whereas leaving the unit unharvested contributes $A_z Q\boldsymbol{\delta}_r$. The tax lots worth harvesting in realization z are therefore exactly those for which the harvested contribution right-dominates the unharvested

contribution:

$$\begin{aligned}\mathcal{H}_z^L :=& \left\{ r \in (0,1] : A_z(\delta_I k_{\mathrm{FUL}}(r)\chi) \succeq_R A_z Q \boldsymbol{\delta}_r \right\} \\ =& \left\{ r \in (0,1] : \begin{array}{l} \delta_I k_{\mathrm{FUL}}(r) \geq 1, \\ \delta_I k_{\mathrm{FUL}}(r)\{z(1-\tau)\mathbb{E}[R]+\tau\} \geq z(1-\tau)\mathbb{E}[R]+\tau/r \end{array} \right\} \\ =& (0, r_z^H].\end{aligned} \tag{11}$$

where the second equality uses $\mathrm{K}_{\mathrm{FUL}}(A_z Q \boldsymbol{\delta}_r) = z(1-\tau)\mathbb{E}[R] + \tau/r$ and the fact that the tail condition reduces to $\delta_I k_{\mathrm{FUL}}(r) \geq 1$. Under the support condition $z\mathbb{E}[R] \geq 1$ for all z, this is a lower interval for each z, with $r_z^H = 0$ if the set is empty. The core harvest region is the intersection $\mathcal{H}^L := \bigcap_{z\in\mathcal{Z}} \mathcal{H}_z^L = (0, r_H]$, $r_H := \min_{z\in\mathcal{Z}} r_z^H$. These are the tax lots that are worth harvesting under every aggregate realization. Tax lots outside $\mathcal{H}^L$ may still be worth harvesting once probabilities and preferences are taken into account.

For a given aggregate realization z, increasing z weakly expands the statewise interval $\mathcal{H}_z^L$: conditional on the tail test $\delta_I k_{\mathrm{FUL}}(r) \geq 1$, a larger aggregate shock raises the continuation value of reinvested harvest proceeds relative to the retained low-return tax lot. Increasing δ_I also weakly expands $\mathcal{H}_z^L$, because reinvested harvest proceeds lose less continuation value; lowering δ_I shrinks it from above and may make it empty. The effect of τ is less uniform. A higher τ raises the current value of harvested losses, so the tail test $\delta_I k_{\mathrm{FUL}}(r) \geq 1$ holds for more tax lots r. But τ also affects the continuation-value comparison through the term τ/r. In the frictionless case $\delta_I = 1$, this ambiguity disappears: the support condition $z\mathbb{E}[R] \geq 1$ implies $\mathcal{H}_z^L = (0,1]$ for every z. The comparative statics of the core $\mathcal{H}^L$ follow by intersecting the statewise intervals.

An LHCL control indexed by h always sells the harvest interval $(0,h]$ and finances current consumption and investment by extending the sale to a cutoff $a > h$. An sLHCL control is an LHCL control with the additional no-roundtrip restriction: if the cutoff lies outside the harvest interval, then investment is zero. Thus define the new core-harvesting correspondence Γ^H, which requires liquidation of the entire core $(0, r_H]$ but imposes no cutoff structure outside it, and recall the indexed LHCL and sLHCL correspondences from Definition 2:

$$\begin{aligned}\Gamma^H(w) &:= \{(\widetilde{w}, I) \in \Gamma(w) : w\mathbf{1}_{(0,r_H]} \leq \widetilde{w}\}, \\ \Gamma_h^L(w) &:= \{(w\mathbf{1}_{(0,a]}, I) : a \in [h,\infty],\ 0 \leq I \leq y + \mathrm{K}(w\mathbf{1}_{(0,a]})\}, \\ \Gamma_h^{sL}(w) &:= \{(w\mathbf{1}_{(0,a]}, I) \in \Gamma_h^L(w) : a > h \Rightarrow I = 0\}.\end{aligned} \tag{12}$$

For every $w \in \mathcal{W}$, the core-endpoint correspondences satisfy $\Gamma_{r_H}^{sL}(w) \subseteq \Gamma_{r_H}^L(w) \subseteq \Gamma^H(w) \subseteq \Gamma(w)$. Proposition 4(i) states that the core can always be harvested without lowering current consumption or any aggregate-state continuation value. Proposition 4(ii) then turns such a core-harvesting control into an LHCL control.

Proposition 4 (Core-harvest LHCL restriction)**.** *Suppose* $\mathrm{K} = \mathrm{K}_{\mathrm{FUL}}$ *and* $z\mathbb{E}[R] \geq 1$ *for every*

$z \in \mathcal{Z}$.

i) For every $w \in \mathcal{W}$ and every feasible control $(\widetilde{w}, I) \in \Gamma(w)$, there is a control $(\widetilde{w}^H, I^H) \in \Gamma^H(w)$ with the same current consumption and $G_z(w, \widetilde{w}^H, I^H) \succeq_R G_z(w, \widetilde{w}, I)$ for every $z \in \mathcal{Z}$. Hence, if V is right-increasing, the suprema defining $\mathbb{T}V$ and $\mathbb{P}V$ are unchanged if $\Gamma(w)$ is replaced by $\Gamma^H(w)$.

ii) For every core-harvesting control $(\widetilde{w}, I) \in \Gamma^H(w)$, there is a cutoff $a \in [r_H, \infty]$ such that $(w\mathbf{1}_{(0,a]}, I) \in \Gamma^L_{r_H}(w)$ has the same current consumption and $G_z(w, w\mathbf{1}_{(0,a]}, I) \succeq_R G_z(w, \widetilde{w}, I)$ for every $z \in \mathcal{Z}$. Hence, if V is right-increasing, the suprema defining $\mathbb{T}V$ and $\mathbb{P}V$ are unchanged if $\Gamma^H(w)$ is replaced by $\Gamma^L_{r_H}(w)$.

In particular, for any right-increasing V, every attained full maximum of either Bellman objective admits a $\Gamma^L_{r_H}$ maximizer; if the full maximizer is unique, it belongs to $\Gamma^L_{r_H}(w)$.

Part (ii) replaces the control while keeping proceeds fixed. Any control in $\Gamma^H(w)$ can be written as $\widetilde{w} = w\mathbf{1}_{(0,r_H]} + \widehat{w}$, where $0 \leq \widehat{w} \leq w\mathbf{1}_{(r_H,\infty)}$. The proposition says that it is preferable to replace this extra liquidation $\widehat{w}$ by extending the cutoff beyond r_H, while preserving total liquidation proceeds and the investment choice.

Recursive LHCL optimality. To formulate the recursive implications of Proposition 4, let $\mathbb{T}^L$ and $\mathbb{P}^L$ denote the discounted-utility and sup-min Bellman operators with the feasible set $\Gamma(w)$ replaced by the LHCL class $\Gamma^L_{r_H}(w)$.

Theorem 3 (Recursive LHCL optimality under linear taxation)**.** *Suppose Assumptions 1 and 3 hold, $\mathrm{K} = \mathrm{K}_{\mathrm{FUL}}$, and $z\mathbb{E}[R] \geq 1$ for every $z \in \mathcal{Z}$. If V is right-increasing, then:*

i) $\mathbb{T}V = \mathbb{T}^L V$ and $\mathbb{P}V = \mathbb{P}^L V$.

ii) Whenever the supremum defining either full Bellman objective at any state w is attained, it is attained by a $\Gamma^L_{r_H}(w)$ control. If the full maximizer is unique, it belongs to $\Gamma^L_{r_H}(w)$.

iii) $\mathbb{T}V$, $\mathbb{T}^L V$, $\mathbb{P}V$, and $\mathbb{P}^L V$ are right-increasing.

The argument combines two comparisons. At a given portfolio, harvesting the core loss region and arranging any remaining liquidation as a cutoff replaces each feasible trade by an LHCL trade with the same current consumption and a right-dominating continuation portfolio in every aggregate state. For any right-increasing continuation value function V, this replacement cannot reduce either Bellman objective, so restricting trade to LHCL leaves the value unchanged. Across two right-dominance ordered starting portfolios, any LHCL trade from the dominated portfolio can likewise be matched from the dominant portfolio with the same consumption and right-dominating continuation portfolios. This second comparison shows that the Bellman update preserves right-increasingness. The full proof is provided in Supplementary Online Appendix B.D.

The theorem is a statement on Bellman operators. The next corollary records what it implies for the dynamic programming objects constructed above: the finite-horizon iterates and limiting value functions are right-increasing, and maximizers can always be found in the LHCL class of controls. For the canonical Bellman operator, initialize discounted value iteration from $V_0^{\mathbb{T}} = V_0^{\mathbb{T},L} := 0$ and set $V_{n+1}^{\mathbb{T}} := \mathbb{T}V_n^{\mathbb{T}}$, $V_{n+1}^{\mathbb{T},L} := \mathbb{T}^L V_n^{\mathbb{T},L}$. For sup-min iteration, initialize the full problem at $V_0^{\mathbb{P}}(w) := \sup_{(\widetilde{w},I)\in\Gamma(w)} c(\widetilde{w},I) = y + \mathrm{K}_{\mathrm{FUL}}(w)$ and the restricted problem at $V_0^{\mathbb{P},L}(w) := \sup_{(\widetilde{w},I)\in\Gamma_{r_H}^L(w)} c(\widetilde{w},I) = y + \mathrm{K}_{\mathrm{FUL}}(w)$. Thus the initial values coincide. Set $V_{n+1}^{\mathbb{P}} := \mathbb{P}V_n^{\mathbb{P}}$, $V_{n+1}^{\mathbb{P},L} := \mathbb{P}^L V_n^{\mathbb{P},L}$.

Corollary 3 (LHCL value iteration and stationary optimality)**.** *Suppose Assumptions 1, 2, and 3 hold,* $\mathrm{K} = \mathrm{K}_{\mathrm{FUL}}$*, and* $z\mathbb{E}[R] \geq 1$ *for every* $z \in \mathcal{Z}$*. Suppose the conditions of Proposition 1(i) hold, so* $\mathbb{T}$ *is a contraction on the relevant* $\mathcal{B}_{W_p}(\mathcal{W})$*.*

i) For all $n \geq 0$*, the full and* $\Gamma_{r_H}^L$*-restricted iterates of* $\mathbb{T}$*,* $\mathbb{P}$ *coincide and are right-increasing.*

ii) The $\mathbb{T}$*- and* $\mathbb{T}^L$*-iterates converge in norm to* $V^{\mathbb{T}} = V^{\mathbb{T},L}$*. The* $\mathbb{P}$*- and* $\mathbb{P}^L$*-iterates converge pointwise to* $V^{\mathbb{P}} = V^{\mathbb{P},L}$*. Both limiting values are right-increasing fixed points of their respective operators.*

iii) At any state w*, for each continuation value function* $V \in \{V_n^{\mathbb{T}}, V_n^{\mathbb{P}}, V^{\mathbb{T}}, V^{\mathbb{P}} : n \geq 0\}$*, the full Bellman maximum admits a maximizer from* $\Gamma_{r_H}^L(w)$*. If the full maximizer is unique, it belongs to* $\Gamma_{r_H}^L(w)$*.*

The iterations begin with values that are right-increasing and identical in the full and restricted problems: zero for discounted utility and immediate liquidation value plus income for the perpetuity problem. At each step, Theorem 3 preserves both properties, allowing the same argument to be repeated at the next horizon. The contraction argument for discounted utility and the decreasing pointwise-convergence argument for the perpetuity problem then carry equality and right-increasingness to the limiting values. Together with the attainment result of Theorem 2, the LHCL replacement yields optimal LHCL trades at every finite horizon and at the stationary limiting values. The full proof is provided in Supplementary Online Appendix B.D.

First note that the LHCL restriction in Corollary 3 does not by itself rule out round trips involving positive investment and sales of appreciated tax lots. Nor does it identify the full liquidation cutoff. Proposition 4 pins down only the preference-independent harvest core $(0, r_H]$; beyond r_H, probabilities, preferences, and frictions determine how far liquidation extends. Generally, harvesting and reinvesting losses beyond $r_H < 1$ involves a trade-off: it shifts holdings toward higher tax lots, but investment costs can reduce liquidation value, so right dominance need not hold in every aggregate state.

When investment is frictionless, $\delta_I = 1$, the harvest core expands to $r_H = 1$, so all losses can be harvested optimally. Selling appreciated tax lots for reinvestment would instead

shift holdings to the left. Optimal controls can therefore be chosen without simultaneous investment and sales of appreciated positions: the sLHCL restriction is valid. The following corollary records this result.

Corollary 4 (Known harvest regions and frictionless sLHCL)**.** *The core-harvest cutoff proposition reduces to sharper restrictions in the frictionless-investment case. If investment is frictionless, $\delta_I = 1$, and $z\mathbb{E}[R] \geq 1$ for every aggregate realization z, then $r_z^H = 1$ for every z, all losses are harvested, and the conclusions of Theorem 3 and Corollary 3 remain valid with Γ_1^{sL} in place of $\Gamma_{r_H}^L$.*

Corollaries 3 and 4 are benchmark restriction results for life-cycle and perpetuity problems under T_{FUL}. They establish when Bellman optimization can be restricted to LHCL—and, in the frictionless-investment case, sLHCL—controls.

IV.D Limited Use of Losses and Carryforward

In this subsection I sketch out the carryforward Bellman modification and then summarize the corresponding trading implication.

The carryforward Bellman operators $\mathbb{T}_{LUL}$ and $\mathbb{P}_{LUL}$ augment the portfolio state by the stock of unused losses, $(w, F) \in \mathcal{S} := \mathcal{W} \times [0, \infty)$, and use the corresponding LUL primitives in eqs. (6) and (7). The dynamic-programming conclusions from the FUL case then extend with little new structure. The extra state variable is scalar, the feasible control set remains compact in the weak-$*$ topology, and T_{LUL}, K_{LUL}, and F_{LUL} are weak-$*$ continuous in the liquidation choice. Thus discounted value iteration has the same contraction logic under the same utility cases; the sup-min iteration has the same decreasing backward sequence; and finite horizon Bellman suprema are attained. In the no-aggregate-risk case, $V_{LUL}^{\mathbb{P}}(w, F)$ again has a perpetuity interpretation: it is the supremum of constant consumption levels feasible forever from (w, F) under the carryforward transition, including carryforward leakage and investment friction.

Carryforward taxation requires modifying both the state and the portfolio order used in the FUL analysis. Current resources depend nonlinearly on realized gains, and unused losses remain available in future periods, so the continuation state is (w, F) rather than w alone. Under perfect carryforward, however, portfolio value and unused credits can be combined. Write $\Psi(w, F) := K_{FUL}(w) + \tau F$ for their joint linear-tax value. Say that $(\bar{w}, \bar{F})$ *carryforward right-dominates* (w, F), written $(\bar{w}, \bar{F}) \succeq_{CFRD} (w, F)$, if $\int_a^\infty [\bar{w}(r) - w(r)]\, dr \geq 0$ for every $a \geq 0$ and $\Psi(\bar{w}, \bar{F}) \geq \Psi(w, F)$. The tail comparison preserves the option value of high-r assets; the Ψ comparison preserves their joint after-tax value. At $a = 0$, the tail test implies $M(\bar{w}) \geq M(w)$. Together with the Ψ comparison, this implies $K_{LUL}(\bar{w}, \bar{F}) \geq K_{LUL}(w, F)$ by

the first of the two identities

$$\mathrm{K}_{\mathrm{LUL}}(w,F) = \min\{\mathrm{M}(w), \Psi(w,F)\}, \tag{13}$$

$$\Psi(w - \widetilde{w} + I\boldsymbol{\delta}_1, F') = \Psi(w,F) - \big(\mathrm{K}_{\mathrm{LUL}}(\widetilde{w},F) - I\big), \quad F' := \big[F - \mathrm{CG}(\widetilde{w})\big]_+. \tag{14}$$

Thus two controls delivering the same current net withdrawal leave the same pre-return Ψ value. A lower-cutoff replacement can therefore improve the upper tails without sacrificing the second component of the order. For clarity, let the perfect-carryforward strong-LHCL correspondence be

$$\Gamma^{sL}_{\mathrm{CF}}(w,F) := \left\{(w\mathbf{1}_{(0,a]}, I) \in \Gamma(w,F) : a \in [1,\infty],\ a > 1 \Rightarrow I = 0\right\}. \tag{15}$$

It harvests every loss; if liquidation extends into gains, it rules out simultaneous reinvestment.

Proposition 5 (Dynamic sLHCL optimality with perfect carryforward)**.** *Suppose $\delta_F = \delta_I = 1$, Assumptions 1 and 3 hold, $\mathbb{E}[R] \geq 1$, and every aggregate realization satisfies $z \geq 1$. Suppose also that the perfect-carryforward Bellman maxima are attained and that discounted value iteration is convergent under the product-state counterpart of Proposition 1. Starting from $V^{\mathbb{T}}_{\mathrm{LUL},0} \equiv 0$, for every finite horizon n and every state (w,F):*

i) $V^{\mathbb{T}}_{\mathrm{LUL},n}$ is increasing in carryforward right dominance;

ii) the full Bellman maximum admits a control in $\Gamma^{sL}_{\mathrm{CF}}(w,F)$, and full and restricted value iteration coincide;

iii) the limiting full and $\Gamma^{sL}_{\mathrm{CF}}$-restricted fixed points coincide, and the stationary full problem admits an sLHCL maximizer. If the full maximizer is unique, it belongs to $\Gamma^{sL}_{\mathrm{CF}}(w,F)$.

The argument differs from the FUL proof at exactly two steps. Equation (13) replaces the linear current-resource comparison, and (14) shows that matching current cash pins down Ψ while the cutoff replacement increases right-tail mass. Idiosyncratic returns preserve both comparisons when $\mathbb{E}[R] \geq 1$, and aggregate returns preserve them state by state when $z \geq 1$. Frictionless reinvestment, $\delta_I = 1$, is also essential: when $\delta_I < 1$, replacing a retained loss with harvested-and-reinvested proceeds can reduce the continuation state, so the sLHCL replacement need not dominate. Bellman induction then gives the proposition. Proposition 8 in the Supplementary Online Appendix records the compactness and convergence verification.

Optimal realization choices determine how statutory tax rates translate into sustainable consumption and government revenue. Section V quantifies these long-run effects using effective tax rates that compare realization-based taxation with taxation upon accrual.

V Effective Taxation

Under realization-based capital-gains taxation, effective taxation depends on which gains are realized, when they are realized, and whether they ever enter the tax base. Recent evidence

documents the importance of gains that reach death unrealized and are forgiven through basis step-up (Msall and Næss, 2026). This section isolates an additional source of persistent deferral: even without forgiveness, optimal trading can continually roll the most appreciated tax lots forward. I first quantify the value of this realization margin to the investor using the *market gains effective tax rate* (MGET). I then use the MGET to construct counterfactual accrual-tax revenue and compute a government-side effective tax rate from collections in a closed cohort economy, including alternative treatments of gains at death.

V.A The Market Gains Effective Tax Rate

The MGET is an indifference rate for an investor who evaluates tax systems by the maximal perpetuity the portfolio can sustain. The MGET asks what accrual-tax rate delivers the same maximal perpetuity as the realization-based system, under which the investor chooses which gains and losses to realize, holding the cross-sectional return distribution fixed and abstracting from aggregate risk. The initial portfolio $w_0 = \chi$ follows from carrying a unit investment for one period.[18]

If the investor faces accrual taxation, the perpetuity from a unit investment is $c = \mathbb{E}[R - 1](1 - \tau)$. Let $\mathcal{C} = \mathcal{C}(\mathrm{T}, f)$ denote the feasible set of perpetuities under tax operator T and return distribution f, and let $\bar{c} := \sup \mathcal{C}$. The MGET is the tax rate τ^* satisfying $\bar{c} = \mathbb{E}[R - 1](1 - \tau^*)$, that is,

$$\tau^* := 1 - \frac{\bar{c}}{\mathbb{E}[R - 1]}. \tag{16}$$

The wedge $1 - \tau^*/\tau$ measures the value of the realization and deferral margin to the investor. The perpetuity problem imposes no terminal-liquidation condition. Its most appreciated tax lots can therefore be rolled forward indefinitely even if every individual asset is eventually sold. Section V.B introduces mortality and alternative death rules to distinguish this portfolio-level deferral from gains forgiven at death.

One can connect the MGET to a revenue statistic by analyzing tax collections in the counterfactual economy in which gains are taxed upon accrual at the investor's indifference rate. Let $X(\tau)$ denote long-run investment exposed to the capital-gains tax when the realization-based statutory rate is τ. Long-run revenue in the corresponding accrual-tax economy is $\mathcal{R}^*(\tau) := X(\tau)\mathbb{E}[R - 1]\tau^*(\tau)$. The factor $\mathbb{E}[R - 1]$ converts invested dollars into expected market-gain units and drops out of elasticities. The local semi-elasticity of this consumption-equivalent accrual-tax benchmark with respect to the statutory rate is therefore

$$\frac{\partial \log \mathcal{R}^*}{\partial \tau} = \frac{1}{X}\frac{\partial X}{\partial \tau} + \frac{1}{\tau^*}\frac{\partial \tau^*}{\partial \tau}. \tag{17}$$

The first term is the long-run investment or participation response, a margin studied in em-

[18] The unit investment is a normalization because $\sup \mathcal{C}(xw) = x \sup \mathcal{C}(w)$ for any $x \geq 0$, compare (8).

pirical work on portfolio choice and asset demand (e.g. Alan et al., 2010, Gabaix et al., 2023, Poterba, 2002). The second term is the tax-timing channel: the statutory rate changes the value of deferring realizations, and the TPM microfounds that channel through the optimal-trading problem summarized by τ^*.

I compute τ^* and its sensitivity to the statutory rate across a range of log-normal return scenarios. For each return environment and statutory rate, I approximate the return distribution on a binomial grid calibrated to match $(\mathbb{E}[R], \mathbb{V}[R])$ and use binary search to recover τ^*. Evaluating this calculation across statutory rates yields the sensitivity $\partial\tau^*/\partial\tau$. I simulate the portfolio forward for 200 periods, liquidating the candidate perpetuity in each period using the recursively optimal strategies of the preceding section. Supplementary Online Appendix Section C.A details the algorithm.

Adjusted MGET: Dividends and Inflation. The baseline definition of $\bar{c}$ assumes real, ex-dividend returns, so that all value growth is subject to the realization margin. Two features break this assumption: firms pay a fraction of returns as dividends, and capital gains taxes apply to nominal gains. Both can be incorporated into the perpetuity set without changing the optimal trading rules.

Let R denote the real cum-dividend gross return, $\Pi := 1 + \pi$ gross inflation, and $\widetilde{R} := \Pi R$ the nominal cum-dividend gross return. Let $\varphi \in [0,1]$ be the fraction of nominal excess returns paid as dividends each period. Dividends are taxed symmetrically with capital gains. In computations under $\mathrm{T}_{\mathrm{LUL}}$, carry-forward credits offset dividend taxes as well as liquidation taxes. Firms first draw nominal dollar returns, pay out the fraction φ of nominal excess returns, and retain the rest. Hence the retained nominal ex-dividend gross return is

$$\widetilde{R}_{\varphi,\Pi} := 1 + (1-\varphi)(\widetilde{R} - 1). \tag{18}$$

Let $d_t(w_t) := \varphi\,(\mathbb{E}[\tilde{R}] - 1)\,\mathrm{M}(w_t)$ denote the gross dividend flow. For FUL, the after-tax dividend flow depends only on total portfolio value and is therefore additively separable from the liquidation problem: $D_t^{\mathrm{net}}(w_t) := (1-\tau)d_t(w_t)$. For $\mathrm{T}_{\mathrm{LUL}}$, however, dividends enter the same taxable base as realized gains. Thus, computations replace $D_t^{\mathrm{net}}(w_t) + \mathrm{K}_{\mathrm{LUL}}(\widetilde{w}_t, F_t)$ by

$$\mathrm{K}^{D}_{\mathrm{LUL}}(\widetilde{w}_t, F_t; w_t) := \mathrm{M}(\widetilde{w}_t) + d_t(w_t) - \tau\left[\mathrm{CG}(\widetilde{w}_t) + d_t(w_t) - \delta_F F_t\right]_+,$$

with dividend-inclusive carry-forward update $F_{t+1} = \left[F_t - \mathrm{CG}(\widetilde{w}_t) - d_t(w_t)\right]_+$.

Inflation enters as a market condition. The investor's real consumption target is constant, so nominal liquidation and consumption targets grow at Π^t. The identity $\widetilde{R} = \Pi R$ gives $\log\widetilde{R} = \log R + \log\Pi$, where $\log\Pi = \log(1+\pi) \approx \pi$. For a fixed real return distribution, inflation therefore shifts log returns without changing their variance and multiplies arithmetic return dispersion by Π. In the numerical comparisons, I instead hold the arithmetic standard deviation of nominal returns, σ, fixed while setting their mean to $\Pi(1+\mu)$, where $\mu :=$

$\mathbb{E}[R] - 1$ is the expected real return. This convention varies inflation while keeping nominal return dispersion constant. The dividend transformation in (18) is then applied to the nominal gross return.

Combining both adjustments, let $\widetilde{Q}_{\varphi,\Pi}$ denote the law of motion induced by the nominal return law for $\widetilde{R}_{\varphi,\Pi}$, and write $\chi_{\varphi,\Pi} := \widetilde{Q}_{\varphi,\Pi}\boldsymbol{\delta}_1$. The adjusted perpetuity set is

$$\mathcal{C}_{\varphi,\Pi}(T,f) := \left\{ c \geq 0 \;\middle|\; \begin{array}{l} \exists (w_t, \widetilde{w}_t, I_t)_{t\geq 0} \text{ with } w_0 = \chi_{\varphi,\Pi}, \\ \widetilde{w}_t \in [0, w_t],\ I_t \geq 0, \quad D_t^{\text{net}}(w_t) + \mathrm{K}(\widetilde{w}_t) = c\Pi^t + I_t, \\ w_{t+1} = \widetilde{Q}_{\varphi,\Pi}(w_t - \widetilde{w}_t) + \delta_I I_t \chi_{\varphi,\Pi} \quad \forall t \in \mathbb{N} \end{array} \right\}. \tag{19}$$

For $\mathrm{T}_{\mathrm{LUL}}$, the resource equation in (19) is interpreted with $\mathrm{K}^D_{\mathrm{LUL}}$ and the dividend-inclusive carry-forward update above. The maximal real perpetuity is $\bar{c}_{\varphi,\Pi} := \sup \mathcal{C}_{\varphi,\Pi}(T,f)$. The adjusted MGET is defined by the unchanged benchmark formula $\bar{c}_{\varphi,\Pi} = \mathbb{E}[R-1](1-\tau^*_{\varphi,\Pi})$, so it still tells what accrual tax on total market gains would generate the same real perpetuity under optimal trading.

MGET Simulation Results

The figures use log-normal returns with expected real return $\mu = 5\%$. I vary nominal return volatility over $\sigma \in \{20\%, 30\%, 40\%\}$ and statutory rates over $\tau \in \{10\%, 50\%\}$. The values for μ span conventional estimates of the equity premium (Mehra and Prescott, 2003), while the volatility values cover moderate and high single-stock dispersion (Campbell et al., 2001). Unreported $\mu = 3\%$ and $\mu = 7\%$ runs preserve the magnitude and qualitative patterns discussed in this section.

Expected real returns, return dispersion, inflation, and dividend payouts vary over time. I view the baseline as the most applicable of several plausible scenarios. I set the baseline statutory rate to 25%, close to the average rate for OECD countries taxing gains upon realization reported in the introduction.

Figure 4 shows how the market environment changes the investor-side effective burden generated by realization-based taxation. I compare $\mathrm{T}_{\mathrm{FUL}}$ to $\mathrm{T}_{\mathrm{LUL}}$ with perfect carryforward, because both systems preserve losses but differ in the timing with which those losses can offset gains. The top row varies dividends and the bottom row inflation; all four panels report the level τ^*/τ. Supplementary Online Appendix Figure 5 reports the corresponding slopes $\partial\tau^*/\partial\tau$. This slope is the timing component in (17): holding long-run investment fixed, a larger slope means that the statutory rate increase passes through more strongly into counterfactual tax revenue.

At the common benchmark $\varphi = 0.30$, $\pi = 2\%$, and $\sigma = 0.30$, the effective-rate ratio under $\mathrm{T}_{\mathrm{LUL}}$ with perfect carryforward is 0.53 at $\tau = 10\%$, 0.56 at $\tau = 25\%$, and 0.60 at $\tau = 50\%$. Thus, optimal realization timing reduces the investor-side effective burden by roughly 40 to 47 percent relative to accrual taxation at the statutory rate. These numbers

Table 1: Baseline MGET levels and statutory-rate pass-through

	T_{LUL}			T_{FUL}		
	$\tau = 10\%$	$\tau = 25\%$	$\tau = 50\%$	$\tau = 10\%$	$\tau = 25\%$	$\tau = 50\%$
Effective-rate ratio, τ^*/τ	0.535	0.557	0.605	0.524	0.547	0.597
Pass-through, $\partial\tau^*/\partial\tau$	0.549	0.597	0.719	0.539	0.590	0.717

Notes: $\mu = 5\%$, $\varphi = 0.30$, $\pi = 2\%$, and $\sigma = 0.30$. Under T_{LUL}, $\delta_F = 1$. Supplementary Online Appendix Figure 5 reports the full comparative statics.

suggest a substantial ability to defer capital gains taxes.

Under T_{LUL}, the corresponding pass-through $\partial\tau^*/\partial\tau$ is 0.55 at $\tau = 10\%$, 0.60 at $\tau = 25\%$, and 0.72 at $\tau = 50\%$. A one-percentage-point increase in the statutory rate therefore raises the effective rate by only 0.55, 0.60, or 0.72 percentage points, holding long-run investment fixed. Pass-through is incomplete at all three rates, although it is stronger when the statutory rate is high. Table 1 also reports the corresponding T_{FUL} estimates, which are quantitatively similar.

To put these effective rates in perspective, I calculate the statutory rate required to match the OECD average tax wedge on labour. In 2025, this wedge was 35.1 percent for a single worker without children earning the national average wage.[19] Holding the market environment and loss-offset provisions at their baseline values, matching this rate requires a statutory rate of approximately 56.6 percent on dividends and realized capital gains. Optimal realization trading therefore permits an investor facing this statutory rate to sustain the same consumption as under an accrual tax of 35.1 percent.

Dividend payout shares raise effective tax rates in both regimes. In Figure 4, moving from no dividends to a payout share above one half raises τ^*/τ by roughly 20–33 percentage points, depending on the statutory rate and dispersion. A dividend is a forced taxable payout: it removes part of market gains from the investor's realization-timing problem, shrinking the base over which trading can defer taxation. Under limited use of losses, the increase is strongest at low dispersion, where the investor has fewer loss-harvesting opportunities. In absolute terms, the increase is larger at higher statutory rates, where the value of forced taxation is larger. These results show that there is a natural tax advantage to retaining earnings and can explain low dividend payout rates (see also Poterba, 2004).

The same dividend force appears in the pass-through panels of Supplementary Online Appendix Figure 5. As φ rises, the curves for $\partial\tau^*/\partial\tau$ move upward. Intuitively, because the tax levied on distributed dividends moves one-for-one with the statutory rate, a higher dividend share ties effective and statutory rates closer together. The result is visible under both T_{FUL}

[19] The tax wedge includes personal income taxes and employee and employer social contributions, net of cash benefits, relative to total labour costs. See OECD (2026, Table 1.2).

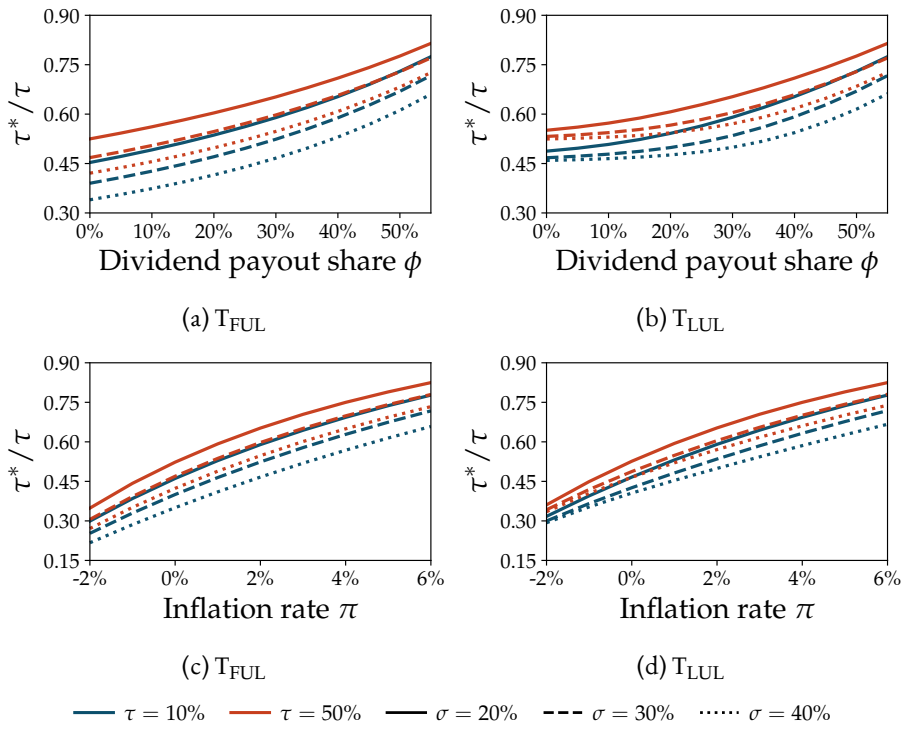


Fig. 4. Investor-side MGET levels. The top row varies the dividend share at $\pi = 2\%$; the bottom row varies inflation at $\varphi = 30\%$. The vertical axis is τ^*/τ. Blue curves use $\tau = 10\%$, orange curves use $\tau = 50\%$; line styles correspond to $\sigma = 20\%, 30\%, 40\%$. Under T_{LUL}, $\delta_F = 1$.

and T_{LUL}, $\delta_F = 1$, though perfect carry-forward compresses part of the regime difference by preserving losses for future use.

Inflation has an even sharper effect. Figure 4 shows that raising nominal inflation substantially increases τ^*/τ. The reason is that capital gains taxes are levied on nominal gains. Inflation shifts the nominal return law upward and creates taxable gains even when the real benchmark has not changed (Feldstein and Slemrod, 1978). This weakens the timing option: more of the portfolio is pushed into taxable-gain states, and a constant real perpetuity requires larger nominal withdrawals over time. A weakening of the timing option also shows in the online-appendix pass-through panels: Higher inflation raises $\partial\tau^*/\partial\tau$, so the effective tax rate becomes more responsive to the statutory rate.

Volatility works in the opposite direction from dividends and inflation in the level panels: ceteris paribus, higher σ lowers τ^*/τ. Cross-sectional dispersion creates more low-r tax lots from which to finance payouts while leaving the most appreciated tax lots invested, so the timing option becomes more valuable. The investor can use depreciated tax lots to fund withdrawals or create carryforward credits while continuing to defer appreciated tax lots. This mechanism is the many-asset version of the tax-loss-harvesting and gain-deferral logic in Constantinides (1983).

Supplementary Online Appendix Section A shows how investors can create this dispersion through portfolio choice. Replacing a common factor loading with a distribution of loadings having the same mean increases cross-sectional return dispersion while preserving the portfolio's aggregate factor exposure. This creates additional opportunities to realize losses and defer gains without increasing exposure to aggregate risk.

Local comparative statics quantify these effects at the baseline. Expressed as percentage-point changes in the effective rate per percentage-point increase in the respective parameter, the derivatives with respect to inflation and the dividend payout share are approximately 1.31 and 0.12. Central-difference estimates for expected real returns are smaller and negative, ranging from -0.07 to -0.05 across the finer grids and step sizes. For cross-sectional dispersion, the semi-elasticity implies that a 1-percent relative increase in dispersion lowers the effective rate by approximately 0.033 percentage points. Each calculation holds the remaining baseline parameters fixed. Inflation thus has a particularly pronounced local effect on effective taxation.

Extension: Leverage

Until now, the MGET benchmark has excluded borrowing. I quantify its effect on effective taxation next. Suppose the investor borrows a fraction x of the initial unit investment at a per-period real rate r_b, rolls the debt at a constant real level, and services interest from liquidations.

Analogously to the unlevered case, the levered MGET $\tau^*(x)$ equates the perpetuity the levered investor achieves, $\bar{c}(x)$, to the perpetuity an accrual tax at rate τ^* would deliver the same investor,

$$\bar{c}(x) = (1+x)\,\mathbb{E}[R-1](1-\tau^*) - x\,r_b + \zeta\,x\,r_b\,\tau^*, \tag{20}$$

where $\zeta = 1$ if interest is deductible and $\zeta = 0$ otherwise.

Under non-deductibility, interest never enters the tax base, so by degree-one homogeneity of all TPM primitives, the debt-financed position is taxed at exactly the zero-leverage rate and the perpetuity obeys the exact closed form

$$\bar{c}(x) \;=\; (1+x)\,\bar{c}_0 - x\,r_b \;=\; \bar{c}_0 + x\big[\mathbb{E}[R-1](1-\tau_0^*) - r_b\big], \tag{21}$$

with $\bar{c}_0$ the unlevered perpetuity and τ_0^* the unlevered MGET. Substituting into (20) with $\zeta = 0$, the interest cancels and the factor $(1+x)$ divides out, leaving $\tau^*(x) = \tau_0^*$: *non-deductible borrowing is neutral to the MGET*. It moves the *level* of the perpetuity — raising it iff the debt cost lies below the deferral-adjusted return, $r_b < b^{nd} := \mathbb{E}[R-1](1-\tau_0^*)$ — but not the effective *rate*, because leverage scales the taxed position and the pre-tax interest identically under realization and accrual taxation. The interesting case is deductible interest, to which I now turn.

Panel A of Table 2 reports the deductible leverage MGET under $\mathrm{T_{LUL}}$ with perfect carry-

Table 2: Leverage and effective taxation under T_{LUL}, $\delta_F = 1$

Panel A. Effective rate, deductible interest

	fair, r_b=5%		cheap, r_b=2.5%	
x	$\bar{c}(x)$	$\tau^*(x)/\tau$	$\bar{c}(x)$	$\tau^*(x)/\tau$
0.00	0.0430	0.557	0.0430	0.557
0.25	0.0442	0.467	0.0490	0.515
0.50	0.0451	0.396	0.0549	0.484
1.00	0.0463	0.295	0.0667	0.441
2.00	0.0477	0.182	0.0901	0.396
3.00	0.0485	0.123	0.1134	0.372

Panel B. Break-even rate, deductible interest

x	$r_b^*(x)$
0.0025	5.64%
0.25	5.57%
0.50	5.50%
1.00	5.38%
2.00	5.26%
3.00	5.19%

Notes: $\bar{c}(x)$ is the maximal real perpetuity under deductible interest, computed at $T_{\max} = 200$, $n_d = 20$, and consumption tolerance 10^{-8} using the sLHCL strategy of Section IV. Both panels set $\mathbb{E}[R] - 1 = 5\%$, $\sigma = 30\%$, $\tau = 25\%$, $\varphi = 30\%$, and $\pi = 2\%$. *Panel A*: the effective rate solves (20) with $\zeta = 1$; non-deductible interest is neutral, $\tau^*(x) = \tau_0^*$ for all x. *Panel B*: $r_b^*(x)$ solves $\bar{c}(x, r_b^*) = \bar{c}_0$; the $x = 0.0025$ row approximates the marginal rate, using consumption tolerance 10^{-10}; the break-even root tolerance is 10^{-6}. Here $b^{nd} = \mathbb{E}[R-1](1-\tau_0^*)$ is the exact, x-independent non-deductible break-even. The $\varphi = \pi = 0$ counterpart is Appendix Table 5.

forward, extending borrowing to three times own capital.[20] With deductibility, the interest deduction enters the tax base. It increases the carryforward F, like a harvested loss credit. The levered position is therefore no longer taxed at τ_0^*, and $\bar{c}(x)$ must be computed rather than recovered from the earlier MGET calculations. The deduction shelters a growing share of the gains that the investor realizes to service the position, so the effective rate *falls* with leverage, and by more the larger the deductible interest. At the fair rate $r_b = \mathbb{E}[R-1]$, it drops from $0.56\,\tau$ at $x = 0$ to $0.12\,\tau$ at $x = 3$; at a cheap rate $r_b = \frac{1}{2}\mathbb{E}[R-1]$, where the interest — and hence the tax shelter — is smaller, only to $0.37\,\tau$. Since $T_{LUL} \geq 0$ the perpetuity is bounded by the zero-tax benchmark ($\bar{c}(x) \leq (1+x)\mathbb{E}[R-1] - x r_b$), so the classic borrow-deduct-defer arbitrage (Stiglitz, 1983) is capped at full tax elimination. Under T_{FUL}, refundability of the deduction restores the closed form (21) with $r_b(1-\tau)$ in place of r_b and makes the arbitrage unbounded, driving $\tau^*(x)$ below zero at high leverage (Appendix Table 6).

How high can the debt cost go before borrowing stops helping? The break-even rate $r_b^*(x)$ is the debt cost at which leverage leaves sustainable consumption unchanged, $\bar{c}(x, r_b^*) = \bar{c}_0$; below it leverage raises the perpetuity, above it lowers it. Evaluated at the margin ($x \to 0$) it is the rate at which an unlevered investor would just begin to borrow. Without deductibility, the closed form (21) pins it down: $\bar{c}(x, r_b) = \bar{c}_0$ requires $r_b = b^{nd}$, which is the same at every x. Deferral lifts b^{nd} above the accrual break-even rate $\mathbb{E}[R-1](1-\tau)$, at the baseline from 3.75% to 4.30%, so realization-based taxation makes borrowing attractive at rates an accrual-taxed

[20] $x = 3$ corresponds to 75% debt-to-assets, attainable under portfolio margin against liquid, low-risk collateral.

investor would refuse.

With deductibility, the deduction adds value, so the break-even is higher and now varies with leverage. Panel B of Table 2 reports it. At the margin it is about 5.6%: above b^{nd}, but below the full-utilization bound $b^{nd}/(1-\tau) = 5.74\%$ (attained under T_{FUL}, where the interest deduction is immediately and fully valued at the statutory rate). Under T_{LUL}, the deduction is instead carried forward and competes with harvested credits. The break-even then declines as leverage rises and the deduction begins to over-shelter, drifting back toward b^{nd}. Since the financing rates available to wealthy investors likely lie below these break-evens, the region where leverage is tax-profitable is exactly where an interior leverage choice would require aggregate risk — which is outside the MGET benchmark — so the reported MGETs are conservative in this respect.

Short selling offers a complementary tax-management margin through "shorting the aggregate box" (Supplementary Online Appendix Section A). Matched long and short positions in distinct diversified portfolios initially offset aggregate exposure, while allowing the investor to realize losses and defer gains.

V.B Revenue in a Closed Cohort Economy

The investor-side MGET measures the value of realization discretion through the consumption a portfolio can sustain. To measure its fiscal consequences, I place cohorts in a closed economy and compare revenue under realization and accrual taxation. Mortality is exogenous and bequests are accidental, in the spirit of Yaari (1965) and Blanchard (1985). Investors die with annual probability $\beta = 3\%$ and, while alive, follow the optimal perpetuity policy used in the MGET calculation. I consider both limited use of losses with perfect carryforward, T_{LUL} with $\delta_F = 1$, and full use of losses, T_{FUL}, with frictionless reinvestment throughout.

This exercise uses mortality to aggregate the previously computed perpetuity paths, rather than solving a new life-cycle problem. The complementary analysis of optimal life-cycle choices is developed by Bennhoff and Scheuer (2026), who build on this framework to solve a quantitative model with many assets and aggregate risk and compare alternative treatments of capital gains at death. Related life-cycle studies examine consumption and investment under capital-gains taxation (Dammon et al., 2001a) and asset location across taxable and tax-deferred accounts (Dammon et al., 2004). Msall and Næss (2026) empirically study the revenue and realization effects of Norway's transition from step-up to carry-over basis.

Taxes and consumption are recycled to entering cohorts, keeping resources within the economy and allowing aggregate wealth to grow at the common gross return $\mathbb{E}[R]$. Within this economy, I compare three treatments of gains at death. Under *step-up of basis*, inherited assets receive a new basis equal to market value, forgiving the embedded gains. Under *carry-over basis*,

heirs retain the decedent's basis, so those gains remain taxable when subsequently realized. Under *constructive realization*, the embedded gains are taxed at death and the inherited assets begin with a new basis. These provisions determine whether death extinguishes, transfers, or settles the outstanding tax liability.

For each loss treatment, I first solve the perpetuity problem for a portfolio with unit initial value and simulate its optimal trading policy forward. At each age a, the resulting path records real consumption $c_a = \bar{c}$, living tax payments tax_a, real accrued gains accr_a, portfolio value, embedded gains, and unused loss credits. The portfolio's tax state also determines the payment dtax_a that would be due upon death. All quantities are expressed in real terms.

The perpetuity problem is homogeneous of degree one in initial wealth: scaling the initial portfolio scales feasible consumption, holdings, and tax payments by the same factor. A cohort entering at date $t - a$ with endowment N_{t-a} therefore generates N_{t-a} times the unit-portfolio path. Under step-up and constructive realization, a fraction $(1-\beta)^a$ survives to age a. Along the economy's constant-growth trajectory, aggregate revenue and the corresponding real accrual flow are consequently

$$\begin{aligned} T_t &:= \sum_{a \geq 0} (1-\beta)^a N_{t-a} \big(\text{tax}_a + \beta\, \text{dtax}_a\big), \\ A_t &:= \sum_{a \geq 0} (1-\beta)^a N_{t-a}\, \text{accr}_a. \end{aligned} \tag{22}$$

The death-tax term is zero under step-up and records the liability settled at death under constructive realization.

Under carry-over, death changes ownership without restarting the portfolio's tax history, including unused loss credits in this benchmark. I therefore aggregate by tax dynasty rather than by the current owner's age. The corresponding sums omit the survival factor and contain no tax payment at death; inherited gains contribute to revenue when heirs realize them. Supplementary Online Appendix Section D derives these aggregations from an economy that begins with a single cohort and adds new cohorts over time.

Along the constant-growth trajectory, entering endowments, revenue, and real accrued gains grow at $\mathbb{E}[R]$. The ratio of revenue to accrued gains is therefore constant. I define the government-side effective tax rate as

$$\tau_G^* := \frac{T_t}{A_t}.$$

It is the accrual-tax rate that would raise the same revenue from the same real accrual base. Taxing that base at the statutory rate would instead raise τA_t, so

$$\frac{\tau_G^*}{\tau} = \frac{T_t}{\tau A_t}$$

measures realization-tax revenue relative to statutory accrual taxation.

The recycling rule gives this comparison an interpretation across entire tax histories. Start-

ing from the same aggregate wealth, an economy taxed upon accrual from the beginning follows the same aggregate wealth path as the realization-tax economy: tax payments and consumption remain within the system, and the gross return is unchanged. Its real accrued returns at date t are therefore also A_t, and its revenue is τA_t. The ratio above thus compares revenue under realization-based and accrual-based tax systems that have been in place since the economy's onset.

Table 3: Cohort revenue relative to statutory accrual-tax revenue, τ_G^*/τ

	T_{LUL}, $\delta_F = 1$			T_{FUL}		
Market environment	Step-up	Carry-over	Constr. real.	Step-up	Carry-over	Constr. real.
Baseline ($\varphi = 30\%$, $\pi = 2\%$)	0.36	0.51	0.69	0.34	0.50	0.67
No dividends or inflation	0.20	0.37	0.51	< 0.01	0.28	0.40

Notes: Entries report the government-side effective rate relative to the statutory rate. The expected real return is 5%, cross-sectional nominal return volatility is 30%, the statutory tax rate is 25%, and annual mortality is 3%. Limited use of losses permits perfect carryforward, $\delta_F = 1$; both loss treatments assume frictionless reinvestment. Taxes and consumption are recycled, so aggregate wealth grows at $\mathbb{E}[R]$ under every tax treatment. Calculations use 200 simulated periods.

Under both loss treatments, step-up raises the least revenue, carry-over is intermediate, and realization at death raises the most. At the baseline, carry-over produces revenue equal to 51 percent of statutory accrual-tax revenue under limited use of losses and 50 percent under full use. These ratios are close to the corresponding investor-side MGET ratios of approximately 56 and 55 percent. Thus, in the baseline carry-over economy, the investor and government measures imply similar effective burdens.

Realization at death raises about 1.9 times as much revenue as step-up under limited use of losses and twice as much under full use. Carry-over closes part of this gap because the embedded liability survives death, but it leaves heirs discretion over when to settle it. Without dividends or inflation, step-up revenue under full use of losses is nearly zero: loss refunds offset revenue collected during life, while gains that remain embedded at death are forgiven.

Even realization at death raises substantially less revenue per period than statutory accrual taxation. A growing economy continually accumulates new unrealized gains, maintaining an outstanding stock of deferred liabilities. Although individual gains are eventually taxed under constructive realization, part of each period's accrual enters this growing stock instead of the current tax base. Deferral can therefore produce a permanent gap between realization and accrual revenue flows without permanently exempting any particular gain. Step-up adds forgiveness to this timing effect.

Dividends and inflation raise the government-side effective rates in Table 3. Dividends

bring part of the return into the current tax base, reducing the share that can remain unrealized. Inflation increases taxable nominal gains relative to the real accrual base and changes the real value of liabilities paid later. These channels complement the treatment of gains at death: the death provision determines whether inherited liabilities are forgiven, carried forward, or settled, while dividends and nominal-basis taxation affect when returns become taxable and the size of the resulting payments relative to real gains. Section V.A discusses the corresponding effects on the investor-side MGET.

The next section expresses the investor's realization advantage in expected-return units.

VI Return Alphas

The MGET expresses effective capital gains taxation in the language of public finance. This section translates the same consumption-equivalence comparison into expected-return units familiar from asset pricing. The resulting return alphas connect tax burdens and investment performance within the same optimal-trading problem, allowing the model to speak to both literatures. They also distinguish the value of deferral within a single asset from the additional value of choosing which gains and losses to realize.

I compare three regimes. Under A, gains are taxed upon accrual at the statutory rate. Under S, gains are taxed upon realization and the investor liquidates every tax lot proportionally, which is equivalent to trading and deferring shares in a single diversified fund. Under M, the investor trades optimally across tax lots. The comparison between A and S isolates single-asset deferral; the comparison between S and M isolates the additional value of tax-lot selection.

Deferral and tax-lot selection are economically complementary. Choosing which assets to sell has tax value because the investor can realize losses or small gains while leaving more appreciated positions invested. Without deferral, gains would enter the tax base as they accrue, and choosing among tax lots would provide no tax benefit.

Write $\mu := \mathbb{E}[R] - 1$ for the expected real return and $\bar{c}_j(\mu)$ for the maximal sustainable real withdrawal under regime $j \in \{\mathrm{A}, \mathrm{S}, \mathrm{M}\}$, suppressing the remaining return and tax parameters. Statutory accrual taxation gives $\bar{c}_{\mathrm{A}}(\mu) = \mu(1-\tau)$. I define three return equivalents by

$$\bar{c}_{\mathrm{A}}\left(\mu + \alpha^{R}_{\mathrm{A}\to\mathrm{S}}\right) = \bar{c}_{\mathrm{S}}(\mu), \tag{23}$$

$$\bar{c}_{\mathrm{S}}\left(\mu + \alpha^{R}_{\mathrm{S}\to\mathrm{M}}\right) = \bar{c}_{\mathrm{M}}(\mu), \tag{24}$$

$$\bar{c}_{\mathrm{A}}\left(\mu + \alpha^{R}_{\mathrm{A}\to\mathrm{M}}\right) = \bar{c}_{\mathrm{M}}(\mu). \tag{25}$$

The *deferral alpha*, $\alpha^{R}_{\mathrm{A}\to\mathrm{S}}$, is the additional expected return an accrual-taxed investment needs to match realization-based deferral in a single fund. The *diversification alpha*, $\alpha^{R}_{\mathrm{S}\to\mathrm{M}}$, is the additional return the single-fund strategy needs to match optimal trading across tax lots. The *total realization advantage*, $\alpha^{R}_{\mathrm{A}\to\mathrm{M}}$, compares statutory accrual taxation directly with optimal

realization trading.

These return equivalents form a chain, but they do not add exactly because sustainable consumption under S is nonlinear in expected returns. I report the interaction $\iota^R := \alpha^R_{A\to M} - (\alpha^R_{A\to S} + \alpha^R_{S\to M})$, which is positive in the specifications considered below.

At the baseline $(\mu, \tau, \sigma, \pi, \varphi) = (5\%, 25\%, 30\%, 2\%, 30\%)$, under limited use of losses with perfect carryforward and frictionless reinvestment, the total realization advantage is 73.8 basis points per year. The deferral alpha is 44.4 basis points, the diversification alpha is 26.3 basis points, and the interaction is 3.1 basis points. Allocating the interaction in proportion to the first two components assigns about 63 percent of the total to single-asset deferral and 37 percent to choosing liquidations across tax lots.

Under full use of losses, the corresponding values are 75.5, 44.4, 27.7, and 3.3 basis points. Immediate loss refunds modestly increase the cross-asset component while leaving single-asset deferral unchanged. Table 4 reports these return equivalents across the same nine specifications used for the MGET calculations.

The two component alphas respond to different features of the environment. Changing cross-sectional volatility leaves the deferral alpha unchanged at 44.4 basis points because a single diversified fund offers no tax-lot selection. Under limited use of losses, the diversification alpha rises from 18.3 to 31.6 basis points as volatility rises from 20 to 40 percent. Dispersion creates a wider menu of gains and losses from which the investor can choose. Inflation and dividends operate mainly through deferral: they move gains into the tax base and sharply reduce both the deferral alpha and the total realization advantage. A higher statutory rate increases every return equivalent because it raises the value of controlling realizations.

The total return equivalent and the MGET express the same indifference comparison in different units. Let $\tau^*_M := \tau^*$ denote the MGET. The MGET holds the expected return fixed and lowers the accrual tax rate; the total return equivalent holds the statutory tax rate fixed and raises the accrual return. Both reproduce sustainable consumption under optimal realization trading:

$$\mu(1-\tau^*_M) = (\mu + \alpha^R_{A\to M})(1-\tau) = \bar{c}_M(\mu). \tag{26}$$

Consequently,

$$\alpha^R_{A\to M} = \frac{\mu(\tau - \tau^*_M)}{1-\tau}. \tag{27}$$

At the baseline, an MGET equal to 55.7 percent of the statutory rate corresponds to 73.8 basis points of additional annual expected return under statutory accrual taxation. The deferral alpha has the same relation to the single-fund effective rate: defining τ^*_S by $\mu(1-\tau^*_S) = \bar{c}_S(\mu)$ gives $\alpha^R_{A\to S} = \mu(\tau - \tau^*_S)/(1-\tau)$.

These return equivalents share the central idea of empirical tax-alpha measures: expressing

Table 4: Return equivalents of realization taxation

Specification	Annual expected-return equivalents (basis points)				Effective rate
	$\alpha^R_{A\to M}$	$\alpha^R_{A\to S}$	$\alpha^R_{S\to M}$	ι^R	τ^*_M/τ
Panel A. Limited use of losses (T_{LUL}, $\delta_F = 1$)					
Baseline	73.85	44.43	26.30	3.12	0.557
Low tax rate	25.85	15.48	9.96	0.40	0.535
High tax rate	197.76	120.15	58.41	19.20	0.604
Low volatility	64.93	44.43	18.33	2.17	0.610
High volatility	79.76	44.43	31.59	3.74	0.521
Low inflation	114.46	93.61	18.72	2.12	0.313
High inflation	43.55	20.62	20.96	1.97	0.739
Low dividend share	85.38	62.94	19.61	2.83	0.488
High dividend share	44.31	21.83	20.65	1.82	0.734
Panel B. Full use of losses (T_{FUL})					
Baseline	75.46	44.43	27.75	3.29	0.547
Low tax rate	26.47	15.48	10.56	0.43	0.524
High tax rate	201.34	120.15	61.10	20.09	0.597
Low volatility	65.07	44.43	18.46	2.19	0.610
High volatility	84.66	44.43	35.97	4.27	0.492
Low inflation	121.75	93.61	25.26	2.87	0.270
High inflation	43.72	20.62	21.11	1.98	0.738
Low dividend share	97.60	62.94	30.29	4.38	0.414
High dividend share	44.31	21.83	20.65	1.82	0.734

Notes: Regime A is statutory accrual taxation, S is realization taxation with proportional liquidation of a single diversified fund, and M is realization taxation with optimal trading across tax lots. The first four columns are annual expected-return basis points. The interaction ι^R is the total return equivalent minus the sum of the deferral and diversification alphas. The last column is the MGET divided by the statutory rate. The baseline sets $(\mu, \tau, \sigma, \pi, \varphi) = (5\%, 25\%, 30\%, 2\%, 30\%)$. Panel A uses T_{LUL} with perfect carryforward ($\delta_F = 1$); Panel B uses T_{FUL}. Both panels assume frictionless reinvestment ($\delta_I = 1$). Low and high specifications set τ to 10 and 50 percent, σ to 20 and 40 percent, π to -2 and 6 percent, or φ to 0 and 55 percent, changing one parameter at a time. Perpetuities are computed over 200 periods using the forward-simulation algorithm of Supplementary Online Appendix Section C.A; M uses sLHCL trades and S uses proportional liquidation. Small differences from Table 1 reflect numerical grid resolution.

a tax advantage as an increment to annual return. Chaudhuri et al. (2020) report a tax-loss-harvesting alpha of 108 basis points per year before transaction costs, falling to 82 basis points when the wash-sale rule is imposed. Mainardi (2026) estimates a 70-basis-point difference in annual tax expenses between the wealthiest and least wealthy investment-account holders in his sample. The total realization advantage estimated here is of a similar magnitude. These empirical measures draw on historical trading strategies or observed tax expenses and can reflect additional features of the tax code, including different rates for short- and long-term gains and wash-sale restrictions. Here, the return equivalents follow from a recursive optimal-trading problem, providing a theoretical foundation for measuring and decomposing the tax advantage under explicit market and tax assumptions.

The complexity of realization taxation described in the introduction creates a practical need for tractable approximations. The consumption-equivalence results provide a theoretical basis

for incorporating the consumption effects of deferral and tax-lot selection without explicitly modelling realization decisions. A life-cycle model, for example, can retain accrual taxation and either increase the expected return by the appropriate return alpha or lower the tax rate to the corresponding effective rate. For a single diversified asset, the appropriate adjustments are the deferral alpha $\alpha^R_{\mathrm{A}\to\mathrm{S}}$ or the effective rate τ^*_{S}. To incorporate both deferral and tax-lot selection, the corresponding adjustments are the total realization advantage $\alpha^R_{\mathrm{A}\to\mathrm{M}}$ or the MGET τ^*_{M}. These are two alternative ways to capture the consumption effects of realization discretion without tracking tax lots. They offer an alternative to the mark-to-market heuristic of Fischer et al. (2013), in which households choose their trades as if gains were taxed upon accrual while taxes are actually imposed upon realization. Here, the benefit of realization taxation is incorporated directly into the expected return or tax rate of an accrual-tax model. Both adjustments exactly reproduce maximal sustainable consumption in the perpetuity setting studied here and provide calibrated approximations for applications to richer life-cycle models.

VII Conclusion

This paper makes optimal portfolio choice usable in the study of taxation and wealth accumulation. The Tax Portfolio Model combines a distributional representation of large portfolios with a recursive characterization of optimal trading, making it possible to evaluate tax systems while allowing investors to choose realizations across many assets over long horizons. The trading results also contribute independently to the theory of tax-aware portfolio choice: they establish when harvesting losses and selling the least appreciated remaining positions is optimal, and show how aggregate downside risk can reverse this prescription.

The quantitative results show that discretion over realizations can substantially change the burden of capital gains taxation. At the baseline, the investor's effective rate is little more than half the statutory rate. The framework makes it possible to evaluate how this gap changes with tax rules, inflation, dividend payouts, and return dispersion. Effective tax rates and return alphas express the same investor advantage in tax-rate and annual-return units, with an exact correspondence between them. The infinite-horizon benchmark expresses the investor and fiscal comparisons as two accrual-equivalent tax rates: one preserves the investor's perpetual real consumption, while the other preserves the fiscal authority's long-run revenue.

These quantitative findings make the treatment of realizations central to evaluating capital gains tax policy. A statutory rate can substantially overstate the effective burden on an optimally managed portfolio, and eliminating forgiveness at death does not by itself eliminate the long-run consequences of deferral. Comparisons of tax systems therefore need to account jointly for statutory rates, realization incentives, and the treatment of losses and inherited gains.

By quantifying both investor value and government revenue under alternative rules, the framework provides inputs for research on tax progressivity and the design of capital taxation.

This connection between portfolio choice and public finance opens further questions about wealth accumulation and inequality. Differences in portfolio composition, liquidity needs, and opportunities for tax management may generate different effective burdens under the same statutory rules. Incorporating these mechanisms into models of household saving would allow researchers to study how realization taxation affects the distribution and transmission of wealth. In ongoing work, Bennhoff and Scheuer (2026) build on this framework to study a quantitative life-cycle model with many assets and aggregate risk, comparing alternative treatments of capital gains at death. More broadly, the framework makes investors' trading responses available for studying how tax reforms affect consumption, wealth accumulation, and government revenue together.

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A Proofs of Main-Text Results

Proof of Theorem 1. For each tax lot r, set

$$w_{\alpha}^{n}(r) := \sum_{i \in I^{n}(r)} \alpha_{i}^{n} \omega_{i}^{n}, \qquad \bar{w}_{\bar{\alpha}}^{n}(r) := \sum_{i \in I^{n}(r)} \bar{\alpha}_{i}^{n} \bar{\omega}_{i}^{n}.$$

Let

$$B_{\text{tot}}^{n} := \sum_{i} \frac{\omega_{i}^{n}}{r_{i}}, \qquad \bar{B}_{\text{tot}}^{n} := \sum_{i} \frac{\bar{\omega}_{i}^{n}}{r_{i}}.$$

Because the set of tax lots is finite and both endowment sequences converge lot by lot to the same limit w, we have $B_{\text{tot}}^{n} - \bar{B}_{\text{tot}}^{n} \to 0$.

Assume statement 1. Then

$$\mathrm{M}_{1}^{n} - \bar{\mathrm{M}}_{1}^{n} \to 0, \qquad \mathrm{CG}_{1}^{n} - \bar{\mathrm{CG}}_{1}^{n} \to 0.$$

Continuity of T_1 gives $c_{1}^{n} - \bar{c}_{1}^{n} \to 0$.

For period 2, write $\mu_Z := Z(1+\mu)$ and define the residuals $\varepsilon_{\alpha}^{n} := \sum_i (1-\alpha_i^n)\omega_i^n (ZR_i^n - \mu_Z)$ and $\varepsilon_{\bar{\alpha}}^{n} := \sum_i (1-\bar{\alpha}_i^n)\bar{\omega}_i^n (Z\bar{R}_i^n - \mu_Z)$. The common-mean condition makes both residuals conditionally mean zero. Moreover, $\mathrm{M}_2^n = \mu_Z(M - \mathrm{M}_1^n) + \varepsilon_{\alpha}^{n}$ and $\bar{\mathrm{M}}_2^n = \mu_Z(M - \bar{\mathrm{M}}_1^n) + \varepsilon_{\bar{\alpha}}^{n}$, where M is the common total market value.

Conditional on Z, the covariance matrix of $Z\boldsymbol{R}^n$ is $Z^2\Sigma^n$, so

$$\begin{aligned}\mathbb{E}[(\varepsilon_{\alpha}^{n})^2 \mid Z] &\le Z^2 \lambda_{\max}(\Sigma^n) \sum_i ((1-\alpha_i^n)\omega_i^n)^2 \\ &\le Z^2 \lambda_{\max}(\Sigma^n) M \max_{i \le n} \omega_i^n.\end{aligned}$$

The analogous bound uses $\bar{\Sigma}^n$ for $\varepsilon_{\bar{\alpha}}^{n}$. Balancedness, diversifiability, and $\mathbb{E}[Z^2] < \infty$ therefore imply that both residuals converge to zero in L^2. It follows that $\mathrm{M}_2^n - \bar{\mathrm{M}}_2^n = -\mu_Z(\mathrm{M}_1^n - \bar{\mathrm{M}}_1^n) + \varepsilon_{\alpha}^{n} - \varepsilon_{\bar{\alpha}}^{n} \to 0$ in L^2.

Next,

$$B_{1}^{n} := \mathrm{M}_{1}^{n} - \mathrm{CG}_{1}^{n} = \sum_{i} \frac{\alpha_{i}^{n} \omega_{i}^{n}}{r_{i}}, \qquad \bar{B}_{1}^{n} := \bar{\mathrm{M}}_{1}^{n} - \bar{\mathrm{CG}}_{1}^{n},$$

so $B_{1}^{n} - \bar{B}_{1}^{n} \to 0$. Since

$$\mathrm{CG}_{2}^{n} = \mathrm{M}_{2}^{n} - (B_{\text{tot}}^{n} - B_{1}^{n}), \qquad \bar{\mathrm{CG}}_{2}^{n} = \bar{\mathrm{M}}_{2}^{n} - (\bar{B}_{\text{tot}}^{n} - \bar{B}_{1}^{n}),$$

we also have $\mathrm{CG}_{2}^{n} - \bar{\mathrm{CG}}_{2}^{n} \to 0$ in L^2. Hence

$$(\mathrm{M}_{2}^{n}, \mathrm{CG}_{2}^{n}, \mathrm{CG}_{1}^{n}) - (\bar{\mathrm{M}}_{2}^{n}, \bar{\mathrm{CG}}_{2}^{n}, \bar{\mathrm{CG}}_{1}^{n}) \to 0 \quad \text{in probability.}$$

By continuity of T_2, $c_{2}^{n} - \bar{c}_{2}^{n} \to 0$ in probability. Therefore

$$u(c_{1}^{n}, c_{2}^{n}) - u(\bar{c}_{1}^{n}, \bar{c}_{2}^{n}) \to 0 \quad \text{in probability.}$$

The maintained boundedness and integrability conditions provide uniform integrability of the utility differences. Their convergence to zero in probability therefore implies $|U^n - \bar{U}^n| \to 0$, proving statement (ii). ■

Proof of Corollary 1. Write $v_i := (\alpha_i^n - \bar{\alpha}_i^n)\omega_i^n / M$. Identical lot-level liquidations imply

$$\sum_i v_i = \sum_i \frac{v_i}{r_i} = 0,$$

so current consumption is identical and, under linear taxes,

$$g_2^n = (1-\tau) Z \sum_i v_i R_i^n.$$

The common-mean assumption gives $\mathbb{E}[g_2^n \mid Z] = 0$. Moreover,

$$\sum_i v_i^2 \le \sum_i \left(\frac{\omega_i^n}{M}\right)^2 \le \frac{\max_i \omega_i^n}{M} \le \frac{C}{n}.$$

The covariance bound therefore yields

$$\mathrm{sd}(g_2^n) \le (1-\tau)\sqrt{\frac{\bar{\lambda} C\, \mathbb{E}[Z^2]}{n}}.$$

Finally, bounded marginal utility in the second argument and identical current consumption imply

$$|U^n - \bar{U}^n| \le LM\, \mathbb{E}|g_2^n| \le LM\, \mathrm{sd}(g_2^n) = O(n^{-1/2}).$$

■

Proof of Lemma 1. Set $x := w - \widetilde{w}$ and $u := Qx + \delta_I I\chi$. Tonelli's theorem and the substitutions $r = sR$ and $r = zs$ give $\mathrm{M}(Qx) = \mathbb{E}[R]\,\mathrm{M}(x)$, $\mathrm{B}(Qx) = \mathrm{B}(x)$, $\mathrm{M}(A_z u) = z\,\mathrm{M}(u)$, and $\mathrm{B}(A_z u) = \mathrm{B}(u)$. Because $\chi = Q\boldsymbol{\delta}_1$, also $\mathrm{M}(\chi) = \mathbb{E}[R]$ and $\mathrm{B}(\chi) = 1$. Substitution yields the first two identities; $\mathrm{CG} = \mathrm{M} - \mathrm{B}$ and $\mathrm{K_{FUL}} = \mathrm{M} - \tau\,\mathrm{CG}$ yield the remaining two. ■

Proof of Proposition 3. Fix $z \in \mathcal{Z}$. For the market-tail component, a change of variables gives

$$\int_a^\infty A_z Q w(r)\, dr = z \int_0^\infty R f(R) \left(\int_{a/(zR)}^\infty w(s)\, ds \right) dR,$$

which is weakly larger than the same expression with $\bar{w}$ whenever w has weakly larger upper tails. Taking $a \downarrow 0$ also gives $\mathrm{M}(w) \ge \mathrm{M}(\bar{w})$. For linear taxes, $\mathrm{K_{FUL}}(A_z Q x) = \mathrm{K_{FUL}}(x) + (1-\tau)(z\mathbb{E}[R] - 1)\,\mathrm{M}(x)$. Hence $\mathrm{K_{FUL}}(A_z Q w) \ge \mathrm{K_{FUL}}(A_z Q \bar{w})$ whenever $z\mathbb{E}[R] \ge 1$ and $w \succeq_R \bar{w}$. Thus $A_z Q w \succeq_R A_z Q \bar{w}$. Adding the common nonnegative portfolio $A_z(I\chi)$ to both sides preserves upper-tail inequalities and K-dominance. ■

Supplementary Online Appendix

A Heterogeneous Factor Exposures

The baseline model gives assets a common marginal return distribution. This section considers a CAPM-style relaxation in which assets share a common idiosyncratic return law but have heterogeneous exposure to an aggregate factor (cf. Al-Najjar, 1998, Khan and Sun, 2003, Lintner, 1965, Sharpe, 1964). Let R_i have the baseline marginal density f, let z denote the realization of a market shock, and let asset i have factor loading β_i. Its period gross return is $(1+\beta_i z)R_i$, where $1+\beta_i z > 0$.

Suppose the investor holds a cross-section of loadings distributed according to q and that this loading distribution is independent of the assets' current tax lots. Conditional on z, the portfolio law of motion becomes

$$w' = \int A_{1+\beta z}(Q(w-\widetilde{w}) + I\delta_I\chi)\, q(\beta)\, \mathrm{d}\beta,$$

and, conditional on z, total market value satisfies

$$\mathrm{M}(w') = \big(1+\mathbb{E}_q[\beta]z\big)\mathbb{E}_R[R]\big(\mathrm{M}(w-\widetilde{w}) + \delta_I I\big).$$

The mean loading therefore controls average aggregate exposure. The full distribution of loadings controls how widely realized returns, and hence value across tax lots, are dispersed.

This distinction creates a simple tax-timing margin. Replacing a common loading $\bar{\beta}$ with a nondegenerate distribution q satisfying $\mathbb{E}_q[\beta] = \bar{\beta}$ leaves average aggregate exposure unchanged but creates more low- and high-return tax lots. Low-return tax lots give the investor more ways to finance withdrawals or generate loss credits, while the most appreciated tax lots can remain invested. The MGET simulations in Section V quantify the same mechanism for exogenous return dispersion: greater cross-sectional dispersion lowers the effective tax burden by expanding the set of realizations from which the investor can choose.

The one-dimensional representation above relies on the same loading distribution q applying at every current tax lot. If factor exposure and accrued return are correlated, the natural state is a joint distribution of portfolio value over (r, β). Thus, this extension allows structured heterogeneity in asset returns while preserving the one-dimensional TPM under an explicit independence condition.

Short-sale access provides a second way to create tax-management opportunities, even without asset-level factor information. Consider a dollar long and a dollar short in disjoint diversified portfolios with the same distribution of factor exposures. Their aggregate exposures offset before liquidation, but each leg contains positions with gains and losses. Conditional on the aggregate state z, let f_z denote their common cross-sectional density of gross returns $\mathcal{R}$.

The market-value densities of the long positions and of the securities owed on the short leg are $w_{\text{long}}(r \mid z) = w_{\text{short}}(r \mid z) = r f_z(r)$. Selling depreciated long positions and covering appreciated securities on the short leg realizes a total loss of

$$\begin{aligned} L(z) &= -\int_0^1 w_{\text{long}}(r \mid z)(1 - 1/r)\,\mathrm{d}r + \int_1^\infty w_{\text{short}}(r \mid z)(1 - 1/r)\,\mathrm{d}r \\ &= \mathbb{E}[|\mathcal{R} - 1| \mid z]\,. \end{aligned} \tag{28}$$

Without aggregate risk this reduces to $L = \mathbb{E}[|R - 1|]$ under the baseline return law.

I call this construction *shorting the aggregate box*. Scaling both legs by a generates $aL(z)$ of realizable losses, which can offset gains realized elsewhere while the winning positions remain open. In the baseline case, a position of size G/L on each leg generates losses equal to a gain G. The construction uses distinct securities on the two legs, whereas shorting against an individual asset offsets that same appreciated position. The calculation isolates the realization margin, abstracting from financing costs, dividends, and margin requirements; selectively closing the losing positions can also change the remaining aggregate exposure. Short-sale access thus expands the set of tax positions the investor can create, complementing the borrowing and interest-deduction margin in the leverage extension of Section V.A.

B Recursive Foundations and Trading Proofs

This appendix records the model-specific verifications behind Section IV and otherwise invokes standard results. The functional-analytic facts come from Aliprantis and Border (2006): the weighted dual representation from their Theorem 13.28, weak-$*$ compactness from Theorem 6.21 and Lemma 8.56, weak-$*$ continuity of adjoints from Theorem 6.43, correspondence results from Theorems 17.16–17.19 and 17.28, and Berge's maximum theorem from Theorem 17.31. The discounted Bellman argument is the weighted sup-norm contraction of Bertsekas (2012, Proposition 2.1), followed by the contraction mapping theorem (Aliprantis and Border, 2006, Theorem 3.48). Thus the proofs below focus on the portfolio transition, tax resources, and the nonstandard sup-min recursion.

B.A Weighted Portfolio Space and Operators

Write

$$\mathcal{L}^1_{1/\phi} := \left\{ \eta : \int_0^\infty \frac{|\eta(r)|}{\phi(r)}\, dr < \infty \right\}, \qquad \langle w, \eta \rangle := \int_0^\infty w(r)\eta(r)\, dr.$$

Multiplication by ϕ identifies $\mathcal{L}^\infty_\phi$ isometrically with ordinary L^∞, while division by ϕ identifies $\mathcal{L}^1_{1/\phi}$ with ordinary L^1. Hence $\mathcal{L}^\infty_\phi$ is the dual of $\mathcal{L}^1_{1/\phi}$ and is a Banach lattice with positive cone $\mathcal{L}^\infty_{\phi,+}$ (Aliprantis and Border, 2006, Section 9.1 and Theorem 13.28). We use the weak-$*$ topology $\sigma(\mathcal{L}^\infty_\phi, \mathcal{L}^1_{1/\phi})$. Its closed norm balls are compact by Banach–Alaoglu, and its norm-bounded order intervals are weak-$*$ compact (Aliprantis and Border, 2006, Theorem 6.21 and Lemma 8.56).

Lemma 2 (Weight-ratio bound). *For all $x, s > 0$, $\phi(x)/\phi(x/s) \le \phi(s)$.*

Proof. If x and x/s lie on the same side of one, the ratio is s^α or s^γ. If $s \le x \le 1$, it is $s^\gamma x^{\alpha-\gamma} \le s^\alpha$; if $1 \le x \le s$, it is $x^{\gamma-\alpha}s^\alpha \le s^\gamma$. ∎

Proposition 6 (Market value and liquidation functionals). *The market-value functional is positive and bounded on $\mathcal{L}^\infty_\phi$, with*

$$|\,\mathrm{M}(w)| \le c_{\mathrm{M}} \|w\|_{\infty,\phi}, \qquad c_{\mathrm{M}} = \frac{1}{1-\alpha} + \frac{1}{\gamma - 1}.$$

If $Kx = \int_0^\infty x(r)k(r)\, dr$, where $k \ge 0$ and $k \in \mathcal{L}^1_{1/\phi}$, then K is positive, linear, bounded, and weak-$*$ continuous, with $|Kx| \le \|k\|_{\mathcal{L}^1_{1/\phi}} \|x\|_{\infty,\phi}$.

Proof. The market-value bound follows by integrating ϕ^{-1} separately on $(0,1]$ and $(1,\infty)$. The second claim is the defining predual pairing; continuity follows directly from the dual representation above. ∎

Proposition 7 (Portfolio operators)**.** *Under Assumption 1, Q and every A_z are positive, linear, bounded, and weak-$*$ continuous on $\mathcal{L}^\infty_\phi$, with*

$$\|Q\| \le m_Q, \qquad \|A_z\| \le \phi(z), \qquad \|A_z Q w\|_{\infty,\phi} \le \phi(z) m_Q \|w\|_{\infty,\phi}.$$

Proof. Lemma 2 gives

$$|Qw(x)|\phi(x) \le \|w\|_{\infty,\phi} \int_0^\infty \frac{\phi(x)}{\phi(x/R)} f(R)\, dR \le m_Q \|w\|_{\infty,\phi},$$

and, after writing $x = zy$, $\|A_z w\|_{\infty,\phi} \le \phi(z)\|w\|_{\infty,\phi}$. For $\eta \in \mathcal{L}^1_{1/\phi}$, the preadjoints are

$$Q^*\eta(y) = \int_0^\infty R f(R)\eta(Ry)\, dR, \qquad A_z^*\eta(y) = z\eta(zy).$$

The same weight-ratio bound gives $\|Q^*\eta\|_{\mathcal{L}^1_{1/\phi}} \le m_Q\|\eta\|_{\mathcal{L}^1_{1/\phi}}$ and $\|A_z^*\eta\|_{\mathcal{L}^1_{1/\phi}} \le \phi(z)\|\eta\|_{\mathcal{L}^1_{1/\phi}}$. The adjoint criterion of Aliprantis and Border (2006, Theorem 6.43) therefore yields weak-$*$ continuity. ∎

B.B Bellman Bounds and Policy Attainment

Lemma 3 (Tax-share drift and current-payoff bounds)**.** *For every $p \in (0,1]$ and $\varepsilon > 0$, the normalization in W_p can be chosen so that every feasible $(w, \widetilde{w}, I)$ satisfies*

$$\sum_z p_z W_p\big(G_z(w,\widetilde{w},I)\big) \le (\lambda_{I,p} + \varepsilon) W_p(w).$$

Moreover, $0 \le c(\widetilde{w}, I) \le y + c_K\|w\|_{\infty,\phi}$.

Proof. Feasibility gives $I \le y + \mathrm{K}(\widetilde{w}) \le y + c_K\|w\|_{\infty,\phi}$. Proposition 7 therefore implies

$$\|G_z(w,\widetilde{w},I)\|_{\infty,\phi} \le \phi(z)\left[\big(m_Q + \delta_I c_K\|\chi\|_{\infty,\phi}\big)\|w\|_{\infty,\phi} + \delta_I y\|\chi\|_{\infty,\phi}\right].$$

Raise this inequality to $p \le 1$, average over z, and use $(a+b)^p \le a^p + b^p$. The first term gives $\lambda_{I,p}\|w\|^p_{\infty,\phi}$; choosing ξ sufficiently large absorbs the additive income term into $\varepsilon W_p(w)$. The consumption bound is immediate from feasibility. ∎

Proof of Proposition 1. The discounted part is an application of the weighted sup-norm theorem in Bertsekas (2012, Proposition 2.1). Under bounded utility take $W_0 \equiv 1$ and obtain modulus β. For log utility, the affine extension removes the singularity at zero and $|\log c| \le C_p(1 + c^p)$ for every sufficiently small $p > 0$; Lemma 3 then gives a modulus approaching β as $p \downarrow 0$. Under sublinear utility, the same lemma gives modulus $\beta(\lambda_{I,p} + \varepsilon)$, which is below one when $\beta\underline{\lambda}_I < 1$ and the free parameters are chosen sufficiently close to their infima. The contraction mapping theorem then gives the asserted fixed point and convergence.

For $\mathbb{P}$, monotonicity of the operator and $\mathbb{P}V_0^{\mathbb{P}} \le V_0^{\mathbb{P}}$ imply $V_{n+1}^{\mathbb{P}} \le V_n^{\mathbb{P}}$. Nonnegativity gives the pointwise limit. To verify the fixed-point claim, fix w and let b_n denote the period objective

on the compact set $\Gamma(w)$. The attainment argument below makes every b_n continuous and gives $b_n \downarrow b$, where b is upper semicontinuous. If a_n maximizes b_n, compactness gives a convergent subnet. Comparing this subnet with every fixed b_N yields $\lim_n \max b_n \leq \max b$; the reverse inequality is immediate from $b_n \geq b$. Thus maxima and limits commute at each state, so $V^{\mathbb{P}} = \mathbb{P}V^{\mathbb{P}}$. ■

Proof of Proposition 2. If $w_1 \leq w_2$, every control feasible at w_1 is feasible at w_2, and positivity of Q and A_z makes every continuation state weakly larger. This proves monotonicity. For concavity, take convex combinations of two states and their feasible controls. Linearity of the tax-share liquidation functional makes the feasible graph convex, current consumption is affine, and the transition is affine. Concavity is preserved by u, by continuation values, and by the minimum of the two concave terms in $\mathbb{P}$. Taking suprema proves the claim. ■

Corollary 5 (CARA utility)**.** *Let $u(c) = \bar{u} - ae^{-\eta c}$ with $a, \eta > 0$, or take any positive affine transformation. Then, under Assumptions 1 and 2, $\mathbb{T}$ is a β-contraction on bounded value functions. The sup-min conclusions of Proposition 1 also apply.*

Corollary 6 (CRRA and log utility)**.** *For $u_\sigma(c) = c^{1-\sigma}/(1-\sigma)$, $0 < \sigma < 1$, take $p = 1 - \sigma$ and require $\beta\underline{\lambda}_I < 1$. For log utility bounded away from zero, choose any sufficiently small $p > 0$. For $\sigma > 1$ and consumption bounded away from zero, utility is bounded and the bounded case applies. The sup-min conclusions are unchanged.*

Proof of Theorem 2. Fix a norm-bounded set of states. Feasible liquidation portfolios then lie in a common weak-$*$ compact ball by Banach–Alaoglu, and each interval $[0, w]$ is weak-$*$ compact (Aliprantis and Border, 2006, Theorem 6.21 and Lemma 8.56). The investment coordinate is bounded by $y + c_K \|w\|_{\infty,\phi}$. The budget condition is closed because K is a predual pairing. Thus Γ has nonempty compact values and a closed graph on the bounded set.

For lower hemicontinuity, if $w_\nu \to w$ and $\widetilde{w} \leq w$, write $\widetilde{w} = \theta w$ for a measurable $0 \leq \theta \leq 1$, set $\widetilde{w}_\nu = \theta w_\nu$, and truncate the investment coordinate at $y + \mathrm{K}(\widetilde{w}_\nu)$. Multiplication by bounded θ is weak-$*$ continuous. The standard closed-graph and lower-hemicontinuity criteria therefore make Γ continuous on bounded state sets (Aliprantis and Border, 2006, Theorems 17.16–17.19 and 17.28).

Proposition 7 makes every transition weak-$*$ continuous. The current payoff is continuous, as are the finite-horizon Bellman objectives whenever the continuation value is. Berge's maximum theorem (Aliprantis and Border, 2006, Theorem 17.31) now gives, by induction, bounded-set weak-$*$ continuity of every finite-horizon value, attainment, compact argmax sets, and upper hemicontinuity on bounded state sets.

For $\mathbb{T}$, weighted-norm convergence is uniform on every norm-bounded state set, so the fixed point is weak-$*$ continuous there. At any fixed state its continuation set is norm bounded; Berge's theorem therefore gives a stationary maximizer and a compact argmax set. For $\mathbb{P}$, the decreasing limit of continuous functions is upper semicontinuous on every bounded state set. The compact maximizer argument in the proof of Proposition 1 gives the fixed point and a stationary maximizer. Its argmax is a closed subset of $\Gamma(w)$ and hence weak-$*$ compact. ■

Proof of Corollary 2. With no aggregate risk, induction on n shows that $V_n^{\mathbb{P}}(w)$ is the largest constant consumption sustainable for the current period and n further periods. Every perpetuity is therefore bounded above by $\inf_n V_n^{\mathbb{P}}(w) = V^{\mathbb{P}}(w)$.

Conversely, fix $c < V^{\mathbb{P}}(w)$ and define $S_n(c) = \{x : V_n^{\mathbb{P}}(x) \geq c\}$. At each $x \in \cap_n S_n(c)$, the controls that deliver at least c now and return the state to $S_n(c)$ form a nonempty nested family of weak-$*$ compact sets. Nonemptiness follows from finite-horizon attainment; compactness follows from Theorem 2. The finite-intersection property gives a control that returns the state to $\cap_n S_n(c)$. Choosing such a control recursively sustains c forever. Letting $c \uparrow V^{\mathbb{P}}(w)$ proves the result. ■

B.C Carryforward Verification

Proposition 8 (Carryforward Bellman foundations)**.** *On the product state (w, F), the discounted and sup-min carryforward problems have the same contraction, backward-iteration, and statewise-attainment properties used in Proposition 5. In the absence of aggregate risk, the sup-min limit has the same perpetuity interpretation as Corollary 2.*

Proof. The only new coordinate is scalar. Since

$$c_{CG} := \int_0^\infty |1 - r^{-1}| \phi(r)^{-1}\, dr = \frac{1}{-\alpha(1-\alpha)} + \frac{1}{\gamma(\gamma-1)} < \infty,$$

both M and CG are predual pairings. The maps T_{LUL}, K_{LUL}, and F_{LUL} are therefore weak-$*$ continuous in liquidation and continuous in F. Order intervals remain compact in the product of the weak-$*$ and Euclidean topologies, so the Banach–Alaoglu and Berge arguments above apply without change.

For the contraction step use $W_p^F(w, F) = \xi + \|w\|_{\infty,\phi}^p + \eta F^p$. Taxes cannot increase proceeds, hence $I \leq y + c_M \|w\|_{\infty,\phi}$, while

$$F_{LUL}(\widetilde{w}, F) \leq F + c_{CG} \|w\|_{\infty,\phi}.$$

For $p \in (0, 1]$, the carryforward term in the next-period weight is therefore bounded by $\eta F^p + \eta c_{CG}^p \|w\|_{\infty,\phi}^p$. The portfolio drift estimate in Lemma 3 applies with c_K replaced by c_M and has coefficient at least one. Hence the term ηF^p is already controlled; choosing η small

makes the additional portfolio term arbitrarily small, while choosing ζ large absorbs additive income. Thus the same strict weighted-drift condition suffices. Proposition 2.1 of Bertsekas (2012) then gives discounted contraction under the same three utility cases. Monotonicity and the compact-maximizer argument give the decreasing sup-min recursion and its fixed point. The nested-compact-set proof of Corollary 2 applies verbatim when aggregate risk is absent. ■

B.D Proofs for Recursive Optimal Trading

Lemma 4 (Tail integral order)**.** *If w has weakly larger upper tails than $\bar{w}$, then for every nonnegative nondecreasing function h for which the integrals are finite, $\int_0^\infty h(r)w(r)\,dr \geq \int_0^\infty h(r)\bar{w}(r)\,dr$.*

Proof. This is the upper-tail formulation of first-order stochastic dominance; see Aliprantis and Border (2006, Section 19.8). For completeness, it holds first for nonnegative step functions by taking linear combinations of the tail inequalities. Approximate h monotonically by such functions and apply monotone convergence. ■

B.D.1 A Broader Tax-Share Sufficient Class

For a general tax-share kernel, the future liquidation value of a formal unit market-value tax lot is

$$H_{k,z}(r) := \mathrm{K}(A_z Q \boldsymbol{\delta}_r) = z \int_0^\infty R f(R) k(zRr)\,dR. \tag{29}$$

Throughout this appendix-only paragraph, write $\succeq_R^k$ for right dominance under the liquidation functional induced by the displayed kernel k. The following compatibility condition is sufficient for right dominance to be stable under idiosyncratic returns and aggregate dilation.

Assumption 4 (Tax-share kernel regularity and aggregate compatibility)**.** *The tax-share kernel k satisfies the following conditions.*

i) $k : (0,\infty) \to (0,\infty)$ *is measurable, belongs to* $\mathcal{L}^1_{1/\phi}$*, and is nonincreasing.*

ii) For every aggregate shock z, there are a scalar $\rho_z \geq 0$ and a nonnegative nondecreasing function h_z such that

$$H_{k,z}(r) = \rho_z k(r) + h_z(r), \qquad r > 0. \tag{30}$$

Proposition 9 (General aggregate propagation)**.** *Suppose Assumption 4(ii) holds. If $w \succeq_R^k \bar{w}$, then $A_z Q w \succeq_R^k A_z Q \bar{w}$ for every aggregate shock z. Consequently, for every $J \geq 0$, $A_z(Qw + J\chi) \succeq_R^k A_z(Q\bar{w} + J\chi)$ for every z.*

Proof. Fix an aggregate realization z. For the market-tail component, a change of variables gives

$$\int_a^\infty A_z Q w(r)\,dr = z\int_0^\infty Rf(R)\left(\int_{a/(zR)}^\infty w(s)\,ds\right)dR,$$

which is weakly larger than the same expression with $\bar{w}$ whenever w has weakly larger upper tails. For the after-tax component, equation (29) gives

$$\mathrm{K}(A_z Q w) - \mathrm{K}(A_z Q \bar{w}) = \int_0^\infty H_{k,z}(r)(w(r)-\bar{w}(r))\,dr.$$

By Assumption 4(ii), this equals

$$\rho_z\{\mathrm{K}(w)-\mathrm{K}(\bar{w})\} + \int_0^\infty h_z(r)(w(r)-\bar{w}(r))\,dr.$$

The first term is nonnegative by (10), and the second is nonnegative by Lemma 4. Thus $A_z Q w \succeq_R^k A_z Q\bar{w}$. Adding the common nonnegative portfolio $A_z(J\chi)$ to both sides preserves upper-tail inequalities and K-dominance. ■

Lemma 5 (Primitive statewise harvest condition)**.** *Fix $r \le 1$ and an aggregate realization z. If $\delta_I k(r) \ge 1$ and*

$$\delta_I k(r) H_{k,z}(1) \ge H_{k,z}(r), \tag{31}$$

then

$$A_z(\delta_I k(r)\chi) \succeq_R^k A_z Q\boldsymbol{\delta}_r. \tag{32}$$

Proposition 10 (Power kernels)**.** *A tractable sufficient class is obtained by taxing a power transform of the gain. For $a,\theta \in [0,1]$, set $k_{a,\theta}(r) := a + (1-a)r^{-\theta}$. Equivalently, for $r \ne 1$ this kernel is generated by the effective tax-rate function $\tau_{a,\theta}(r) := (1-a)(1-r^{-\theta})/(1-r^{-1})$, with removable value $\tau_{a,\theta}(1) = (1-a)\theta$. The usual linear tax with rate τ is the case $a = 1-\tau$ and $\theta = 1$.*

Let $\mu_q := \mathbb{E}[R^q]$ and suppose the moments μ_1 and $\mu_{1-\theta}$ are finite. For $k = k_{a,\theta}$, the following hold.

i) The tax-rate function satisfies $0 \le \tau_{a,\theta}(r) \le 1-a \le 1$, and k is positive and nonincreasing.

ii) For every aggregate shock z,

$$H_{k,z}(r) = az\mu_1 + (1-a)z^{1-\theta}\mu_{1-\theta}r^{-\theta}. \tag{33}$$

Hence Assumption 4(ii) holds if $a = 0$ or $z^\theta\mu_1 \ge \mu_{1-\theta}$ for every z. In particular, without aggregate risk this condition follows from $\mu_1 \ge 1$ by Jensen's inequality.

iii) For each z, the harvest test (31) is equivalent to

$$\delta_I\left[a + (1-a)r^{-\theta}\right]\left[az\mu_1 + (1-a)z^{1-\theta}\mu_{1-\theta}\right] \ge az\mu_1 + (1-a)z^{1-\theta}\mu_{1-\theta}r^{-\theta}. \tag{34}$$

The tax lots satisfying this K-part of the harvest test form a lower interval on $(0,1]$. The

additional tail condition $\delta_I k(r) \geq 1$ is also lower-interval, and intersecting over the finite aggregate support preserves the lower-interval property.

Lemma 6 (Lower-interval residual dominance)**.** *Under* $\mathrm{T}_{\mathrm{FUL}}$*, fix* $w \in \mathcal{W}$ *and* $x \in [0, \mathrm{K}(w)]$*, and choose* $a \in [0, \infty]$ *such that* $\mathrm{K}(w\mathbf{1}_{(0,a]}) = x$*, with* $a = \infty$ *meaning sale of the whole portfolio. If* $0 \leq \widetilde{w} \leq w$ *and* $\mathrm{K}(\widetilde{w}) = x$*, then* $w\mathbf{1}_{(a,\infty)} \succeq_R w - \widetilde{w}$*. Moreover, if* $w \succeq_R \bar{w}$*,* $x \leq \mathrm{K}(\bar{w})$*, and* $a, \bar{a}$ *satisfy* $\mathrm{K}(w\mathbf{1}_{(0,a]}) = \mathrm{K}(\bar{w}\mathbf{1}_{(0,\bar{a}]}) = x$*, then* $w\mathbf{1}_{(a,\infty)} \succeq_R \bar{w}\mathbf{1}_{(\bar{a},\infty)}$.

Proof of Lemma 6. The cutoff a exists because the map $b \mapsto \mathrm{K}(w\mathbf{1}_{(0,b]})$ is continuous, starts at zero, and converges to $\mathrm{K}(w)$. For the single-portfolio claim, the K component holds with equality because $\mathrm{K}(w\mathbf{1}_{(a,\infty)}) = \mathrm{K}(w) - x = \mathrm{K}(w - \widetilde{w})$. It remains to compare upper tails. First approximate w and $\widetilde{w}$ by portfolios that are constant on finitely many ordered cells $0 < r_1 < \cdots < r_N$, and write l_j for the market value liquidated from cell j. If there are indices $i < j$ such that $l_i < w_i$ and $l_j > 0$, move an after-tax cash amount ε of liquidation from cell j to cell i. Since $k(r_i) \geq k(r_j)$, this changes sold market value by $+\varepsilon/k(r_i)$ in the lower cell and $-\varepsilon/k(r_j)$ in the higher cell. The liquidation target is unchanged, and every upper-tail residual weakly increases: tails above r_j are unchanged, tails between r_i and r_j gain $\varepsilon/k(r_j)$, and tails below r_i gain $\varepsilon/k(r_j) - \varepsilon/k(r_i) \geq 0$. Iterating until no such inversion remains gives the lower-interval liquidation. Passing from finite partitions to measurable portfolios follows by monotone approximation of the cumulative liquidation and tail integrals.

Now suppose $w \succeq_R \bar{w}$ and $x \leq \mathrm{K}(\bar{w})$. The K component is $\mathrm{K}(w\mathbf{1}_{(a,\infty)}) = \mathrm{K}(w) - x \geq \mathrm{K}(\bar{w}) - x = \mathrm{K}(\bar{w}\mathbf{1}_{(\bar{a},\infty)})$. For upper tails, removing the first x units of after-tax $k(r)w(r)\,dr$-mass from the left tail preserves first-order right-tail ordering: a portfolio with weakly larger upper tails cannot have smaller residual upper tails after the same left-tail cash extraction. Equivalently, use the exchange argument above on common finite partitions and pass to the limit. Hence $w\mathbf{1}_{(a,\infty)} \succeq_R \bar{w}\mathbf{1}_{(\bar{a},\infty)}$. ■

Proof of Lemma 5. Fix the aggregate realization z. For upper tails and $r \leq 1$,

$$\int_a^\infty A_z Q\boldsymbol{\delta}_r(s)\,ds = z\int_{a/(zr)}^\infty Rf(R)\,dR \leq z\int_{a/z}^\infty Rf(R)\,dR = \int_a^\infty A_z\chi(s)\,ds.$$

Multiplication by $\delta_I k(r) \geq 1$ gives the market-tail inequality for $A_z(\delta_I k(r)\chi)$ against $A_z Q\boldsymbol{\delta}_r$. For the after-tax component, $\mathrm{K}(A_z(\delta_I k(r)\chi)) = \delta_I k(r) H_{k,z}(1)$ and $\mathrm{K}(A_z Q\boldsymbol{\delta}_r) = H_{k,z}(r)$, so (31) gives K-dominance. Hence r satisfies the state-z harvest condition (32).

■

Proposition 11 (Statewise lower-harvest cutoff replacement)**.**

Suppose $\mathrm{K} = \mathrm{K}_{\mathrm{FUL}}$ *and* $z\mathbb{E}[R] \geq 1$ *for every* $z \in \mathcal{Z}$*. For every* $z \in \mathcal{Z}$*, let* $\mathcal{H}_z^L = (0, r_z^H]$*. Then, for every* $z \in \mathcal{Z}$ *and every feasible control* $(\widetilde{w}, I) \in \Gamma(w)$*, there is a cutoff* $a \in [r_z^H, \infty]$ *and*

an investment I^L *such that* $(w\mathbf{1}_{(0,a]}, I^L) \in \Gamma(w)$, $y + \mathrm{K}(w\mathbf{1}_{(0,a]}) - I^L = y + \mathrm{K}(\widetilde{w}) - I$, *and*

$$A_z\big(Q(w - w\mathbf{1}_{(0,a]}) + \delta_I I^L \chi\big) \succeq_R A_z\big(Q(w - \widetilde{w}) + \delta_I I \chi\big). \tag{35}$$

If all harvest intervals coincide, $\mathcal{H}_z^L = (0, r_H]$ *for every* $z \in \mathcal{Z}$, *then* $(0, r_H]$ *is the common maximal dominance-inducing harvest interval. In that case the same replacement can be chosen in* $\Gamma^L_{r_H}(w)$ *and right-dominates the original continuation state for every* $z \in \mathcal{Z}$.

Proof of Proposition 11. Fix $z \in \mathcal{Z}$ and put $h := r_z^H$. Let $x_0 := \mathrm{K}(\widetilde{w})$ and $C := x_0 - I$, so current consumption is $y + C$. First harvest every unharvested tax lot in $(0, r_z^H]$ and add the resulting after-tax proceeds to investment. If a unit of market value at $r \in (0, r_z^H]$ is changed from kept to harvested-and-reinvested, the state-z continuation contribution changes from $A_z Q \boldsymbol{\delta}_r$ to $A_z(\delta_I k(r)\chi)$; since $(0, r_z^H] = \mathcal{H}_z^L$, the latter right-dominates the former in aggregate state z. Right dominance is additive, so harvesting all remaining tax lots in $(0, r_z^H]$ weakly improves the state-z continuation state and preserves current consumption. Denote the resulting total after-tax liquidation proceeds by x_1 and the resulting investment by $I_1 = I + x_1 - x_0$. Feasibility is preserved because $I_1 \le y + x_1$, and $x_1 \ge \mathrm{K}(w\mathbf{1}_{(0,r_z^H]})$.

Holding x_1 and I_1 fixed, choose $a_1 \in [h, \infty]$ with $\mathrm{K}(w\mathbf{1}_{(0,a_1]}) = x_1$ and replace the liquidated portfolio by $w\mathbf{1}_{(0,a_1]}$. Lemma 6 gives $w\mathbf{1}_{(a_1,\infty)} \succeq_R w - \widetilde{w}_1$, where $\widetilde{w}_1$ is the post-harvest liquidation. Proposition 3 then gives $A_z Q(w\mathbf{1}_{(a_1,\infty)}) \succeq_R A_z Q(w - \widetilde{w}_1)$, and adding the common investment portfolio $A_z(\delta_I I_1 \chi)$ preserves dominance. Set $(\widetilde{w}^L, I^L) := (w\mathbf{1}_{(0,a_1]}, I_1)$. Since $a_1 \ge r_z^H$, this control has the required statewise lower-harvest cutoff form, has the same current consumption $y + C$, and satisfies (35) for the fixed aggregate state z.

If all harvest intervals coincide, set $h = r_H$. The construction above then no longer depends on the aggregate realization. Since $(0, r_H]$ is the statewise harvest interval for every $z \in \mathcal{Z}$, the same control belongs to $\Gamma^L_{r_H}(w)$ and the preceding argument gives right-dominance for every $z \in \mathcal{Z}$. ∎

Proof of Proposition 4. Let $H := (0, r_H]$ and let $h := (w - \widetilde{w})\mathbf{1}_H$ be the part of the core harvest region that the original control leaves unsold. Set $\widetilde{w}^H := \widetilde{w} + h$ and $I^H := I + \mathrm{K}(h)$. Then $(\widetilde{w}^H, I^H) \in \Gamma^H(w)$, feasibility is preserved because $I^H \le y + \mathrm{K}(\widetilde{w}) + \mathrm{K}(h) = y + \mathrm{K}(\widetilde{w}^H)$, and current consumption is unchanged.

Fix $z \in \mathcal{Z}$. Since $H \subseteq \mathcal{H}_z^L$, each unit tax lot $\boldsymbol{\delta}_r$ with $r \in H$ satisfies $A_z(\delta_I k_{\mathrm{FUL}}(r)\chi) \succeq_R A_z Q \boldsymbol{\delta}_r$. Integrating this comparison against the nonnegative measure $h(r)\,dr$ and adding the unchanged part of the continuation portfolio gives

$$G_z(w, \widetilde{w}^H, I^H) \succeq_R G_z(w, \widetilde{w}, I).$$

Thus the constructed core-harvesting control weakly improves every aggregate-state continuation while preserving current consumption. If V is right-increasing, both the discounted objec-

tive and the sup-min objective weakly increase under this replacement. Since $\Gamma^H(w) \subseteq \Gamma(w)$, the suprema over $\Gamma(w)$ and $\Gamma^H(w)$ are equal.

Let $(\widetilde{w}, I) \in \Gamma^H(w)$ and set $x := \mathrm{K}(\widetilde{w})$. Since the control harvests the core, $x \geq \mathrm{K}(w\mathbf{1}_{(0,r_H]})$. Choose $a \in [r_H, \infty]$ such that $\mathrm{K}(w\mathbf{1}_{(0,a]}) = x$. Then $(w\mathbf{1}_{(0,a]}, I) \in \Gamma^L_{r_H}(w)$ and current consumption is unchanged. By Lemma 6, $w\mathbf{1}_{(a,\infty)} \succeq_R w - \widetilde{w}$. Proposition 3 implies

$$A_z Q(w\mathbf{1}_{(a,\infty)}) \succeq_R A_z Q(w - \widetilde{w}) \qquad \text{for every } z \in \mathcal{Z}.$$

Adding the common investment portfolio $A_z(\delta_I I\chi)$ preserves right dominance, so

$$G_z(w, w\mathbf{1}_{(0,a]}, I) \succeq_R G_z(w, \widetilde{w}, I) \qquad \text{for every } z \in \mathcal{Z}.$$

If V is right-increasing, this proves equality of the suprema over $\Gamma^H(w)$ and $\Gamma^L_{r_H}(w)$.

Applying the first replacement and then the second turns any feasible full control into a $\Gamma^L_{r_H}(w)$ control with the same current consumption and weakly better continuation states. Hence any attained full Bellman maximum admits a $\Gamma^L_{r_H}$ maximizer; uniqueness of the full maximizer forces that maximizer to belong to $\Gamma^L_{r_H}(w)$. ■

Lemma 7 (Order-compatible core-harvest feasibility)**.** *Suppose* $\mathrm{K} = \mathrm{K}_{\mathrm{FUL}}$ *and* $z\mathbb{E}[R] \geq 1$ *for every* $z \in \mathcal{Z}$*. If* $w \succeq_R \bar{w}$*, then every* $\Gamma^L_{r_H}$ *control at* $\bar{w}$ *can be matched by a* $\Gamma^L_{r_H}$ *control at* w *with the same current consumption and with a right-dominating continuation state for every* $z \in \mathcal{Z}$*.*

Proof of Lemma 7. Let $(\bar{\widetilde{w}}^L, \bar{I}) = (\bar{w}\mathbf{1}_{(0,\bar{a}]}, \bar{I}) \in \Gamma^L_{r_H}(\bar{w})$ and put $\bar{x} := \mathrm{K}(\bar{\widetilde{w}}^L)$. Since $w \succeq_R \bar{w}$, $\bar{x} \leq \mathrm{K}(\bar{w}) \leq \mathrm{K}(w)$. Choose $a \in [0, \infty]$ with $\mathrm{K}(w\mathbf{1}_{(0,a]}) = \bar{x}$. The full control $(w\mathbf{1}_{(0,a]}, \bar{I})$ is feasible at w and delivers the same current consumption as the $\Gamma^L_{r_H}$ control at $\bar{w}$. By Lemma 6, $w\mathbf{1}_{(a,\infty)} \succeq_R \bar{w}\mathbf{1}_{(\bar{a},\infty)} = \bar{w} - \bar{\widetilde{w}}^L$. Proposition 3 implies that, for every $z \in \mathcal{Z}$, the continuation state generated by $(w\mathbf{1}_{(0,a]}, \bar{I})$ right-dominates the state generated by $(\bar{\widetilde{w}}^L, \bar{I})$. Apply Proposition 4 first to harvest the core of the feasible control $(w\mathbf{1}_{(0,a]}, \bar{I})$ and then to complete it as a lower-cutoff control. The final control lies in $\Gamma^L_{r_H}(w)$, has the same current consumption, and weakly improves the continuation state in every aggregate realization. Transitivity of $\succeq_R$ gives the desired match. ■

Proof of Proposition 10. For $r \neq 1$, the identity $1 - \tau_{a,\theta}(r)(1 - r^{-1}) = a + (1-a)r^{-\theta}$ is immediate, and the value at $r = 1$ follows from l'Hopital's rule. Since $0 \leq \theta \leq 1$, the ratio $(1 - r^{-\theta})/(1 - r^{-1})$ lies in $[0, 1]$ on both sides of $r = 1$, so $0 \leq \tau_{a,\theta} \leq 1 - a$. Positivity of k is clear, and $k'(r) = -(1-a)\theta r^{-\theta-1} \leq 0$ wherever the derivative is taken.

Using (29),

$$H_{k,z}(r) = z \int_0^\infty Rf(R)\{a + (1-a)(zRr)^{-\theta}\}\, dR = az\mu_1 + (1-a)z^{1-\theta}\mu_{1-\theta}r^{-\theta},$$

which proves (33). If $a = 0$, this is $z^{1-\theta}\mu_{1-\theta}k(r)$, so compatibility holds with $h_z = 0$. If $a > 0$, write

$$H_{k,z}(r) = z^{1-\theta}\mu_{1-\theta}k(r) + a\{z\mu_1 - z^{1-\theta}\mu_{1-\theta}\}.$$

The residual term is a nonnegative constant whenever $z^\theta\mu_1 \geq \mu_{1-\theta}$, giving Assumption 4(ii). With no aggregate risk this condition is $\mu_1 \geq \mu_{1-\theta}$, which follows from Jensen's inequality and $\mu_1 \geq 1$.

Substituting (33) into (31) gives (34). For fixed z, write $q = r^{-\theta} \geq 1$. The difference between the left- and right-hand sides of (34) is affine in q and is nonpositive at $q = 1$ unless $\delta_I = 1$. If its slope is nonpositive, the test is empty on $(0, 1]$ except for possible boundary cases; if its slope is positive, the test is of the form $q \geq q_i^*$. Thus the tested tax lots form a lower interval in r. A finite intersection of lower intervals is a lower interval. ■

Proof of Theorem 3. If V is right-increasing and $(\widetilde{w}, I) \in \Gamma(w)$, apply the two replacements in Proposition 4: first harvest the core, then complete the control as a lower cutoff. This gives a control $(\widetilde{w}^L, I^L) \in \Gamma^L_{r_H}(w)$ with the same current consumption and

$$G_z(w, \widetilde{w}^L, I^L) \succeq_R G_z(w, \widetilde{w}, I) \qquad \text{for every } z \in \mathcal{Z}.$$

Therefore $V(G_z(w, \widetilde{w}^L, I^L)) \geq V(G_z(w, \widetilde{w}, I))$ state by state. The discounted and sup-min objectives under the $\Gamma^L_{r_H}$ replacement are weakly larger than under the original control, so $\mathbb{T}V \leq \mathbb{T}^L V$ and $\mathbb{P}V \leq \mathbb{P}^L V$. The reverse inequalities are immediate from $\Gamma^L_{r_H}(w) \subseteq \Gamma(w)$.

If a full Bellman supremum is attained at $(\widetilde{w}, I)$, the same replacement gives a $\Gamma^L_{r_H}(w)$ control whose objective value is weakly higher. Since the full and restricted suprema are equal, this replacement is also a full maximizer. If the full maximizer is unique, it must already belong to $\Gamma^L_{r_H}(w)$.

To show preservation of right-increasingness, let $w \succeq_R \bar{w}$ and take any $\Gamma^L_{r_H}$ control at $\bar{w}$. Lemma 7 gives a $\Gamma^L_{r_H}$ control at w with the same current consumption and statewise right-dominating continuation states. Since Assumption 3 makes u increasing, the corresponding discounted objective at w is at least as large as the one at $\bar{w}$; the same is true for the sup-min objective because the map $(c, v) \mapsto \min\{c, v\}$ is increasing in both arguments. Taking suprema over $\Gamma^L_{r_H}$ controls at $\bar{w}$ proves that $\mathbb{T}^L V$ and $\mathbb{P}^L V$ are right-increasing. Since the full and restricted operators agree at right-increasing V, $\mathbb{T}V$ and $\mathbb{P}V$ are right-increasing. ■

Proof of Corollary 3. The discounted initial value $V_0^{\mathbb{T}} = 0$ is right-increasing. The sup-min initial value $V_0^{\mathbb{P}}(w) = y + \mathrm{K}_{\mathrm{FUL}}(w)$ is right-increasing by (10); it is also the one-period value for the full and restricted problems because selling the whole portfolio and setting $I = 0$ is feasible in both $\Gamma(w)$ and $\Gamma^L_{r_H}(w)$.

Consider first the discounted iteration. If $V_n^{\mathbb{T}} = V_n^{\mathbb{T},L}$ and the common value is right-increasing, then Theorem 3 implies

$$V_{n+1}^{\mathbb{T}} = \mathbb{T}V_n^{\mathbb{T}} = \mathbb{T}^L V_n^{\mathbb{T},L} = V_{n+1}^{\mathbb{T},L},$$

and the next common value is right-increasing. Induction gives finite-horizon equality and right-increasingness. Since $\mathbb{T}$ is a contraction, $V_n^{\mathbb{T}} \to V^{\mathbb{T}}$ in norm. The restricted operator satisfies the same contraction estimate as $\mathbb{T}$, so $V_n^{\mathbb{T},L}$ converges in norm to its restricted fixed point. The finite-horizon equality forces the two limits to coincide. Pointwise inequalities are preserved under norm convergence, so the common fixed point is right-increasing.

The same induction, now using the sup-min part of Theorem 3, gives $V_n^{\mathbb{P}} = V_n^{\mathbb{P},L}$ and right-increasingness for every n. The full sup-min sequence is decreasing by Proposition 1, so the restricted sequence has the same pointwise limit. Proposition 1 gives $V^{\mathbb{P}} = \mathbb{P}V^{\mathbb{P}}$ for the full limit. Since the limit is right-increasing, Theorem 3 gives $\mathbb{P}V^{\mathbb{P}} = \mathbb{P}^L V^{\mathbb{P}}$, so the common limit is also a fixed point of $\mathbb{P}^L$. Pointwise limits preserve right-dominance inequalities, hence $V^{\mathbb{P}}$ is right-increasing.

It remains to verify the maximizer statements. Theorem 2 gives attainment of the finite-horizon full Bellman maxima and of the stationary full maxima at $V^{\mathbb{T}}$ and $V^{\mathbb{P}}$. For any finite-horizon full maximizer, the $\Gamma_{r_H}^L$ replacement constructed in Theorem 3 has the same current consumption and weakly better continuation states. It is therefore also a full maximizer, and it lies in $\Gamma_{r_H}^L$.

For the stationary claim, fix a state w and let $\mathbb{H} \in \{\mathbb{T}, \mathbb{P}\}$, with limiting value V^* equal to $V^{\mathbb{T}}$ or $V^{\mathbb{P}}$. Since V^* is right-increasing, Theorem 3 gives $\mathbb{H}V^* = \mathbb{H}^L V^*$. Hence

$$\sup_{(\widetilde{w},I)\in\Gamma(w)} B_{V^*}^{\mathbb{H}}(w,\widetilde{w},I) = V^*(w) = \sup_{(\widetilde{w},I)\in\Gamma_{r_H}^L(w)} B_{V^*}^{\mathbb{H}}(w,\widetilde{w},I),$$

where $B_{V^*}^{\mathbb{H}}$ denotes the one-period objective of the relevant Bellman operator. The finite-horizon replacement argument applied at V^* gives existence of a $\Gamma_{r_H}^L$ stationary maximizer. Any restricted stationary maximizer attains the common value above and is therefore full-problem optimal. If the full stationary maximizer is unique, it must coincide with the restricted maximizer and hence belongs to $\Gamma_{r_H}^L(w)$. ■

Proof of Corollary 4. If $\delta_I = 1$, the formula for (11) gives $\mathcal{H}_z^L = (0,1]$ for every $z \in \mathcal{Z}$ under $z\mathbb{E}[R] \geq 1$. Hence $r_H = 1$, and all loss lots are in the common harvest region.

It remains to rule out round trips above 1. Fix $r > 1$ and $z \in \mathcal{Z}$. Since $\delta_I = 1$, $k_{\mathrm{FUL}}(r) \leq 1$. For any upper-tail threshold q,

$$z\int_{q/(zr)}^{\infty} Rf(R)\,dR \geq k_{\mathrm{FUL}}(r)z\int_{q/z}^{\infty} Rf(R)\,dR,$$

so the market-tail part of $A_z Q\boldsymbol{\delta}_r \succeq_R A_z(k_{\mathrm{FUL}}(r)\chi)$ holds. For the liquidation component,

$$\mathrm{K}_{\mathrm{FUL}}(A_z Q\boldsymbol{\delta}_r) - \mathrm{K}_{\mathrm{FUL}}(A_z(k_{\mathrm{FUL}}(r)\chi)) = \tau(1-r^{-1})\{z(1-\tau)\mathbb{E}[R] + \tau - 1\} \geq 0,$$

where the inequality uses $r > 1$ and $z\mathbb{E}[R] \geq 1$. Thus keeping any gain lot $r > 1$ right-dominates selling it and reinvesting its after-tax proceeds.

Now let $(w\mathbf{1}_{(0,a]}, I) \in \Gamma_1^L(w)$ and put $x := \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,a]}) - I$, the liquidation-and-investment contribution to current consumption apart from y. If $x \leq \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,1]})$, set $b = 1$ and $I' := \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,1]}) - x$. Original feasibility gives $x \geq -y$, so $I' \leq y + \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,1]})$ and $(w\mathbf{1}_{(0,1]}, I') \in \Gamma_1^{sL}(w)$. If $x > \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,1]})$, choose $b \in [1,a]$ with $\mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,b]}) = x$ and set $I' = 0$. In both cases, $\mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,b]}) - I' = \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(0,a]}) - I$, so current consumption is unchanged, and $I - I' = \mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(b,a]})$.

For each $z \in \mathcal{Z}$, integrate the Dirac comparison over $r \in (b,a]$. Since $\mathrm{K}_{\mathrm{FUL}} = \mathrm{K}$ is the linear functional with kernel k_{FUL}, additivity of right dominance gives

$$A_z Q(w\mathbf{1}_{(b,a]}) \succeq_R A_z\big(\mathrm{K}_{\mathrm{FUL}}(w\mathbf{1}_{(b,a]})\chi\big) = A_z\big((I - I')\chi\big).$$

Adding the common portfolio $A_z\{Q(w\mathbf{1}_{(a,\infty)}) + I'\chi\}$ to both sides yields

$$G_z(w, w\mathbf{1}_{(0,b]}, I') \succeq_R G_z(w, w\mathbf{1}_{(0,a]}, I).$$

If V is right-increasing, the replacement weakly raises the continuation term and leaves current consumption fixed. Hence the suprema over $\Gamma_1^L(w)$ and $\Gamma_1^{sL}(w)$ coincide in both Bellman objectives. Combining this equality with the full-to-Γ_1^L restriction in Theorem 3 and Corollary 3 gives the corresponding full-to-Γ_1^{sL} restriction and the stated value-iteration consequences. ■

C Effective-Taxation Computation and Robustness

C.A MGET Computation: Numerical Algorithm

The MGET computation proceeds as follows. Fix a tax operator $\mathrm{T} \in \{\mathrm{T}_{\mathrm{FUL}}, \mathrm{T}_{\mathrm{LUL}}\}$ and a log-normal return distribution f calibrated to the target mean $\mathbb{E}[R]$ and variance $\mathbb{V}[R]$. Dividends and inflation are incorporated as in Section V.A.

i) **Grid construction.** Approximate f by a binomial tree with N steps per period. The portfolio w_t is defined on the resulting grid of cumulative returns.

ii) **Initialization.** Set $w_0(r) = rf(r)$, corresponding to a unit investment at $t = 0$.

iii) **Forward simulation.** For a candidate consumption level $\bar{c}$:

   i) At each $t = 1, \ldots, \bar{T}$, compute the optimal liquidation portfolio $\tilde{w}_t \in [0, w_t]$ using the relevant trading rule from Sections IV.C and IV.D; in the linear tax-share family,

Table 5: Leverage and effective taxation under T_{LUL}, $\delta_F = 1$: no dividends, no inflation

Panel A. Effective rate, deductible interest
$\mathbb{E}[R] - 1 = 5\%, \sigma = 30\%, \tau = 25\%, \varphi = \pi = 0$

	fair, r_b=5%		cheap, r_b=2.5%	
x	$\bar{c}(x)$	$\tau^*(x)/\tau$	$\bar{c}(x)$	$\tau^*(x)/\tau$
0.00	0.0449	0.411	0.0449	0.411
0.25	0.0459	0.327	0.0510	0.371
0.50	0.0467	0.261	0.0571	0.343
1.00	0.0479	0.167	0.0693	0.304
2.00	0.0492	0.066	0.0935	0.261
3.00	0.0497	0.025	0.1175	0.239

Panel B. Break-even rate $r_b^*(x)$
deductible; $\sigma = 30\%, \tau = 25\%, \varphi = \pi = 0$

x	$r_b^*(x)$
0.0025	5.59%
0.25	5.51%
0.50	5.44%
1.00	5.34%
2.00	5.23%
3.00	5.17%

Notes: As Table 2 but at $\varphi = \pi = 0$; definitions, normalizations, and layout are identical. The deductible break-even lies in $(b^{nd}, b^{nd}/(1-\tau)) = (4.49\%, 5.98\%)$ and decreases over the reported leverage range.

this is the LHCL restriction in Theorem 3, subject to $\mathrm{K}(\tilde{w}_t) = \bar{c} + I_t$.

ii) Update the portfolio via $w_{t+1} = Q(w_t - \tilde{w}_t) + I_t \delta_I \chi$.

iii) Track the residual liquidation value sequence $\{\mathrm{K}(w_t - \tilde{w}_t)\}_{t=1}^{\bar{T}}$.

iv) **Stability check.** If the residual liquidation value sequence is non-decreasing (or bounded away from zero) over $\bar{T} = 200$ periods, declare $\bar{c}$ feasible.

v) **Bisection.** Search over $\bar{c}$ by bisection to find $\bar{c} = \sup \mathcal{C}$.

vi) **MGET.** Compute $\tau^* = 1 - \bar{c}/\mathbb{E}[R-1]$.

For further details on the forward simulation of the law of motion on binomial grids, consult the replication files.

C.B Leverage: the $\varphi = \pi = 0$ benchmark and full loss offset

Table 2 in the main text reports leverage at the baseline market environment. Table 5 repeats it at $\varphi = \pi = 0$ — no dividends, no inflation — which isolates the leverage/tax interaction from those two margins; the two tables differ only through $\bar{c}_0$ and the effective rates it induces. Here the deductible break-even $r_b^*(x)$ (Panel B) decreases over the reported leverage range, from approximately 5.59% near the margin toward the non-deductible break-even $b^{nd} = 4.49\%$.

Under T_{FUL} (full loss offset), loss refunds monetize deductions in cash, so the closed form (21) holds in both deductibility regimes, with $r_b^{net} = r_b(1 - \tau \cdot \mathbf{1}_{\{\text{deductible}\}})$ in place of r_b. Two consequences distinguish FUL from the LUL benchmark (Tables 2 and 5). First, with deductible interest at the fair rate, $r_b^{net} = \mathbb{E}[R-1](1-\tau) < \mathbb{E}[R-1](1-\tau_0^*)$ always, so leverage is profitable without bound: $\tau^*(x)$ falls linearly and turns negative — the unbounded borrow-deduct-defer arbitrage. Second, both break-even rates are exact and *independent of leverage*: $b^{nd} = \bar{c}_0$ and $b^d = \bar{c}_0/(1-\tau)$ at every x, with b^d attained rather than merely

Table 6: Leverage under T_{FUL} (FUL): deductible-interest effective rate and break-even debt cost

Panel A. Effective rate deductible interest

x	$\bar{c}(x)$	$\tau^*(x)/\tau$
0.00	0.0459	0.324
0.25	0.0481	0.155
0.50	0.0502	−0.014
1.00	0.0544	−0.352
2.00	0.0628	−1.027
3.00	0.0713	−1.703

Panel B. Break-even rates exact, x-independent

τ	10%	25%	40%	50%
$b^{nd} = \bar{c}_0$	4.85%	4.59%	4.30%	4.07%
$b^d = \bar{c}_0/(1-\tau)$	5.39%	6.13%	7.16%	8.14%

Notes: Computed at $T_{\max} = 200$, $n_d = 20$, tolerance 10^{-8}; parameters as in Table 5 ($\mathbb{E}[R] - 1 = 5\%$, $\varphi = \pi = 0$, $\sigma = 30\%$, fair $r_b = 5\%$), with T_{FUL} in place of T_{LUL}. Panel A sets $\tau = 25\%$; Panel B varies τ. Non-deductible interest is neutral ($\tau^*(x) = \tau_0^*$) and omitted. Under FUL the deductible closed form is exact, $\bar{c}(x) = (1+x)\bar{c}_0 - x\, r_b(1-\tau)$, so all cells follow from $\bar{c}_0$; the deductible break-even $b^d = \bar{c}_0/(1-\tau)$ is exact and independent of x (full refundability implies full utilization), and deductible fair-rate leverage is profitable without bound (the effective rate falls linearly through zero).

bounded, because full refundability cashes the deduction at τ regardless of the investor's gains position. Panel B therefore reports them against τ alone.

C.C Statutory-Rate Pass-Through to the MGET

Figure 5 reports the slope panels corresponding to the level comparisons in Figure 4. The panels provide the full dividend, inflation, dispersion, and tax-system comparative statics behind the baseline pass-through estimates reported in the main text.

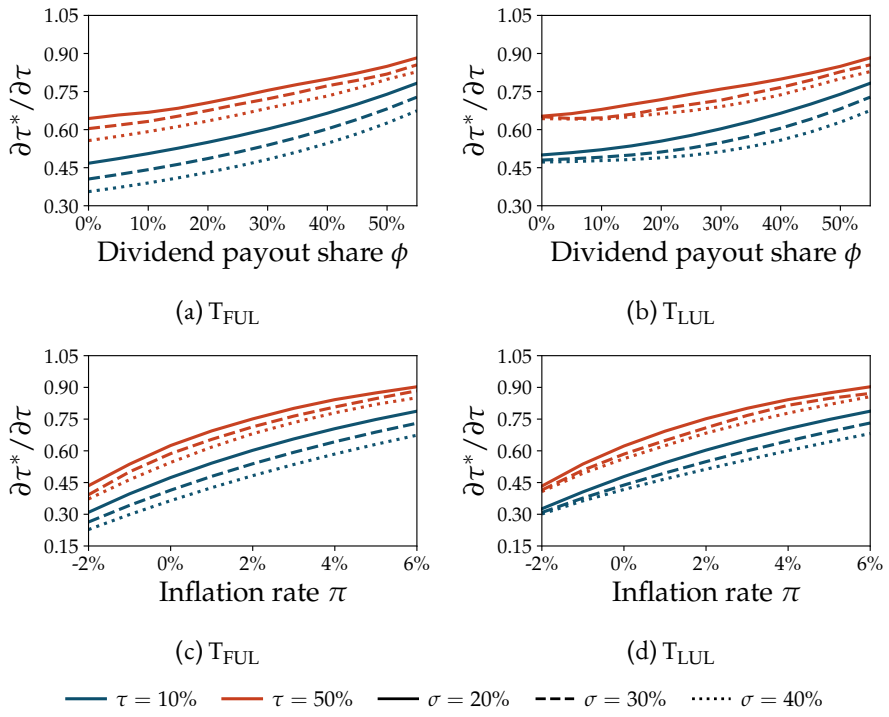


Fig. 5. Pass-through of statutory rates into the investor-side MGET. The vertical axis is $\partial\tau^*/\partial\tau$; parameters, colors, and line styles match Figure 4.

D Derivation of the Cohort Economy

This appendix derives the cohort aggregations used in Section V.B and the interpretation of their accrual-tax benchmark. The calculation holds the perpetuity trading policy fixed and introduces no new life-cycle optimization. I write $g = \mathbb{E}[R]$ for the common gross real return and $\Pi = 1 + \pi$ for gross inflation.

For each loss treatment, consider the unit-endowment perpetuity path. Let v_a denote real wealth at the start of age a, m_a wealth after returns, consumption, and living taxes but before death, and $c_a = \bar{c}$ the constant real consumption flow. With $v_0 = 1$ and $v_{a+1} = m_a$ conditional on continuation, the unit-path budget and real accrual flow are

$$m_a = g v_a - c_a - \text{tax}_a, \qquad \text{accr}_a = (g-1)v_a.$$

Living taxes include taxes on dividends as well as realized capital gains. The tax state also records embedded gains and unused loss credits. Under constructive realization, the death payment is τ times embedded gains under FUL and τ times positive embedded gains net of available loss credits under LUL. Under step-up and carry-over, the payment at death is zero.

All flows are measured in real consumption units. If the reference path is recorded in nominal units relative to prices at entry, age-a flows and end-of-period stocks are divided by Π^{a+1}, while opening wealth is divided by Π^a. This timing is also used for the accrual-tax comparison.

Under step-up and constructive realization, death resets the inherited portfolio's basis and begins a new tax history. The previous history's unused credits expire after any offset against the constructive-realization liability. Under carry-over, the inherited portfolio continues the dynasty's tax history, including its unused credits in this benchmark. Entry N_t therefore means funds beginning a new tax history; an inheritance under carry-over is not new entry.

Let $\kappa = 1 - \beta$ under step-up or constructive realization and $\kappa = 1$ under carry-over. Set $\text{dtax}_a = 0$ in the step-up and carry-over cases. Starting with a single unit-profile cohort of size N_0, the finite-date aggregates are

$$\begin{pmatrix} C_t \\ T_t \\ A_t \end{pmatrix} = \sum_{a=0}^{t} \kappa^a N_{t-a} \begin{pmatrix} c_a \\ \text{tax}_a + \beta \text{dtax}_a \\ \text{accr}_a \end{pmatrix}. \tag{36}$$

Homogeneity justifies scaling each reference profile by its entering endowment. The survival factor applies to reset cohorts. Under carry-over, the same sum instead counts continuing tax histories and is independent of changes in ownership.

Recycling determines the entrant endowments. For the two treatments that reset basis,

new entrants receive consumption, tax receipts, and estates net of death taxes:

$$N_{t+1} = C_t + T_t + \beta \sum_{a=0}^{t} (1-\beta)^a N_{t-a}(m_a - \mathrm{dtax}_a).$$

For carry-over, inherited portfolios remain in their existing tax dynasties, while recycled consumption and tax receipts finance new histories, so $N_{t+1} = C_t + T_t$. In both cases all transfers are reinvested before the next period's return. Combining these entry rules with the unit-path budget gives

$$W_t = \sum_{a=0}^{t} \kappa^a N_{t-a} v_a, \qquad W_{t+1} = N_{t+1} + \sum_{a=0}^{t} \kappa^{a+1} N_{t-a} m_a = g W_t.$$

Aggregate wealth therefore grows at g from the outset. Entrant endowments need not grow geometrically during the initial transition: they depend on the economy's current composition of tax histories.

On a constant-growth trajectory, $N_{t-a} = N_t / g^a$. Writing $q = \kappa / g$, the corresponding aggregate flows per unit of current entry are

$$\frac{1}{N_t} \begin{pmatrix} C_t \\ T_t \\ A_t \end{pmatrix} = \sum_{a=0}^{\infty} q^a \begin{pmatrix} \bar{c} \\ \mathrm{tax}_a + \beta \mathrm{dtax}_a \\ \mathrm{accr}_a \end{pmatrix}. \tag{37}$$

For bounded real reference profiles and $g > 1$, these sums converge under both types of aggregation. In particular, $C_t / N_t = \bar{c} / (1-q)$, and

$$\tau_G^* = \frac{\sum_{a \geq 0} q^a (\mathrm{tax}_a + \beta \mathrm{dtax}_a)}{\sum_{a \geq 0} q^a \mathrm{accr}_a}, \qquad \frac{\tau_G^*}{\tau} = \frac{T_t}{\tau A_t}. \tag{38}$$

The numerical calculation approximates these age sums using 200-period reference paths. Geometric age weights describe the constant-growth comparison; the finite-date construction instead uses the recycling recursion above.

The recycling rule also identifies the accrual counterfactual. Start both tax systems from the same aggregate wealth W_0 and recycle consumption and tax payments under both. The common gross return then implies

$$W_t^{\text{realization}} = W_t^{\text{accrual}} = g^t W_0, \qquad A_t = (g-1) g^t W_0.$$

An economy taxed on real accruals at rate τ from its outset consequently collects τA_t at date t. The denominator is therefore the revenue of that counterfactual economy, not merely a tax imposed on the realization economy's current accrual base. This comparison holds during the transition as well as on a constant-growth trajectory.

Finally, gain accounting explains why deferral can affect revenue flows permanently. Let H_t^n be the outstanding nominal stock of embedded gains at the start of period t, G_t^n nominal price appreciation during the period, L_t^n net capital gains realized through living trades, and D_t^n

embedded gains removed by basis resets at death. New investments begin with no embedded gain, so

$$H^n_{t+1} = H^n_t + G^n_t - L^n_t - D^n_t.$$

The death-resolution term applies under step-up and constructive realization. It is zero under carry-over, because transferring ownership leaves the embedded gains in the system.

Let P_t be the price level, $H_t = H^n_t / P_t$, R_t living realized capital gains plus dividends divided by P_{t+1}, and $D_t = D^n_t / P_{t+1}$. Dividends enter both accrued and realized income but do not enter the embedded-gain stock. With $\Pi = P_{t+1}/P_t$, the corresponding identity for real economic accruals is

$$A_t = R_t + D_t + (H_{t+1} - H_t) - (1 - \Pi^{-1})(W_t - H_t). \tag{39}$$

The final term accounts for the difference between real economic accruals and nominal gains measured in real units; $W_t - H_t$ is the real value of historical nominal basis. Without inflation it vanishes. Along a constant-growth trajectory, the embedded-gain stock grows with aggregate wealth. Part of current accrual can therefore add to that stock instead of entering current taxable realizations, even when constructive realization eventually taxes each gain. Under LUL, translating the gain identity into tax payments additionally requires the recorded loss-credit flows.